

Representation theory

Xiaoyang Chen
2025 Winter
compiled on 2026.10.06

Contents

1. The geometric Satake equivalence	3
1.1. The affine Grassmannian	3
1.1.1. Basic constructions	3
1.1.2. The affine Weyl group	5
1.2. The proof of the Satake equivalence	7
1.2.1. The Satake category	7
1.2.2. The geometric Satake transform	7
2. The Bezrukavnikov-Finkelberg equivalence	10
2.1. Step 1	10
2.1.1. Computations of equivariant cohomology	10
2.1.2. The involved categories	11
3. Koszul duality	11
3.1. A first glance: Koszul duality for finite type groups	12
3.1.1. Highest weight categories	12
3.1.2. Where is Koszul duality?	19
3.2. Koszul Duality for Kac–Moody groups	19
3.2.1. Kac–Moody lie algebras and Kac–Moody groups	20
3.2.2. A realization result	25
3.2.3. Application 1: The equivariant (irreducible) side	32
3.2.4. Basics on (completed) monodromic categories	41
3.2.5. Application 2: The monodromic (tilting) side	50
4. The Bezrukavnikov-Arkhipov equivalence	50
5. Mixed hodge modules	51
5.1. Pure and mixed hodge structures	51
5.1.1. Preliminaries on filtered objects	51
5.1.2. Basic definitions	51
6. Nori motives	52
7. Appendix	53
7.1. Filtered objects	53
Bibliography	54

1. The geometric Satake equivalence

1.1. The affine Grassmannian

1.1.1. Basic constructions

Notation 1.1.1.1. Let k be a field, and $F = k((t))$ the local function field. Then let $\mathcal{O} = k[[t]]$ denote the ring of integers. We let $D_R = \text{Spec } R[[t]]$ and $D_R^* = \text{Spec } R((t))$ denote the (punctured) formal disc.

Definition 1.1.1.1. Let G be a smooth affine k -group. We let Gr_G , the affine Grassmannian of G be the functor of points given by $\text{Gr}_{G(R)} = \{(\mathcal{E}, \beta)\}$, where $\mathcal{E}$ is a G -torsor on D_R and $\beta : \mathcal{E}|_{D_R^*} \rightarrow \mathcal{E}^0|_{D_R^*}$ be a trivialization of $\mathcal{E}$ on the punctured disc (here $\mathcal{E}^0$ is the trivial bundle). We shall call the data β a modification of the bundle $\mathcal{E}$, sometimes denoted by $\beta : \mathcal{E} \rightarrow \mathcal{E}^0$

Note that this definition makes sense for general group schemes $\underline{G}$ over $\mathcal{O}$ or D_k (whereas in the above case $\underline{G} = G_{\mathcal{O}}$), which we also denote by $\text{Gr}_{\underline{G}}$.

Example 1.1.1.2. As a first example, take $G = \text{GL}_n$. Then $\text{Gr}_{G(R)}$ admits another functor of points description, given by the set of lattices $\Lambda \hookrightarrow R((t))^n$, where Λ is a projective $R[[t]]$ -module such that $\Lambda \otimes_{R[[t]]} R((t)) = R((t))^n$. To see that this is equivalent to the above definition, notice that a GL_n -principal bundle is equivalent to an n -dimensional vector bundle, and the projective module is the data for $\mathcal{E}$, and the second isomorphism is the trivialization away from 0.

Proposition 1.1.1.2. $\text{Gr}_{\underline{G}}$ is represented by an ind-scheme ind-of-finite-type over k . Moreover, if $\underline{G}$ is reductive, then $\text{Gr}_{\underline{G}}$ is ind-projective.

Proof: We only prove the case where G is constant, since the general fact relies on the fact that for any group $\underline{G}$ over $k[[t]]$, there is a faithful representation $\underline{G} \hookrightarrow \text{GL}_n$ for some n such that $\frac{\text{GL}_n}{\underline{G}}$ is quasi-affine, and when $\underline{G}$ is reductive, it is affine. In the constant case, we can obviously embed G into GL_n over k with the above properties and base change to $k[[t]]$. In this case, we have a locally closed embedding $\text{Gr}_G \hookrightarrow \text{Gr}_{\text{GL}_n}$ which is closed in the reductive case. It thus remains to show that Gr_{GL_n} is ind-projective.

To show this, we note that for every lattice $\Lambda \hookrightarrow R((t))^n$, there is some N such that $t^{-N}R[[t]]^n \hookrightarrow \Lambda \hookrightarrow t^N R[[t]]^n$. By definition, the functor of points for lattices Λ between $t^{-N}R[[t]]^n$ and $t^N R[[t]]^n$ is $\text{Gr}(2nN)$, the usual grassmannian. By the Plucker embedding, $\text{Gr}(2nN)$ is projective, and by definition $\text{Gr}_{\text{GL}_n} = \text{colim}_N \text{Gr}(2nN)$. Moreover, note that the transition maps are closed embeddings, so we have Gr_{GL_n} ind-projective. $\square$

We then give another characterization of the affine Grassmannian.

Definition 1.1.1.3. Let X denote a presheaf over $\mathcal{O}$. We define the n -th jet space of X $L^n X$ to be the presheaf over k such that $L^n X(R) = X\left(\frac{R[[t]]}{t^n}\right)$. Moreover, we define the positive loop space of X $L^+ X$ to be the presheaf over k such that $L^+ X(R) = X(R[[t]])$.

Then let X denote a presheaf over K . We define the loop space LX to be the presheaf over k such that $LX(R) = X(R((t)))$.

For X a presheaf over k , we write $L^n X$, $L^+ X$, LX short for $L^n(X \otimes_k \mathcal{O})$, $L^+(X \otimes_k \mathcal{O})$, and $L(X \otimes_k K)$.

Example 1.1.1.3. In particular, for $\underline{G}$ a group scheme over $\mathcal{O}$, we may define the loop group LG and the positive loop group L^+G , which would be of fundamental importance in the theory.

Theorem 1.1.1.4. $\text{Gr}_{\underline{G}} = LG/L^+G$, where the right hand side is identified with the fpqc quotient stack.

Proof: We first construct a natural action $LG \curvearrowright \text{Gr}_{\underline{G}}$, given on R -points by $g \cdot (\mathcal{E}, \beta) = (\mathcal{E}, g\beta)$. Notice that we have a natural identification of $LG(R)$ with the set of data $(\mathcal{E}, \beta, \varepsilon)$, where $(\mathcal{E}, \beta) \in \text{Gr}_{\underline{G}}(R)$, and $\varepsilon : \mathcal{E} \rightarrow \mathcal{E}^0$ is a global trivialization. We have this isomorphism by mapping $g \in LG(R)$ to $g \mapsto (\mathcal{E}^0, g, \text{id})$ and the inverse $(\mathcal{E}, \beta, \varepsilon) \mapsto \beta \cdot \varepsilon$. Thus the projection $LG \twoheadrightarrow \text{Gr}_{\underline{G}}$ to the first two factors clearly identifies LG as a L^+G -torsor over $\text{Gr}_{\underline{G}}$. $\square$

We now introduce the global version of the affine Grassmannian.

Definition 1.1.1.5. Let X be a reduced connected curve over k (not necessarily smooth), and let $x \in |X|$ be a closed point. Let $\underline{G}$ be a smooth affine group scheme over X , and we define $\text{Gr}_{\underline{G}, x}$ to be the presheaf over k sending R to pairs $(\mathcal{E}, \beta)$, where $\mathcal{E}$ is a $\underline{G}$ -torsor on X_R , and $\beta : \mathcal{E}|_{X_R^*} \rightarrow \mathcal{E}^0|_{X_R^*}$ be a trivialization of $\mathcal{E}$ away from x (here X_R^* is the the base change of $X^* = X - x$ to R).

Definition 1.1.1.6. Let $x \in |X|$ be a closed point in a reduced connected curve over k . We let $\mathcal{O}_x$ denote the completed local ring at x (since we will not use the usual stalk in this section), and let $D_{x, R}$ denote $\text{Spec } \mathcal{O}_x \hat{\otimes} R$.

Definition 1.1.1.7. Let $\underline{G}_x = \underline{G} \times_X x$. Moreover, let $\text{res} : \text{Gr}_{\underline{G}, x} \rightarrow \text{Gr}_{\underline{G}_x}$ induced by restricting a $\underline{G}$ -torsor on X_R to $D_{x, R}$.

Proposition 1.1.1.8. The above restriction map is an isomorphism.

To prove [Proposition 1.1.1.8](#), we need the Beauville-Laszlo's descent theorem.

Theorem 1.1.1.9 (Beauville-Laszlo). Let $p : A \rightarrow \tilde{A}$ be a map of commutative rings. We fix a non-zero-divisor $f \in A$ and assume that $p(f)$ is also not a zero divisor. Moreover, we assume that the natural map $A/f \rightarrow \tilde{A}/p(f)$ is an isomorphism. Let $\mathbf{Mod}_A^f$ denote the f -torsion-free A -modules, and

$\text{Desc}(A, \tilde{A}, f)$ denote the category of the descent datum (M_1, M_2, φ) where $M_1 \in \mathbf{Mod}_{A[f^{-1}]}$, $M_2 \in \mathbf{Mod}_{\tilde{A}}^{p(f)}$, and $\varphi : M_1 \otimes_{A[f^{-1}]} \tilde{A}[p(f)^{-1}] \rightarrow M_2 \otimes_{\tilde{A}} \tilde{A}[p(f)^{-1}]$ is an isomorphism.

Now, by flatness of localization, we have $\varphi : M[f^{-1}] \otimes_{A[f^{-1}]} \tilde{A}[p(f)^{-1}] \simeq (M \otimes_A \tilde{A})[p(f)^{-1}]$, so $M \mapsto (M[f^{-1}], M \otimes_A \tilde{A}, \varphi)$ gives a functor $\mathbf{Mod}_A^f \rightarrow \text{Desc}(A, \tilde{A}, f)$. This functor is then an equivalence, and finiteness, flatness, and finite projectivity are preserved under descent.

Proof: We now prove [Proposition 1.1.1.8](#). We directly construct an inverse to res , namely, given an $\underline{G}_x$ -torsor on $D_{x,R}$ with a trivialization β on $D_{x,R}^*$, we glue it with the trivial $\underline{G}$ -torsor on X_R^* by β via [Theorem 1.1.1.9](#) (we should find a vector bundle on X such that it is the associated bundle of our principal bundles, but I'm temporarily not quite sure on how to do this). $\square$

Corollary 1.1.1.10. The loop group $L\underline{G}_x$ represents the k -presheaf sending R to triples $(\mathcal{E}, \alpha, \beta)$, $\mathcal{E}$ is a $\underline{G}$ -torsor on X , α a trivialization of $\mathcal{E}|_{D_{x,R}}$, and β a trivialization of $\mathcal{E}|_{X_R^*}$.

remark 1.1.1.4. There is an alternate construction of loopgroups and affine Grassmannians over the mixed characteristic world, sometimes called the Witt Grassmannian. Namely, this time, $L^+G(R) = G(W(R))$, $LG(R) = G(W(R)[p^{-1}])$, $\text{Gr}_G = LG/L^+G$. They are sheaves over the perfect algebras over $\mathbb{F}_p$.

1.1.2. The affine Weyl group

Notation 1.1.2.1. From now on, we assume G is a reductive group over k . We fix a Borel subgroup $B \subset G$, and its maximal torus T . Denote the weight (character) lattice and coweight (cocharacter) lattice by $X^\bullet(T) = \text{Hom}(T, \mathbb{G}_m)$ and $X_\bullet(T) = \text{Hom}(\mathbb{G}_m, T)$ respectively. We let $\Phi \subset X^\bullet(T)$ denote the set of roots, and $\Phi^\vee \subset X_\bullet(T)$ denote the coroots. We denote the positive roots by Φ^+ , and the dominant coweights by $X_\bullet(T)^+$.

For $\mu \in X_\bullet(T)$ a coweight, we denote t^μ by the image of $t \in \mathbb{G}_m(K)$ by μ in $T(K)$.

Theorem 1.1.2.1. We have a natural isomorphism $G(\mathcal{O}) \backslash G(F) / G(\mathcal{O}) \simeq X_\bullet(T) / W = X_\bullet(T)^+$.

Proof: Recall that we have the Cartan decomposition by linear algebra

$$G(F) = \coprod_{\mu \in X_\bullet(T)} G(\mathcal{O}) t^\mu G(\mathcal{O})$$

which gives the desired decomposition. $\square$

Construction 1.1.2.2. Let $\mathcal{E}_1$ and $\mathcal{E}_2$ be two trivial G -torsors over D with trivializations $\varphi_i : \mathcal{E}_i \rightarrow \mathcal{E}^0$, and let $\beta : \mathcal{E}_1|_{D^*} \rightarrow \mathcal{E}_2|_{D^*}$ be an isomorphism over D^* . Then, the composition $\varphi_2 \beta \varphi_1^{-1} \in \text{Aut}(\mathcal{E}^0|_{D^*}) = G(F)$, so obviously its image in $G(\mathcal{O}) \backslash G(F) / G(\mathcal{O}) = X_\bullet(T)^+$ does not depend on the trivializations φ_i , denoted by $\text{Inv}(\beta)$, the relative position of the modification β .

Now, given the Bruhat decomposition, as in the classical case, we may define the Schubert varieties. The Schubert varieties defined via the dominant coweights by the spherical Schubert varieties. But before we start, we need a lemma on group actions and orbits.

Lemma 1.1.2.3 (orbit lemma). Let G be a smooth group scheme and X an algebraic variety. Let $G \curvearrowright X$, then for any $x \in X$, the orbit $G \cdot x \subset X$ is locally closed in X .

We omit the proof, but the key of the proof is using that the image of maps of varieties is constructible, and finally using a dense open set to deduce locally closedness.

Definition 1.1.2.4. Recall $L^+G \curvearrowright LG$ hence on the quotient LG/L^+G (note that we take quotient on the right here, so the action is not trivial). We let Gr_μ denote the L^+G -orbit (naturally equipped with the reduced structure) of the image of the point $t^\mu \in LG \twoheadrightarrow \text{Gr}$. Then, we let

$$\text{Gr}_{\leq \mu} = \coprod_{\lambda \leq \mu} \text{Gr}_\lambda$$

We call $\text{Gr}_{\leq \mu}$ the spherical Schubert varieties and Gr_μ the spherical Schubert cell.

Alternatively, one may directly describe the locally closed subset Gr_μ via the modification.

Proposition 1.1.2.5. Gr_μ can also be identified as the subset of Gr

$$\text{Gr}_\mu = \{(\mathcal{E}, \beta) \in \text{Gr} \mid \text{Inv}(\beta) = \mu\}$$

Proof: Firstly, note that $t^\mu = (\mathcal{E}, \beta) \in \text{Gr}$ has obviously $\text{Inv}(\beta) = \mu$ by the definition of relative position and the projection $LG \rightarrow \text{Gr}$. Then, note that for $(\mathcal{E}', \beta') = g(\mathcal{E}, \beta)$ for some $g \in L^+G$, we have $\text{Inv}(\beta') = \text{Inv}(\beta)$ since the action of g on $(\mathcal{E}, \beta)$ is given by left multiplication on β , and in the definition of relative position, we have a quotient of $G(\mathcal{O})$ on the left. Thus, in a single L^+G -orbit, the relative position of β is always the same. Finally, by [Theorem 1.1.2.1](#), we have

$$\text{Gr} = \coprod (L^+G \backslash LG / L^+G) / L^+G = \coprod \text{Gr}_\mu$$

so each Gr_μ is exactly the orbit such that the invariant is equal to μ . □

remark 1.1.2.2. There is a slight subtlety in the above proof by confusing the set $G(F)$ ($G(\mathcal{O})$) with the ind-scheme LG (L^+G). However, since everything is reduced, and we are working over an algebraically closed field, no serious problem arises since any closed reduced subscheme over an algebraically closed field is determined by its closed points.

Proposition 1.1.2.6. Gr_μ is a quasi-projective variety of dimension $\langle 2\rho, \mu \rangle$. Moreover, we have $\text{Gr}_{\leq \mu}$ the Zariski closure of Gr_μ , so is a (not necessarily smooth) projective variety.

Proof: By definition, the stabilizer for $L^+G \curvearrowright \text{Gr}$ of t^μ is $L^+G / (L^+G \cap t^\mu L^+G t^{-\mu})$, which is firstly finite dimensional by calculating the dimension of its tangent space, which is given by

$$\mathfrak{g}(\mathcal{O})/(\mathfrak{g}(\mathcal{O}) \cap \mathrm{Ad}_{t^\mu} \mathfrak{g}(\mathcal{O})) = \bigoplus_{\langle \alpha, \mu \rangle \geq 0} \mathfrak{g}_\alpha(\mathcal{O})/t^{\langle \alpha, \mu \rangle} \mathfrak{g}_\alpha$$

so $\dim \mathrm{Gr}_\mu = \langle 2\rho, \mu \rangle$. Then, since it is finite dimensional locally closed subscheme of an ind-projective variety Gr , it is quasi-projective.

The second assertion is more difficult. We show that the Zariski closure of Gr_μ contains Gr_λ for any $\lambda \leq \mu$, and assume the spherical Schubert variety $\mathrm{Gr}_{\leq \mu}$ is closed in Gr , so identifying $\mathrm{Gr}_{\leq \mu}$ as the closure of Gr_μ . Then, recall that for $\lambda < \mu$, there is a sequence μ_i such that $\lambda = \mu_0 < \mu_1 < \dots < \mu_n = \mu$, where μ_i and μ_{i-1} differ by a positive simple root. Thus, it suffices to show that $t^{\mu-\alpha}$ lies in the closure of Gr_μ . To show this, we construct a curve C connecting the two points $t^{\mu-\alpha}$ □

Lemma 1.1.2.7. Θ is an effective Cartier divisor corresponding to the $\mathcal{O}(1)$ -bundle.

1.2. The proof of the Satake equivalence

1.2.1. The Satake category

1.2.2. The geometric Satake transform

Now that we have proven that the functor of taking hypercohomology $H^* : \mathrm{Sat}_G \rightarrow \mathbf{Vect}_{\overline{\mathbb{Q}}_\ell}$ is a neutral Tannakian category, we may identify Sat_G with the representation category $\mathbf{Rep}_{\tilde{G}}$, where the Tannakian group $\tilde{G}$ is the algebraic group of automorphisms $\mathrm{Aut}^\otimes H^*$. We now (and the proof of theorem suffices to) identify $\tilde{G}$ with the dual group of G over $\overline{\mathbb{Q}}_\ell$.

Definition 1.2.2.1. Recall a reductive group is completely determined by its root datum $(X_\bullet(T), \Phi^\vee, X^\bullet(T), \Phi)$. For a reductive group G over an algebraically closed field with its root datum given by $(X_\bullet(T), \Phi^\vee, X^\bullet(T), \Phi)$, its dual group $\hat{G}$ to be the reductive group with root datum $(X^\bullet(T), \Phi, X_\bullet(T), \Phi^\vee)$.

Notice that the base field of $\hat{G}$ can be arbitrary.

Example 1.2.2.1. GL_n has dual GL_n , and SL_n has dual PGL_n .

Lemma 1.2.2.2. $\tilde{G}$ is a connected reductive group over $\overline{\mathbb{Q}}_\ell$.

Proof: Firstly, we claim that the sheaves IC_μ for μ the fundamental coweights (the weights dual to the simple roots) form a set of tensor generators for Sat_G . Notice that the convolution Grassmannian $\mathrm{Gr}_{\leq \mu} \tilde{\times} \mathrm{Gr}_{\leq \nu} \rightarrow \mathrm{Gr}_{\leq \mu+\nu}$ is birational. Now, given a proper birational map $f : X \rightarrow Y$, we prove IC_Y is a direct summand of $f_* \mathrm{IC}_X$. Now, take an open set U contained in the smooth locus of Y such that f is an isomorphism on U . By definition, $f_* \mathrm{IC}_X|_U = \mathrm{IC}_Y|_U = k_U$. By the perverse continuation principle applied to the sheaves $f_* \mathrm{IC}_X|_U$ and $\mathrm{IC}_Y|_U$, we have a map $\mathrm{IC}_Y \rightarrow f_* \mathrm{IC}_X$ non-zero on U , so is non-zero. Thus since IC_Y is simple, it must be a subobject of $f_* \mathrm{IC}_X$. Thus applied to our previous settings, $\mathrm{IC}_{\mu+\nu}$ is a subobject hence a direct summand of the convolution sheaf $\mathrm{IC}_\mu * \mathrm{IC}_\nu$ by the decomposition theorem.

This provides that $\tilde{G}$ is connected and algebraic (i.e. the algebra $\mathcal{O}_{\tilde{G}}(\tilde{G})$ is of finite type) because $\bigotimes_{\mu} \mathrm{IC}_{\mu}$ for μ running over fundamental coweights is a tensor generator for Sat_G . Then, since Sat_G is semisimple, we have furthermore that $\tilde{G}$ is reductive. $\square$

Theorem 1.2.2.3. Letting $(X^{\bullet}(T), \Phi^{\vee}, X_{\bullet}(T), \Phi)$ denote the root datum of G , we have the Tannakian group $\tilde{G}$ has root datum $(X_{\bullet}(T), \Phi, X^{\bullet}(T), \Phi^{\vee})$, so $\tilde{G}$ can be identified with the dual group of G .

The proof of [Theorem 1.2.2.3](#) is by first proving the easy case where G is a torus, then reducing general cases to tori.

Now, in general, the affine Grassmannians can be very non-reduced, but since we are talking about ℓ -adic sheaves over the spaces, we may temporarily get rid ourselves of the nuisances by taking $(\mathrm{Gr}_G)_{\mathrm{red}}$. Thus, from now on, Gr_G will naturally mean the reduced versions without further notices.

Lemma 1.2.2.4. [Theorem 1.2.2.3](#) holds if $G = T$ is a torus.

Proof: We first discuss the structure of Gr_T . By definition, let $T = \mathbb{G}_m^n$, then $LT(R) = T(R((t))) = (R((t))^{\times})^n$, and $L^+T(R) = (R[[t]]^{\times})^n$. Then, by definition, we have $R((t))^{\times}/R[[t]]^{\times}$ identified by $\mathbb{Z}$, identified to the lowest exponent of t in each component. Thus this tuple of integers is given equivalently by a cocharacter $\lambda \in X_{\bullet}(T) \simeq \mathbb{Z}^n$. Therefore, Gr_T is a discrete set with points identified with $X_{\bullet}(T)$. Thus, Sat_T is the category of $X_{\bullet}(T)$ -graded finite dimensional vector spaces, and H^* is the functor forgetting the grading.

Now, since $X_{\bullet}(T)$ -graded vector spaces is tensor generated by 1_{μ} where μ are fundamental coweights, so an R -point of the Tannakian groups is given by maps $1_{\mu} \rightarrow R^{\times}$, hence by the tensor-automorphism condition, are given by $\mathrm{Hom}_{\mathrm{Ab}}(X_{\bullet}(T), R^{\times})$. And by definition, this is the functor of points of the dual torus. $\square$

Now it suffices to reduce the case to $G = T$. To do this, we construct the categorical Satake transform

$$\mathrm{CT} : \mathrm{Sat}_G \rightarrow \mathrm{Sat}_T$$

which is a symmetric monoidal functor from the Satake category of G to the one of its maximal torus T , which firstly fits in the following commutative diagram

$$\begin{array}{ccc} \mathrm{Sat}_G & \xrightarrow{\mathrm{CT}} & \mathrm{Sat}_T \\ & \searrow H^* & \swarrow H^* \\ & \mathrm{Vect}_{\overline{\mathbb{Q}}_{\ell}} & \end{array}$$

Construction 1.2.2.5. The main idea of constructing CT is via parabolic induction. We fix a borel subgroup $i : B \subset G$ and let $q : B \rightarrow T$ denote the projection to the abstract Cartan subgroup. The maps induces maps on the affine Grassmannians

$$\mathrm{Gr}_T \xleftarrow{q} \mathrm{Gr}_B \xrightarrow{i} \mathrm{Gr}_G$$

Then, we have $\mathrm{Gr}_T \simeq X_\bullet(T)$, and for $\lambda \in X_\bullet(T)$, we denote $S_\lambda = i(q^{-1}(\lambda)) \subset \mathrm{Gr}_G$, which, by definition, can also be identified with $LU \cdot t^\lambda$, where $U \subset B$ is the unipotent subgroup. Moreover, we define

$$S_{\leq \lambda} = \bigcup_{\lambda' \leq \lambda} S_{\lambda'}$$

To construct and investigate the properties of the parabolic inductions, we start with some topological (and finally, cohomological) properties of S_λ .

Proposition 1.2.2.6. $S_{\leq \lambda}$ is closed with $S_\lambda \subset S_{\leq \lambda}$ an open dense subset.

Proof: We first show that $S_{\leq \lambda}$ is closed. We consider its moduli description. Let χ be a dominant integral weight, and V_χ denote the highest weight representation of G , and ℓ_χ denote its highest weight space. Then we have

$$S_{\leq \lambda} = \{(\mathcal{E}, \beta) \in \mathrm{Gr} | \beta^{-1}(\ell_\chi) \subset t^{-(x, \lambda)}(V_{\chi, \mathcal{E}})\}$$

where $V_{\chi, \mathcal{E}}$ is the associated bundle of $\mathcal{E}$, namely, $\mathcal{E} \times^G V_\chi$, which is a vector bundle over D_k . Then, these conditions can be translated to a polynomial equation by passing to matrices, so is a closed subscheme of Gr_G .

Then, for the density of $S_\lambda \subset S_{\leq \lambda}$, it actually follows from the construction of the curve in [Proposition 1.1.2.6](#). □

Lemma 1.2.2.7. The set $S_{< \lambda}$ is the intersection of $\overline{S}_\lambda$ with a hyperplane section of the bundle $\mathcal{O}(1)$.

Proof: Let $2\rho^\vee$ denote the sum of positive roots of G . Then we choose some embedding $T \subset B$, and consider the map

$$\mathbb{G}_m \xrightarrow{2\rho^\vee} T \subset G \subset L^+G$$

Then the L^+G action on Gr_G restricts to a $\mathbb{G}_m$ -action on Gr_G via the above inclusion. Notice that the fixed points of this action is given by Gr_T by the Cartan decomposition, and we can identify the attractor of this action with Gr_B . □

Theorem 1.2.2.8. We have an isomorphism of functors

$$H^*(-) = \bigoplus_{\lambda} H_c^*(S_\lambda, -) : \mathrm{Sat}_G \rightarrow \mathbf{Vect}_{\overline{\mathbb{Q}}_\ell}$$

Proof:

□

2. The Bezrukavnikov-Finkelberg equivalence

We attempt to give a rather complete sketch of the infamous (at least among my friends) article of Bezrukavnikov-Finkelberg. The main theorem of the article is the following:

Theorem 2.1. $D_{\text{perf}}^{\check{G}}(U_h^{\square}) \simeq D_{L^+G \rtimes \mathbb{G}_m}(\text{Gr})$, and $D_{\text{perf}}^{\check{G}}(\text{Sym}^{\square}(\check{\mathfrak{g}})) \simeq D_{L^+G}(\text{Gr})$.

where the algebras $U_h^{\square}$ and $\text{Sym}^{\square}(\check{\mathfrak{g}})$ are to be defined later. The strategy of the proof is given by the following steps.

- (1) Construct functors $\kappa_h : \mathcal{HC}_h \rightarrow \text{QCoh}^{\mathbb{G}_m} \left(M_{(\check{\mathfrak{t}}/W)^2 \Delta} \right)$ and $\kappa : \text{Coh}^{\check{G} \times \mathbb{G}_m}(\check{\mathfrak{g}}^*) \rightarrow \text{Coh}^{\mathbb{G}_m} \left(T(\check{\mathfrak{t}}^*/W) \right)$.
The functors are fully faithful on the subcategory of free objects, namely $\mathcal{HC}_h^{\text{fr}}$ and $\text{Coh}_{\text{fr}}^{\check{G} \times \mathbb{G}_m}(\check{\mathfrak{g}}^*)$.
- (2) Extend the Satake functor $S : \mathbf{Rep}(\check{G}) \rightarrow \mathbf{Perv}_{L^+G \rtimes \mathbb{G}_m}(\text{Gr})$ to fully faithful functors

$$S_h : \mathcal{HC}_h^{\text{fr}} \rightarrow D_{L^+G \rtimes \mathbb{G}_m}(\text{Gr}), \text{ and } S_{qc} : \text{Coh}_{\text{fr}}^{\check{G} \times \mathbb{G}_m}(\check{\mathfrak{g}}^*) \rightarrow D_{L^+G}(\text{Gr})$$

These functors are constructed via the fully faithful cohomology functors

$$H_{L^+G \rtimes \mathbb{G}_m}^* : \mathbf{Perv}_{L^+G \rtimes \mathbb{G}_m}(\text{Gr}) \rightarrow \text{Coh}^{\mathbb{G}_m} \left(M_{(\check{\mathfrak{t}}/W)^2 \Delta} \right), H_{L^+G}^* : \mathbf{Perv}_{L^+G}(\text{Gr}) \rightarrow \text{Coh}^{\mathbb{G}_m} \left(T(\check{\mathfrak{t}}^*/W) \right)$$

and by analysing the images, we have $\kappa_h|_{\mathcal{HC}_h^{\text{fr}}}$ and $\kappa|_{\text{Coh}_{\text{fr}}^{\check{G} \times \mathbb{G}_m}(\check{\mathfrak{g}}^*)}$ factors through the cohomology functors by S_h and S_{qc} , i.e., $\kappa_h = H_{L^+G \rtimes \mathbb{G}_m}^* \circ S_h$ and $\kappa = H_{L^+G}^* \circ S_{qc}$.

- (3) Extend S_h and S_{qc} to our final equivalences via the fully faithful inclusions $\mathcal{HC}_h^{\text{fr}} \subset D_{\text{perf}}^{\check{G}}(U_h^{\square})$ and $\text{Coh}_{\text{fr}}^{\check{G} \times \mathbb{G}_m}(\check{\mathfrak{g}}^*) \subset D_{\text{perf}}^{\check{G}}(\text{Sym}^{\square}(\check{\mathfrak{g}}))$.

We will expand the three steps one by one.

2.1. Step 1

2.1.1. Computations of equivariant cohomology

Before we start, we must compute the equivariant cohomologies of points.

Definition 2.1.1.1. Let X be a scheme and $Z \subset X$ be a closed subscheme classified by the quasi-coherent sheaf of ideals $\mathcal{I}$. We construct the deformation to the normal cone as following. Firstly, take $\mathbb{A}^1 = \text{Spec } k[\hbar]$. Then, consider the following extended Rees $\mathcal{O}_X$ -algebra $\mathcal{R}$ given by the sheaf of subalgebras in $\mathcal{O}_X[\hbar^{\pm}]$ generated by $\hbar$ and $f\hbar^{-1}$ where $f \in \mathcal{I}$. We define the deformation to the normal cone $M_X Z = \text{Spec}_X \mathcal{R}$.

Definition 2.1.1.2. For groups G with $H_G^*(\text{pt})$ is an integral domain, we define the localized equivariant cohomology $H_G^*(X)_{\text{loc}} = H_G^*(X) \otimes_{H_G^*(\text{pt})} \text{Frac}(H_G^*(\text{pt}))$.

Theorem 2.1.1.3. There is an equivalence

$$\alpha : \mathcal{O} \left(N_{(\check{\mathfrak{t}}/W)^2 \Delta} \right) \simeq \bigoplus_{\pi_1(G)} H_{L^+G \rtimes \mathbb{G}_m}^*(\text{Gr})$$

Here Δ denote the diagonal embedding.

Proof: We first treat the case where G is simply connected.

Firstly, notice $H_{L^+G \rtimes \mathbb{G}_m}^*(\mathrm{Gr}) = H_{\mathbb{G}_m}^*(L^+G \backslash LG / L^+G)$ admit a bimodule structure of $H_{L^+G}^*(\mathrm{pt})$. Since we have the homotopy equivalence $L^+G \simeq G$ by contractibility of $k[[t]]$, we have $H_{L^+G}^*(\mathrm{pt}) = H_G^*(\mathrm{pt}) = \mathcal{O}(\mathfrak{t}/W)$ by Chern-Weil theory. Thus, we have $H_{L^+G \rtimes \mathbb{G}_m}^*(\mathrm{Gr})$ is a $H_{L^+G}^*(\mathrm{pt}) \otimes H_{L^+G}^*(\mathrm{pt}) \otimes H_{\mathbb{G}_m}^*(\mathrm{pt}) = \mathcal{O}((\mathfrak{t}/W)^2 \times \mathbb{A}^1)$. This gives a structure map

$$\alpha : \mathcal{O}((\mathfrak{t}/W)^2 \times \mathbb{A}^1) \rightarrow H_{L^+G \rtimes \mathbb{G}_m}^*(\mathrm{Gr})$$

Then we show it extends to the extended Rees algebra $\mathcal{R}$, that is, we show that for $f \in \mathcal{I}_\Delta$, we have $\alpha(f)$ contains a factor of $\alpha(\hbar)$ where we regard $H_{\mathbb{G}_m}^*(\mathrm{pt}) = k[\hbar]$, i.e., the two maps $H_{L^+G}^*(\mathrm{pt}) \rightarrow H_{L^+G \rtimes \mathbb{G}_m}^*(\mathrm{Gr})/\hbar$ coincide. This is because $H_{L^+G \rtimes \mathbb{G}_m}^*(\mathrm{Gr})/\hbar = H^*(L^+G \backslash LG / L^+G)$. Thus we obtain a map

$$\alpha : \mathcal{O}(M_{(\mathfrak{t}/W)^2} \Delta) \rightarrow H_{L^+G \rtimes \mathbb{G}_m}^*(\mathrm{Gr})$$

We then show it is an equivalence. For injectivity, note $H_{L^+G \rtimes \mathbb{G}_m}^*(\mathrm{pt}) = \mathcal{O}(\mathfrak{t}/W \times \mathbb{A}^1)$. Notice by definition that $\mathcal{O}(M_{(\mathfrak{t}/W)^2} \Delta)$ is an integral domain, so it suffices to prove the localized morphism

$$\alpha_{\mathrm{loc}} : \mathcal{O}(M_{(\mathfrak{t}/W)^2} \Delta) \otimes_{\mathcal{O}(\mathfrak{t}/W \times \mathbb{A}^1)} \mathrm{Frac}(\mathcal{O}(\mathfrak{t} \times \mathbb{A}^1)) \rightarrow H_{L^+G \rtimes \mathbb{G}_m}^*(\mathrm{Gr}) \otimes_{\mathcal{O}(\mathfrak{t}/W \times \mathbb{A}^1)} \mathrm{Frac}(\mathcal{O}(\mathfrak{t} \times \mathbb{A}^1))$$

is an injection. This is because the right hand side can be identified with the localized equivariant cohomology $H_{T \times \mathbb{G}_m}^*(\mathrm{Gr})_{\mathrm{loc}}$. $\square$

2.1.2. The involved categories

Definition 2.1.2.1. Let $U = U(\check{\mathfrak{g}})$, and let $U_{\hbar}$ denote the graded enveloping algebra, explicitly, the graded $k[\hbar]$ -algebra generated by $\check{\mathfrak{g}}$ under the relations $xy - yx = \hbar[x, y]$. Equivalently, $U_{\hbar}$ is the Rees algebra of the filtered algebra U .

These enveloping algebras are equipped with the structure of $\check{G}$ -representations via the adjoint action $\check{G} \curvearrowright \check{\mathfrak{g}}$.

Definition 2.1.2.2. We define the category of $\hbar$ -Harish-Chandra bimodules $\widetilde{\mathcal{HC}}_{\hbar}$ to be the category with the following objects with the requirements:

- (1) An object M is a graded $U_{\hbar} \otimes_{k[\hbar]} U_{\hbar}$ -module equipped with the structure of a $\check{G}$ -representation $\rho : \check{G} \curvearrowright M$, such that:
- (2) the action map $U_{\hbar} \otimes_{k[\hbar]} U_{\hbar} \otimes M \rightarrow M$ is $\check{G}$ -equivariant, and,
- (3) for $x \in \check{\mathfrak{g}}$, the action of $1 \otimes x + x \otimes 1$ on M is $\hbar \cdot d\rho(x)$.

The intuition to this definition is some sort of Harish-Chandra modules of Harish-Chandra pairs.

We define $\mathcal{HC}_{\hbar} \subset \widetilde{\mathcal{HC}}_{\hbar}$ to be the full subcategory spanned by the modules M such that M is finitely generated as $U_{\hbar} \otimes 1$ -modules.

3. Koszul duality

In this part, we discuss the application of the general theory of Koszul duality to representation theory. In particular, we will be interested in the Koszul duality of the stack $\mathrm{Spec} A/\mathbb{G}_m$ where A is some quasi-hereditary algebra. In this section, all $\mathbf{Mod}_A$ or $\mathbf{Mod}_A^{\heartsuit}$ will mean $\mathbf{Mod}_A^{\omega}$ or $\mathbf{Mod}_A^{\heartsuit, \omega}$ for conventional simplicity.

3.1. A first glance: Koszul duality for finite type groups

3.1.1. Highest weight categories

In this section, we introduce a special type of Abelian categories which behaves like the category of perverse sheaves over a suitable stratified space or the BGG category $\mathcal{O}$. The history of the developments of such categories is complicated, resulting in various definitions of such categories (some of which are simple generalizations of others). We will begin with the first definition.

Definition 3.1.1.1. Let $\mathcal{A}$ be an Abelian category over some field k . We say $\mathcal{A}$ is highest weight in the sense of [CPS88] if the following conditions are met.

- (1) $\mathcal{A}$ satisfies the condition AB5 and is locally Artinian. That is, every object is generated by finite-length objects by filtered colimits. In particular, every object admit a (possibly infinite) composition series.
- (2) There is an interval finite poset Λ (that is, the intervals $[\mu, \nu] = \{\lambda \mid \mu \leq \lambda \leq \nu\} \subset \Lambda$ are finite) with the set of all non-isomorphic simple objects $\mathrm{IC}_\lambda \in \mathcal{A}$ indexed by Λ .
- (3) There is a collection of costandard objects ∇_λ indexed by Λ .
- (4) These objects contain $\mathrm{IC}_\lambda \subset \nabla_\lambda$ as a subobject, and the quotient $\nabla_\lambda / \mathrm{IC}_\lambda$ contains only IC_μ with $\mu > \lambda$ in the composition series. Moreover, $[\nabla_\mu : \mathrm{IC}_\nu]$ and $\dim_k \mathrm{Hom}(\nabla_\mu, \nabla_\nu)$ are finite.
- (5) IC_λ admit an injective envelope I_λ in $\mathcal{A}$. This injective I_λ admit another complete filtration

$$0 = F_0 \subset F_1 \subset \cdots \subset I_\lambda$$

such that $F_1 = \nabla_\lambda$, and $F_i / F_{i-1} = \nabla_\mu$ where $\mu > \lambda$ for $i > 1$. Moreover, each ∇_μ appears only finitely many times.

This is not the main focus of this section, so we only present the main theorem of the article.

Theorem 3.1.1.2 (Theorem 3.9, [CPS88]). Finite length highest weight categories with finitely many simple objects are always of the form as a right module category $\mathbf{Mod}_A^{\omega, \heartsuit}$ for some right quasi-hereditary algebra A .

This is mainly used in the paper [BGS96]. In their paper, they only focused on finite length categories with finitely many simple objects, namely, the category of perverse sheaves with respect to a nice enough finite stratification. Their definitions went as follows.

Definition 3.1.1.3. Let $\mathcal{A}$ be an Abelian category over some field k . We say $\mathcal{A}$ is highest weight in the sense of [BGS96] if the following conditions hold.

- (1) $\mathcal{A}$ is of finite length.
- (2) $\mathcal{A}$ has simple objects indexed by a finite poset Λ , denoted by IC_λ . Moreover, $\mathrm{End}(\mathrm{IC}_\lambda) = k$. The poset Λ will be referred to as the weight poset.
- (3) There are standard and costandard objects Δ_λ and ∇_λ indexed by Λ , equipped with structure morphisms $\Delta_\lambda \rightarrow \mathrm{IC}_\lambda$ and $\mathrm{IC}_\lambda \rightarrow \nabla_\lambda$.
- (4) Let $\Lambda' \subset \Lambda$ be any subposet closed under taking smaller elements. We define $\mathcal{A}_{\Lambda'}$ to be the subcategory of $\mathcal{A}$ spanned by objects with simple components $\mathrm{IC}_{\lambda'}$ for $\lambda' \in \Lambda'$. Then, for any maximal element $m \in \Lambda'$, we have $\Delta_m \rightarrow \mathrm{IC}_m$ a projective cover and $\mathrm{IC}_m \rightarrow \nabla_m$ an injective hull in $\mathcal{A}_{\Lambda'}$.
- (5) $\ker(\Delta_\lambda \rightarrow \mathrm{IC}_\lambda)$ and $\mathrm{coker}(\mathrm{IC}_\lambda \rightarrow \nabla_\lambda)$ lie in $\mathcal{A}_{<\lambda}$.
- (6) $\mathrm{Ext}^2(\Delta_\mu, \nabla_\nu) = 0$.

Lemma 3.1.1.4. We have the following orthogonality and Ext vanishing conditions:

$$\dim_k \operatorname{Hom}(\Delta_\mu, \nabla_\nu) = \delta_{\mu,\nu}, \operatorname{Ext}^i(\Delta_\mu, \nabla_\nu) = 0$$

Proof: For the first orthogonality condition, note $\Delta_\mu \rightarrow \operatorname{IC}_\mu$ is a projective cover in $\mathcal{A}_{\leq \mu}$, so every quotient of Δ_μ must have IC_μ as a factor. Similarly, every subobject of ∇_ν must intersect IC_ν . Thus, if there is a non-trivial map $\Delta_\mu \rightarrow \nabla_\nu$, since the image and coimage is non-zero, we have

$$[\Delta_\mu : \operatorname{IC}_\nu] \neq 0, [\Delta_\nu : \operatorname{IC}_\mu] \neq 0$$

which implies $\nu \leq \mu$ and $\mu \leq \nu$, so $\mu = \nu$. Then, to compute $\dim_k \operatorname{Hom}(\Delta_\mu, \nabla_\mu)$, by the fifth axiom, we have

$$\operatorname{Hom}(\Delta_\mu, \nabla_\mu) = \operatorname{End}(\operatorname{IC}_\mu) = k$$

For the Ext^1 vanishing, we put $\Lambda' = \{\leq \mu\} \cup \{\leq \nu\}$. Then, either μ or ν or both is maximal in Λ' , so the exact sequence

$$0 \rightarrow \nabla_\nu \rightarrow M \rightarrow \Delta_\mu \rightarrow 0$$

splits by either projectivity of ∇_ν or injectivity of Δ_μ in $\mathcal{A}_{\Lambda'}$.

For higher Ext vanishing, we prove by descending induction on μ . If μ is maximal in Λ , then Δ_μ is projective and there is nothing to prove. In general, consider $P_\mu \rightarrow \operatorname{IC}_\mu$, which has a standard filtration with Δ_μ the last term. Thus considering the long exact sequence of $\operatorname{Hom}(-, \nabla_\nu)$ applied to

$$0 \rightarrow K \rightarrow P_\mu \rightarrow \Delta_\mu \rightarrow 0$$

we have

$$\operatorname{Ext}^{i-1}(K, \nabla_\nu) \rightarrow \operatorname{Ext}^i(\Delta_\mu, \nabla_\nu) \rightarrow \operatorname{Ext}^i(P_\mu, \nabla_\nu)$$

exact with the first term and last term 0 by induction. Thus the claim follows. $\square$

Proposition 3.1.1.5. Such categories have enough projective objects, with each projective admitting a finite filtration with standard associated graded pieces. The dual argument applies for injective objects.

Proof: We prove the following statement by induction: There are enough projectives in $\mathcal{A}_{\Lambda'}$, and each one admit a standard filtration.

We prove by induction on the cardinality of Λ' . The case where $\#\Lambda' = 1$ is trivial. Then, suppose Λ' and $\Lambda' \cup \{\lambda\}$ are both closed under taking smaller elements. We will prove the statement for $\Lambda' \cup \{\lambda\}$ from the statement for Λ' .

Now, for any $\mu \in \Lambda' \cup \{\lambda\}$, we attempt to construct a projective cover of IC_μ in $\mathcal{A}_{\Lambda' \cup \{\lambda\}}$. Since Λ' is closed under taking smaller elements, it is necessary that λ is an maximal element. Thus, $\Delta_\lambda \rightarrow \operatorname{IC}_\lambda$ is a desired projective cover. Then, for $\mu \in \Lambda'$, we wish to upgrade the projective cover $P \rightarrow \operatorname{IC}_\mu$ in $\mathcal{A}_{\Lambda'}$ to a projective cover in $\mathcal{A}_{\Lambda' \cup \{\lambda\}}$. We put $E = \operatorname{Ext}^1(P, \Delta_\lambda)$. Since P is not yet

projective, E somehow measures the extent that P is non-projective. Then consider the unit element in

$$1 \in E \otimes E^* = \text{Ext}^1(P, E^* \otimes \Delta_\lambda)$$

This gives rise to an extension

$$0 \longrightarrow E^* \otimes \Delta_\lambda \longrightarrow \tilde{P} \longrightarrow P \longrightarrow 0$$

such that the composition

$$E \rightarrow \text{Hom}(E^* \otimes \Delta_\lambda, \Delta_\lambda) \xrightarrow{\delta} \text{Ext}^1(P, \Delta_\lambda)$$

is the identity. Then, since the first map is an isomorphism because $\text{End}(\Delta_\lambda) = \text{End}(\text{IC}_\lambda) = k$, the boundary map is also an isomorphism. Then, by projectivity of Δ_λ , $\text{Ext}^1(\Delta_\lambda, \Delta_\lambda) = 0$.

Hence, the long exact sequence

$$\text{Hom}(E^* \otimes \Delta_\lambda, \Delta_\lambda) \longrightarrow \text{Ext}^1(P, \Delta_\lambda) \longrightarrow \text{Ext}^1(\tilde{P}, \Delta_\lambda) \longrightarrow \text{Ext}^1(E^* \otimes \Delta_\lambda, \Delta_\lambda)$$

gives $\text{Ext}^1(\tilde{P}, \Delta_\lambda) = 0$. Now, it suffices to prove that $\tilde{P}$ is a projective cover for IC_μ . This is equivalent to saying

$$\dim_k \text{Hom}(\tilde{P}, \text{IC}_\nu) = \delta_{\mu, \nu}, \text{Ext}^1(\tilde{P}, \text{IC}_\nu) = 0$$

where the second condition ensures that $\tilde{P}$ is projective and the first condition ensures that $\tilde{P} \rightarrow \text{IC}_\mu$ is a cover. Consider the exact sequence

$$\text{Ext}^1(P, \text{IC}_\nu) \longrightarrow \text{Ext}^1(\tilde{P}, \text{IC}_\nu) \longrightarrow \text{Ext}^1(E^* \otimes \Delta_\lambda, \text{IC}_\nu)$$

the first term vanishes by the projectivity of P in $\mathcal{A}_{\Lambda'}$, and the third vanishes by projectivity of Δ_λ . So the second group vanishes. By the same reason, $\dim_k \text{Hom}(\tilde{P}, \text{IC}_\nu) = \delta_{\mu, \nu}$. Thus, it suffices to prove the statement for IC_λ . Then, to compute $\text{Hom}(\tilde{P}, \text{IC}_\lambda)$, consider the exact sequence

$$0 \longrightarrow K \longrightarrow \Delta_\lambda \longrightarrow \text{IC}_\lambda \longrightarrow 0$$

Then, $\text{Hom}(\tilde{P}, \text{IC}_\lambda) = 0$ because K lies in $\mathcal{A}_{\Lambda'}$ and run the exact sequence, and Ext^1 vanishes by the vanishing condition of Ext^2 .

Hence, the original statement follows from induction. □

Corollary 3.1.1.6. BGG reciprocity holds, that is, we have $(P_\mu : \Delta_\nu) = [\nabla_\nu : L_\mu]$

Proof: By the Hom orthogonality, the first number is equal to $\dim_k \text{Hom}(P_\mu, \nabla_\nu)$. For the second, consider the Jordan-Holder series of ∇_ν

$$0 = F_0 \subset \cdots \subset F_n = \nabla_\nu$$

By the projectivity of P_μ , we have

$$0 \longrightarrow \text{Hom}(P_\mu, F_{i-1}) \longrightarrow \text{Hom}(P_\mu, F_i) \longrightarrow \text{Hom}(P_\mu, \text{IC}_\lambda) \longrightarrow 0$$

exact, where IC_λ is a simple factor. But then, since $P_\mu \rightarrow \text{IC}_\mu$ is a projective cover, $P_\mu \rightarrow \text{IC}_\lambda$ is nontrivial iff $\lambda = \mu$, so the second number is equal to $\dim_k \text{Hom}(P_\mu, \nabla_\nu)$. □

Proposition 3.1.1.7. Let $X = \coprod_{\lambda \in \Lambda} X_\lambda$ be an affine stratification. Then, $\mathbf{Perv}_\Lambda(X)$ is a highest weight category in the sense of [BGS96].

Proof: We verify the axioms. The first three axioms are obtained directly by the properties of perverse sheaves on an affine stratified space. For the fourth property, it suffices to prove that $*$ - or $!$ -restriction to the open strata is exact (which is standard), because

$$\mathrm{Hom}(\Delta, -) = \mathrm{Hom}(j_!1, -) = \mathrm{Hom}(1, j^*(-)) = j^*(-)$$

and the dual holds for ∇ . Moreover, these objects are indecomposable, so they are the projective cover and injective hull. The fifth property also follows from the property of intermediate extensions. The final property is slightly more difficult, and is proven by the following property of perverse sheaves: firstly, by an analysis of extensions we have

$$\mathrm{Ext}_{\mathbf{Perv}_\Lambda}^2(\Delta_\mu, \nabla_\nu) \subset \mathrm{Hom}_{D_\Lambda}^2(\Delta_\mu, \nabla_\nu)$$

and for the latter, we have

$$\begin{aligned} \mathrm{Hom}_{D_\Lambda}^2(\Delta_\mu, \nabla_\nu) &= \mathrm{Hom}_{D_\Lambda}(j_{\mu!}1_\mu[\dim X_\mu], j_{\nu*}1_\nu[\dim X_\nu + 2]) \\ &= \mathrm{Hom}_{D_\Lambda}(j_\nu^*j_{\mu!}1_\mu[\dim X_\mu], 1_\nu[\dim X_\nu + 2]) \end{aligned}$$

Then, the first term is 0 for $\nu \neq \mu$, and if $\nu = \mu$, then the above Hom is an ext group of vector spaces, which is 0. □

Corollary 3.1.1.8. We have the realization functor an equivalence.

$$D(\mathbf{Perv}_\Lambda(X)) \xrightarrow{\cong} D_\Lambda(X)$$

Proof: Note that P_λ and ∇_λ separately generates both $D(\mathbf{Perv}_\Lambda(X))$ and $D_\Lambda(X)$. It suffices to show that

$$\mathrm{Ext}_{\mathbf{Perv}_\Lambda(X)}^i(P_\mu, \nabla_\nu) = \mathrm{Hom}_{D_\Lambda(X)}^i(P_\mu, \nabla_\nu)$$

By projectivity of P_μ , the left hand side vanishes unless $i = 0$. The right side also vanishes unless $i = 0$ by the standard filtration of P and applying the property of $!$ -pushforwards. The statement for $i = 0$ is trivial. □

However, one ought to be careful about the stratified spaces. For example, if there are stacky components in the strata, the resulting category of sheaves is often not highest weight, for example, we have

$$D(I \backslash LG/I) \neq D(\mathbf{Perv}(I \backslash LG/I))$$

which is an evidence of $\mathbf{Perv}(I \backslash LG/I)$ not being highest weight.

Now, we may wish to generalize the above properties to possibly infinite stratifications. This is where the modern definition of highest weight categories come in.

Definition 3.1.1.9. Let $\mathcal{A}$ be an Abelian category over some field k . We say $\mathcal{A}$ is highest weight in the sense of [Ric17] if it is highest weight in the sense of [BGS96], but with the condition of the finite poset Λ replaced by for any $\lambda \in \Lambda$, the set $\{\mu | \mu \leq \lambda\} \subset \Lambda$ is finite.

Properties such as the orthogonality and Ext vanishing still holds. However, the existence of projective (injective) object fails by construction. However, this will prove not to be a major difficulty, since the categories we are dealing with such as $D(I \setminus LG/I)$ is given by the 2-colimit on the finite stage objects

$$D(I \setminus \text{Fl}) = \text{colim}_{w \in W^{\text{ext}}} D(I \setminus \text{Fl}_{\leq w})$$

and each finite stage object will allow the previous properties. Later on, we will adopt this definition of highest weight categories.

Lemma 3.1.1.10. Let $\Lambda' \subset \Lambda$ be a subposet closed under taking smaller objects, and let $\mathcal{A}$ be a highest weight category with weight poset Λ . Then, the Serre quotient $\pi : \mathcal{A} \twoheadrightarrow \mathcal{A}/\mathcal{A}_{\Lambda'}$ is a highest weight category with weight poset $\Lambda - \Lambda'$. Its simple, standard, and costandard objects are the image of the projection π of the original simple, standard, and costandard objects.

Proof: Axioms (1), (2), (4), (5) are obvious. Then, for (4), we claim that we have an isomorphism

$$\text{Hom}_{\mathcal{A}}(\Delta_{\lambda}, N) = \text{Hom}_{\mathcal{A}/\mathcal{A}_{\Lambda'}}(\pi(\Delta_{\lambda}), \pi(N))$$

for $\lambda \in \Lambda - \Lambda'$. This is because a map $\pi(\Delta_{\lambda}) \rightarrow \pi(N)$ is represented by a map $M' \rightarrow N/N'$ in $\mathcal{A}$, where $M' \subset \Delta_{\lambda}$ and $N' \subset N$ such that Δ_{λ}/M' and N' is in $\mathcal{A}_{\Lambda'}$. Then, since every quotient of Δ_{λ} necessarily contains $\text{IC}_{\lambda} \notin \mathcal{A}_{\Lambda'}$, we have $M' = \Delta_{\lambda}$. Then, since $N' \in \mathcal{A}_{\Lambda'} \subset \mathcal{A}_{\leq \lambda}$, and that Δ_{λ} is projective in $\mathcal{A}_{\leq \lambda}$, we have $\text{Ext}^1(\Delta_{\lambda}, N') = 0$, which implies that the map $\Delta_{\lambda} \rightarrow N/N'$ factors through $N \rightarrow N/N'$. Hence, we have the above map is surjective. Moreover since $M' = \Delta_{\lambda}$ necessarily, and any map $\Delta_{\lambda} \rightarrow N/N'$ does not annihilate IC_{λ} , it is non-zero. Thus the above map is an isomorphism. This implies that Δ_{λ} is projective in $(\mathcal{A}/\mathcal{A}_{\Lambda'})_{\Lambda''}$ for λ maximal in Λ'' .

Finally for axiom (6), it suffices to prove the case where Λ is finite because to compute $\text{Ext}^2(\pi(\Delta_{\mu}), \pi(\nabla_{\nu}))$, it suffices to compute this in the category $\mathcal{A}_{\leq \mu, \nu}$, which is finite by assumption. However, in the finite case, we have enough projectives, which allows us to prove the claim as follows. We consider the object $\pi(P_{\mu}) \twoheadrightarrow \pi(\Delta_{\mu})$. The kernel of this map has a filtration by $\pi(\Delta_{\lambda})$ where $\lambda > \mu$. Then, if $\pi(P_{\mu})$ is projective, the above vanishing of Ext^2 follows by the long exact sequence given by

$$\text{Ext}^1(K, \pi(\nabla_{\nu})) \rightarrow \text{Ext}^2(\pi(\Delta_{\mu}), \pi(\nabla_{\nu})) \rightarrow \text{Ext}^2(\pi(P_{\mu}), \pi(\nabla_{\nu}))$$

where the last term is 0 by projectivity and the first is 0 by analysing extensions. Thus it suffices to prove projectivity. If not, consider a non-split surjection $\pi(M) \twoheadrightarrow \pi(P_{\mu})$. This surjection is represented by a map $M' \rightarrow P_{\mu}/N'$ where M/M' and N' lies in $\mathcal{A}_{\Lambda'}$. Moreover, by surjectivity, this map must have cokernel lying in $\mathcal{A}_{\Lambda'}$. However, since $P_{\mu} \twoheadrightarrow \text{IC}_{\mu}$ is a projective cover with $\mu \notin \Lambda'$, the cokernel must necessarily vanish, so the representing map is itself surjective. Thus, $M' \twoheadrightarrow P_{\mu}$ splits, which contradicts $\pi(M) \twoheadrightarrow \pi(P_{\mu})$ not split. $\square$

Proposition 3.1.1.11. Let $\mathcal{A}$ be a highest weight category with weight poset Λ . Then we have the following.

- (1) The embedding $\iota : D(\mathcal{A}_{\Lambda'}) \hookrightarrow D(\mathcal{A})$ is fully faithful.
- (2) The embedding $\iota : D(\mathcal{A}_{\Lambda'}) \hookrightarrow D(\mathcal{A})$ admit left and right adjoints, denoted by ι^L and ι^R . Moreover, the projection $\varpi : D(\mathcal{A}) \rightarrow D(\mathcal{A})/D(\mathcal{A}_{\Lambda'})$ admit left and right adjoints, denoted by ϖ^L and ϖ^R . These adjoints provide a recollement of $D(\mathcal{A})$.

Proof: For (1), note that the category $D(\mathcal{A})$ and $D(\mathcal{A}_{\Lambda'})$ are generated by the standard and costandard objects separately. Thus, it suffices to prove that for $\mu, \nu \in \Lambda'$, we have

$$\mathrm{Ext}_{\mathcal{A}_{\Lambda'}}(\Delta_\mu, \nabla_\nu) \rightarrow \mathrm{Ext}_{\mathcal{A}}(\Delta_\mu, \nabla_\nu)$$

an isomorphism. This is by [Lemma 3.1.1.4](#).

For (2), consider the category $D(\mathcal{A})_{\Lambda-\Lambda'}^\nabla$, given by the subcategory of $D(\mathcal{A})$ generated by ∇_λ for $\lambda \in \Lambda - \Lambda'$. Now, there exists a decomposition (fiber sequence) for any $M \in D(\mathcal{A})$

$$M' \longrightarrow M \longrightarrow M''$$

where $M' \in D(\mathcal{A}_{\Lambda'})$ and $M'' \in D(\mathcal{A})_{\Lambda-\Lambda'}^\nabla$. The construction is analogous to the process of decomposing a sheaf into closed and open parts, and is unique and functorial. Then, this gives $\varpi|_{D(\mathcal{A})_{\Lambda-\Lambda'}^\nabla} : D(\mathcal{A})_{\Lambda-\Lambda'}^\nabla \simeq D(\mathcal{A})/D(\mathcal{A}_{\Lambda'})$. The inverse of this functor gives ϖ^R , and projecting M to M' gives ι^R .

Conversely, considering $D(\mathcal{A})_{\Lambda-\Lambda'}^\Delta$ gives ι^L and ϖ^L . □

Corollary 3.1.1.12. We have an equivalence of categories

$$D(\mathcal{A})/D(\mathcal{A}_{\Lambda'}) = D(\mathcal{A}/\mathcal{A}_{\Lambda'})$$

Proof: Again, both categories are generated by Δ_λ 's and ∇_λ 's for $\lambda \in \Lambda - \Lambda'$ separately. Note that there is a natural functor $D(\mathcal{A})/D(\mathcal{A}_{\Lambda'}) \rightarrow D(\mathcal{A}/\mathcal{A}_{\Lambda'})$. It now suffices to prove that

$$\mathrm{Hom}_{D(\mathcal{A})/D(\mathcal{A}_{\Lambda'})}(\varpi(\Delta_\mu), \varpi(\nabla_\nu)) = \mathrm{Hom}_{D(\mathcal{A}/\mathcal{A}_{\Lambda'})}(\pi(\Delta_\mu), \pi(\nabla_\nu))$$

But by adjunction, we have

$$\mathrm{LHS} = \mathrm{Hom}_{D(\mathcal{A})}(\Delta_\mu, \varpi^R \varpi(\nabla_\nu)) = \mathrm{Hom}_{D(\mathcal{A})}(\Delta_\mu, \nabla_\nu)$$

where the second inequality follows from the construction of ϖ^R . The analogous fact for RHS follows likewise. □

Definition 3.1.1.13. Let $M \in \mathcal{A}$. We say it is a tilting object if it has both a standard and costandard filtration.

By categorical nonsense, M admit a filtration by standard objects iff it has $\mathrm{Ext}^1(\Delta_\lambda, M) = 0$, iff it has $\mathrm{Ext}^i(\Delta_\lambda, M) = 0$ for all $i > 0$. Moreover, the number of ∇_λ 's appearing in M is computed via $\dim \mathrm{Hom}(\Delta_\lambda, M)$. The dual statement for costandard filtration is also true.

Proposition 3.1.1.14. For any $\lambda \in \Lambda$, there is a unique tilting object T_λ such that $[T_\lambda : \text{IC}_\lambda] = 1$, and $[T_\lambda : \text{IC}_\mu] = 0$ for $t \not\leq s$.

Proof: We claim that there is a tilting object T with an injection $\Delta_\lambda \hookrightarrow T$ with cokernel admitting a standard filtration by Δ_μ for $\mu < \lambda$. We first use the claim to derive the final statement, and then prove the claim.

Notice that T admits an indecomposable direct summand T_λ with $(T_\lambda : \Delta_\lambda) = 1$, which is given by taking the limit along all quotient objects of T containing Δ_λ . Then, by the property of T , we have the composition $\Delta_\lambda \rightarrow T \rightarrow T_\lambda$ still injective. The cokernel of $\Delta_\lambda \hookrightarrow T_\lambda$ is still equipped with a costandard filtration by Δ_μ for $\mu < \lambda$, so such T_λ satisfies indeed the conditions required in the property.

Then, we prove that any other T'_s subject to the requirements are isomorphic to T_λ . In fact, we will prove that for T an indecomposable tilting object, it must be isomorphic to some T_λ . We take the first term of the standard filtration of T , say $\Delta_\mu \hookrightarrow T$. Consider the cokernel $C = \text{coker}(\Delta_\mu \hookrightarrow T)$. Since T_μ is tilting, we have $\text{Ext}^1(C, T_\mu) = 0$. Thus, the long exact sequence gives

$$\text{Hom}(T, T_\mu) \rightarrow \text{Hom}(\Delta_\mu, T_\mu) \rightarrow \text{Ext}^1(C, T_\mu) = 0$$

so, we find a preimage of the structural morphism $\Delta_\mu \hookrightarrow T_\mu$, which gives $\varphi : T \rightarrow T_\mu$ restricting to the identity on Δ_μ . Conversely, there is a $\psi : T_\mu \rightarrow T$ restricting to the identity on Δ_μ . Thus, $\psi \circ \varphi : T \rightarrow T$ is a non-nilpotent element of the Artinian local ring $\text{End}(T)$, hence is invertible. Thus, φ and ψ are isomorphisms.

Now, it suffices to prove the previous claim. We will construct such T by induction on the size of the poset Λ . Firstly, if λ is minimal, we can take $T = \Delta_\lambda = \nabla_\lambda$. Otherwise, pick μ minimal with $\mu \in \Lambda$. By induction, there is an object M in $\mathcal{A}/\mathcal{A}_\mu$ satisfying the properties, and we can take $M' = \pi_\mu^L(M)$. By adjunction, there is an embedding $\Delta_s \hookrightarrow M'$. Again by adjunction, for any $\nu \neq \mu$, we have $\text{Ext}^1(\Delta_\nu, M') = \text{Ext}^1(\Delta_\nu, M) = 0$. Now, take $E = \text{Ext}^1(\Delta_\mu, M')$. By definition, we can identify $E^\vee \otimes E = \text{Ext}^1(E \otimes \Delta_\mu, M')$, and consider the object $T \in \text{Ext}^1(E \otimes \Delta_\mu, M')$ representing the identity on E . Hence, we obtain an exact sequence

$$0 \rightarrow M' \rightarrow T \rightarrow E \otimes \Delta_\mu \rightarrow 0$$

Now, T admits an injection $\Delta_\lambda \hookrightarrow T$ with cokernel admitting a costandard filtration by Δ_μ with $\mu < \lambda$. So now it suffices to prove that T is tilting, i.e. it also admits a standard filtration.

However, this amounts to prove that $\text{Ext}^1(\Delta_\nu, T) = 0$ for all ν . If $\nu \neq \mu$, this follows from the ext-vanishing by adjunction. If $\nu = \mu$, consider the long exact sequence associated to the short exact sequence defining T

$$\text{Hom}(\Delta_\mu, E \otimes \Delta_\mu) \rightarrow \text{Ext}^1(\Delta_\mu, M') \rightarrow \text{Ext}^1(\Delta_\mu, T) \rightarrow \text{Ext}^1(\Delta_\mu, E \otimes \Delta_\mu)$$

By construction, the first map is an isomorphism, and the fourth group is trivial, so we obtain the desired vanishing. □

Corollary 3.1.1.15. The natural functor $K(\text{Tilt}(\mathcal{A})) \rightarrow D(\mathcal{A})$ is an equivalence of categories.

Proof: $D(\mathcal{A})$ is clearly generated by the tilting objects by the presence of T_λ 's. Thus, it suffices to show that the above functor is fully faithful. However, for $T, T' \in \text{Tilt}(\mathcal{A})$, consider the

costandard filtration on T and standard filtration on T' , we have $\text{Ext}^i(T, T') = 0$ for all $i > 0$. Hence full faithfulness holds. $\square$

We can give a further enhancement of the formalism introduced above.

Definition 3.1.1.16. We define a graded highest weight category to be an Abelian category $\mathcal{A}$ over some field k , such that every requirement in [Definition 3.1.1.9](#) holds except for (2) and (4). For the modification of (2), this time $\mathcal{A}$ has simple objects indexed by $\Lambda \times \mathbb{Z}$, with simple objects denoted by $\text{IC}_\lambda \langle n \rangle$. There is an additional requirement that $D(\mathcal{A})$ is a stable ∞ -category equipped with a t -structure and a weight structure, where all simple objects $\text{IC}_\lambda \langle n \rangle$ has weight n .

Clearly, forgetting the grading $\langle n \rangle$ and identifying the corresponding simple objects gives a highest weight category.

For the modification of (4), the projective objects and injective objects do not exist in $\mathcal{A}$, but in the underlying highest weight category.

todo 3.1.1.1. Is this definition correct?

We list but not prove the following properties below, since everything is a simple generalization of the previous proofs.

Proposition 3.1.1.17. Let $X = \coprod_{\lambda \in \Lambda} X_\lambda$ be a finite affine stratification over a finite field k . Then, $\mathbf{Perv}_{\Lambda, m}(X)$ is a graded highest weight category.

Corollary 3.1.1.18. We have the realization functor an equivalence $D(\mathbf{Perv}_{\Lambda, m}(X)) \rightarrow D_{\Lambda, m}(X)$.

3.1.2. Where is Koszul duality?

3.2. Koszul Duality for Kac–Moody groups

In this chapter, we give a review of Bezrukavnikov–Yun’s paper [\[YB14\]](#). For X a stack over any finite field k , we will use the notation $D_m(X)$ ($\mathbf{Perv}_m(X)$) to denote the category of (perverse) sheaves over X , but with the space of morphisms given by $R \text{Hom}(\iota \mathcal{F}, \iota \mathcal{G})$, where $\iota : D_m(X) \rightarrow D_m(X \otimes \bar{k})$. In particular, this space admit a Frobenius action since $\mathcal{F}$ and $\mathcal{G}$ are Frobenius fixed. We will write $\underline{\text{Hom}}(\mathcal{F}, \mathcal{G})$ if we mean the usual Hom space in $D_m(X)$ instead. This convention indicates that what we should be caring about is the category of mixed sheaves $D_m(X \otimes \bar{k})$, but in the applications of representation theory, we only care about the image of the map $D(X) \rightarrow D_m(X \otimes \bar{k})$, leaving the objects we are considering not too complicated. Moreover, we have the following relation on the Hom groups

$$\underline{\text{Hom}}(\mathcal{F}, \mathcal{G}) = \text{Hom}(\iota \mathcal{F}, \iota \mathcal{G})^{h\text{Fr}}$$

The issue that $D(X) \rightarrow D_m(X \otimes \bar{k})$ is not surjective can be fixed using the formalism of graded sheaves. Via this, we can define the category of mixed sheaves over $X \otimes \bar{k}$ where all possible reductions of X over finite fields yield the same category.

3.2.1. Kac–Moody lie algebras and Kac–Moody groups

Definition 3.2.1.1. We define a generalized Cartan matrix of size ℓ to be a matrix $A \in \text{Mat}_{\ell \times \ell}(\mathbb{Z})$, such that the following conditions are satisfied:

- (1) $a_{i,i} = 2$;
- (2) $a_{i,j} \leq 0$ for $i \neq j$;
- (3) $a_{i,j} = 0$ iff $a_{j,i} = 0$.

We define a realization of a generalized Cartan matrix to be a triple $(\mathfrak{h}, \pi, \pi^\vee)$, where $\mathfrak{h}$ is a complex $(2\ell - \text{rank } A)$ -dimensional vector space, $\pi = \{\alpha_1, \dots, \alpha_\ell\} \subset \mathfrak{h}^*$ a set of roots, and $\pi^\vee = \{\alpha_1^\vee, \dots, \alpha_\ell^\vee\} \subset \mathfrak{h}$ a set of coroots, subject to the relation $\alpha_j(\alpha_i^\vee) = a_{i,j}$.

By linear algebra, such realization exists and is unique up to isomorphism.

Construction 3.2.1.2. Given a generalized Cartan matrix A , we construct its associated Kac–Moody Lie algebra $\mathfrak{g}(A)$ by the following presentation:

- It is generated by $\mathfrak{h}$ and symbols $\{e_1, \dots, e_\ell, f_1, \dots, f_\ell\}$.
- The relations are given by
 - (1) $[\mathfrak{h}, \mathfrak{h}] = 0$.
 - (2) $[h, e_i] = \alpha_i(h)e_i$, $[h, f_i] = -\alpha_i(h)f_i$.
 - (3) $[e_i, f_j] = \delta_{i,j}\alpha_i^\vee$.
 - (4) $(\text{ad } e_i)^{1-a_{i,j}}(e_j) = 0$ for $i \neq j$, and $(\text{ad } f_i)^{1-a_{i,j}}(f_j) = 0$.

We denote $\mathfrak{n}$ and $\mathfrak{n}^-$ by the subalgebra generated by $\{e_i\}$ and $\{f_i\}$ respectively. We define $\mathfrak{b} = \mathfrak{n} + \mathfrak{h}$ and $\mathfrak{b}^- = \mathfrak{n}^- + \mathfrak{h}$ the standard Borel subalgebras. We also define $\tilde{\mathfrak{g}}$ to be the Lie algebra generated by the same generators but only subject to the relations (1), (2), and (3).

Definition 3.2.1.3. Let $\alpha \in \mathfrak{h}^*$. We define the root space $\mathfrak{g}_\alpha \subset \mathfrak{g}$ to be the subspace

$$\mathfrak{g}_\alpha = \{x \in \mathfrak{g} \mid [h, x] = \alpha(h)x, \forall h \in \mathfrak{h}\}$$

We define $Q = \bigoplus \mathbb{Z}\alpha_i$, and $Q^+ = \bigoplus \mathbb{Z}_{\geq 0}\alpha_i$ the root lattice and nonnegative root lattice. Q^- is defined likewise. We say $\alpha \in Q^+$ is a positive root if $\mathfrak{g}_\alpha \neq 0$. We denote the set of positive roots by R^+ . We likewise define R and R^- .

For a root α , we define $|\alpha| = \sum n_i$, where $\alpha = \sum n_i\alpha_i$.

Theorem 3.2.1.4. We have the following decompositions for $\mathfrak{g}$:

- (1) $\mathfrak{g} = \mathfrak{n} \oplus \mathfrak{h} \oplus \mathfrak{n}^-$ as vector spaces.

$$(2) \quad \mathfrak{n}^\pm = \bigoplus_{\alpha \in Q^\pm} \mathfrak{g}_\alpha$$

Moreover, each root space is finite dimensional.

Proof: Omitted. □

Construction 3.2.1.5. Then, we slightly enlarge the above constructed Lie algebra. We define the complete version of the Lie algebra

$$\hat{\mathfrak{n}} = \prod_{\alpha \in R^+} \mathfrak{g}_\alpha$$

equipped with the natural Lie bracket. This is well-defined because each root space is finite dimensional. Then, notice the family of ideals

$$\hat{\mathfrak{n}}(k) = \prod_{\substack{\alpha \in R^+ \\ |\alpha| \geq k}} \mathfrak{g}_\alpha$$

gives $\hat{\mathfrak{n}}$ a structure of a pro-Lie algebra. Moreover, it is pro-nilpotent. Thus, its exponential gives a well-defined pro-algebraic group $\mathcal{U}$, namely given by

$$\mathcal{U} = \lim_k \exp(\hat{\mathfrak{n}}/\hat{\mathfrak{n}}(k))$$

Then, one defines the Kac–Moody group G by adding the weight 0 and negative part one by one, with no completion involved in the process.

However, the above construction has two caveats: firstly, it works well only for characteristic 0 fields. To work over base fields of arbitrary characteristic, we need to slightly modify the above and arrive at the following integral construction. Secondly, it lacks the additional root datum as in the finite dimensional case to define a group, and will arrive at only the simply connected Kac–Moody groups.

Definition 3.2.1.6. We define a Kac–Moody root datum of a generalized Cartan matrix A to be a finite rank free Abelian group X and morphisms of lattices $\alpha : Q \rightarrow X$ and $\alpha^\vee : Q^\vee \rightarrow X^\vee$, such that the pairing is preserved under the morphisms. Explicitly, we choose $\{\alpha_i\} \subset X$ and $\{\alpha_i^\vee\} \subset X^\vee$ such that $\langle \alpha_j, \alpha_i^\vee \rangle = a_{i,j}$. We call this datum $\mathcal{D}$.

Construction 3.2.1.7 (Mathieu). We construct the Kac–Moody Lie algebra $\mathfrak{g}(\mathcal{D})$ associated to the Kac–Moody root datum $\mathcal{D}$. In fact, the construction is completely analogous to [Construction 3.2.1.2](#), but with the naive $\mathfrak{h}$ replaced by $X^\vee \otimes \mathbb{Q}$, and everything works rationally. We will not copy the generators and relations again.

We then define the Chevalley–Tits integral enveloping algebra $\mathcal{U}_{\mathbb{Z}}(\mathfrak{g}) \subset U_{\mathbb{Q}}(\mathfrak{g})$. We take

$$e_i^{(n)} = \frac{e_i^n}{n!}, f_i^{(n)} = \frac{f_i^n}{n!}, \binom{x^\vee}{n} = \frac{x^\vee \cdot (x^\vee - 1) \cdots (x^\vee - n + 1)}{n!}$$

We then take $\mathcal{U}_{\mathbb{Z}}(\mathfrak{g})$ to be the integral lattice spanned by the above generators.

We can likewise define $\mathcal{U}_{\mathbb{Z}}^+ = \mathcal{U}_{\mathbb{Z}}(\mathfrak{n}^+) = U_{\mathbb{Q}}(\mathfrak{n}^+) \cap \mathcal{U}_{\mathbb{Z}}(\mathfrak{g})$, or equivalently generated by the elements $e_i^{(n)}$, by an integral version of the PBW theorem. More generally, take $\Psi \subset \Delta_+$ a subset of positive roots closed under addition. We take

$$\mathfrak{g}_\Psi = \bigoplus_{\alpha \in \Psi} \mathfrak{g}_\alpha$$

and $\mathcal{U}_{\mathbb{Z}}(\Psi) = U(\mathfrak{g}_\Psi) \cap \mathcal{U}_{\mathbb{Z}}(\mathfrak{g})$.

Then, we define the restricted gradewise dual of the above Hopf algebra

$$\mathbb{Z}[U_{\Psi}^{\max}] := \bigoplus_{\gamma \in \mathbb{N}\Psi} \mathcal{U}_{\mathbb{Z}}(\Psi)_{\gamma}^{\vee}$$

where $\mathcal{U}_{\mathbb{Z}}(\Psi)_{\gamma}$ denote the graded piece of $\mathcal{U}_{\mathbb{Z}}(\Psi)$ with weight γ . Then since $\mathcal{U}_{\mathbb{Z}}(\Psi)$ is a cocommutative Hopf algebra, we have $\mathbb{Z}[U_{\Psi}^{\max}]$ a commutative Hopf algebra, whose spectrum defines a group scheme

$$U_{\Psi}^{\max} = \text{Spec } \mathbb{Z}[U_{\Psi}^{\max}]$$

We can see that this is a pro-unipotent group via the following height filtration:

$$U_{\Psi}^{\max} = \lim U_{\Psi}^{\max} / U_{\Psi, \geq n}^{\max}$$

where $(\Psi, \geq n)$ is the subset of Ψ containing of positive roots α such that $|\alpha| \geq n$. We can also compute its k -points for any k . Take

$$\widehat{U}_k(\Psi) = \prod_{\gamma \in \mathbb{N}\Psi} (\mathcal{U}_{\mathbb{Z}}(\Psi)_{\gamma} \otimes k)$$

and we have $U_{\Psi}^{\max}(k)$ the group-like elements in $\widehat{U}_k(\Psi)$, which is another evidence of pro-unipotence. For $\Psi = \Delta_+$, we define $U_+^{\max} = U_{\Psi}^{\max}$.

Construction 3.2.1.8 (Mathieu). Now we construct the whole group. For this, we first need the exponential map. Let $x \in \mathfrak{g}_{\alpha, \mathbb{Z}}$. We define an exponential sequence for x to be $(x^{[n]})_{n \geq 0}$, where $x^{[n]} \in \mathcal{U}_{\mathbb{Z}, n\alpha}^+$, subject to the conditions $x^{[0]} = 1$, $x^{[1]} = x$, $x^{[n]} - x^{(n)} \in F_{n-1}U_{\mathbb{Q}}(\mathfrak{g})$, where F is the PBW filtration (this means their difference is measured by polynomials of degree at most $n-1$).

Moreover, we require the formula

$$\Delta(x^{[n]}) = \sum_{a+b=n} x^{[a]} \otimes x^{[b]}, \varepsilon(x^{[n]}) = 0, n > 0$$

For any x , an exponential sequence always exists: firstly one finds integral lifts of $x^{(n)}$, and modulo low PBW filtered terms to adjust to the final formula.

Notice that for α a real root, we may choose $x^{[n]} = x^{(n)}$ because α is conjugate to some simple root α , so $x^{(n)}$ is known to be integral. The above formulas are obviously satisfied.

We now use this exponential sequence to construct an exponential map. Let R be any commutative ring. For $\lambda \in R$, we define $[\exp](\lambda x) := \sum \lambda^n x^{[n]} \in \widehat{U}_R^+$. The above formulas ensure that $[\exp](\lambda x)$ is a group-like element, so lies in $U_+^{\max}(R)$. Then, this map is functorial in R , so assembles to a map $E_x : \mathbb{A}_{\mathbb{Z}}^1 \rightarrow U_+^{\max}$. Note that by unwinding definitions, this map is a group homomorphism iff

$$x^{[m]}x^{[n]} = \binom{m+n}{m} x^{[m+n]}$$

which holds for α a real root, but not necessarily holds for imaginary roots.

We take H to be the torus with root lattice X and coroot lattice $X^{\vee}$. Then, we have $H \curvearrowright U_+^{\max}$ via $h \cdot [\exp](\lambda x) \cdot h^{-1} = [\exp](\alpha(h)\lambda x)$ for $x \in \mathfrak{g}_{\alpha}$. We left it to the readers that this action is indeed well-defined. With this property, we may define

$$B_+^{\max} = U_+^{\max} \rtimes H$$

We then extend this definition to negative roots. For each simple root α_i , the $\mathfrak{sl}_2$ -triple $e_i, f_i, \alpha_i^\vee$ integrates to a reductive group G_i with positive part U_i , and take $G_i *_{U_i} B_+^{\max}$ gives a minimal parabolic group $P_i \supset B_+^{\max}$, with the quotient $P_i/B_+^{\max}$ given by $\mathbb{P}^1$.

Then let W denote the Weyl group of the Kac–Moody algebra (i.e., the subgroup of $\mathrm{GL}(X \otimes \mathbb{Q})$ generated by the reflections $s_i : \alpha \mapsto \alpha - \langle \alpha, \alpha_i^\vee \rangle \alpha_i$). Take a word presentation $\mathbf{s} = s_{i_1} \cdots s_{i_n}$. We define $E(\mathbf{s}) = P_{i_1} \times^B P_{i_2} \times^B \cdots \times^B P_{i_n}$ (in this construction, we abbreviate $B_+^{\max}$ by B). Then, we define the Bott–Samuelson variety $D(\mathbf{s}) = E(\mathbf{s})/B$, where B acts on the right of P_{i_n} . If we take $\mathbf{s}' = s_{i_1} \cdots s_{i_{n-1}}$, we have that $D(\mathbf{s}) \twoheadrightarrow D(\mathbf{s}')$ by forgetting the final factor. This is actually a $\mathbb{P}^1$ -bundle since $P_{i_n}/B = \mathbb{P}^1$. Then, $D(\emptyset) = \mathrm{pt}$, so $D(\mathbf{s})$ is smooth, projective, and geometrically integral. We remark here that $D(\mathbf{s})$ is the compactification of

$$Bs_{i_1}B \times^B Bs_{i_2}B \times^B \cdots \times^B Bs_{i_n}B/B = \mathbb{A}^n := D(w)^\circ$$

We then construct the Schubert cells. Take a regular dominant weight λ and its highest weight module $L(\lambda)$ (finite dimensional) with highest weight vector v_λ . Then, for $w \in W$ we pick a reduced word presentation (which we abuse notations and also denote by w). We have a map

$$\pi_w : D(w) \rightarrow \mathbb{P}(L(\lambda)), [p_1, \dots, p_n] \mapsto [p_1 \cdot p_2 \cdots p_n \cdot v_\lambda]$$

The image is well-defined since B -action preserves the highest weight line in $L(\lambda)$. Then, picking a set of Weyl lifts $\dot{s}_{i_j}$, $\pi_w([\dot{s}_{i_1}, \dots, \dot{s}_{i_n}])$ is the line $[L(\lambda)_{w\lambda}]$. We define $S_w = \overline{B[L(\lambda)_{w\lambda}]} \subset \mathbb{P}(E_w(\lambda))$ where $E_w(\lambda) = U(\mathfrak{b})L(\lambda)_{w\lambda}$ is upward part of $L(\lambda)$ over $L(\lambda)_{w\lambda}$ (sorry for the unfortunate notation). This image has the image of a projective scheme, induced by $\pi_{w,*}\mathcal{O}_{D(w)}$, hence we obtain a map of projective schemes $\pi_w : D(w) \twoheadrightarrow S_w$. After we have finished constructed G and G/B , we should think $S_w \approx \overline{BwB/B}$, and the projection $\pi_w : D(w) \twoheadrightarrow S_w$ as the multiplication map

$$P_{i_1} \times^B \cdots \times^B P_{i_n}/B \rightarrow \overline{BwB/B}$$

Moreover, we have π_w an isomorphism on the open cells $\pi_w : D(w)^\circ \xrightarrow{\sim} C_w = \mathrm{im}(D(w)^\circ)$.

Finally, we construct $B(w)$, which is a B -torsor over S_w , which we should think of the closed cell $\overline{BwB}$. We first let $\mathcal{L}(M)$ be the constant quasi-coherent sheaf over $D(w)$ associated to the B -module M , and $\mathcal{L}_w(M) = \pi_{w,*}\mathcal{L}(M)$. We define $B(w) = \mathrm{Spec}_{S_w} \mathcal{L}_w(k[B])$, which gives an affine morphism $\nu_w : B(w) \rightarrow S_w$. Actually, we have the non-trivial fact that $B(w) = \mathrm{Spec} \Gamma(E(w), \mathcal{O}_{E(w)})$. However the core property we want to use is that $B(w) \rightarrow S_w$ is a B -torsor.

For $u, v \in W$, recall that we can define $u * v \in W$ to be the element defined recursively: pick a reduced word presentation $s_{i_1} \cdots s_{i_r} = v$, we define $u * v := u * s_{i_1} * s_{i_2} * \cdots * s_{i_r}$, and for a simple reflection s , we define $u * s = \max(u, us)$. Then, the above construction gives a multiplication $E(u) \times E(v) \rightarrow B(u * v)$, which descends to $B(u) \times B(v) \rightarrow B(u * v)$ (to be thought of $\overline{BuB} \times \overline{BvB} \subset \overline{B(u * v)B}$) by affinization. With these structural maps, we take

$$G = \mathrm{colim}_{w \in W} B(w)$$

which gives a ind-group structure.

Construction 3.2.1.9. We define $\mathrm{Fl} = G/B = \mathrm{colim} S_w$ to be the affine flag variety, which is also canonically an ind-scheme. The open Schubert cell $\mathrm{Fl}_w = BwB/B = \mathbb{A}^{\ell(w)}$, which can also be thought of the image of C_w constructed previously. We hence have

$$\mathrm{Fl} = \coprod_{w \in W} \mathrm{Fl}_w, S_w = \coprod_{v \leq w} \mathrm{Fl}_w$$

Likewise, we may define the enhanced flag variety $\widetilde{\mathrm{Fl}} = G/U$, which is a H -torsor over $\mathrm{Fl} = G/B$.

Construction 3.2.1.10. If we let $\Sigma \subset W$ denote the subset of simple reflections, and take $\Theta \subset \Sigma$ such that the subgroup of W generated by Θ is finite, likewise to the construction of P_i 's in [Construction 3.2.1.8](#) we can construct a parabolic subgroup P_Θ . Then, we define L_Θ the Levi subgroup, and U_Θ the unipotent radical of P_Θ . We let B_Θ be the Borel of L_Θ . Finally, we define $U_{L_\Theta}^-$ to be the negative unipotent radical of L_Θ .

In the following chapter, we will adopt the left flag variety notation. That is, we take $\mathrm{Fl} = B \backslash G$, and for any Θ satisfying the condition above, take ${}_\Theta \mathrm{Fl} = P_\Theta \backslash G$. Moreover, we let W_Θ denote the Weyl group of L_Θ , and $[W_\Theta \backslash W] \subset W$ the set of minimal length representatives of the cosets. It inherits a length function and a poset structure from W . We denote the maximal length element $w_\Theta \in W_\Theta$ of length ℓ_Θ . Then, we have the strata of $P_\Theta \backslash G/B$ indexed by $\bar{w} \in [W_\Theta \backslash W]$. We make the following definitions

$$i_{\bar{w}} : {}_\Theta \mathrm{Fl}_{\bar{w}} \hookrightarrow {}_\Theta \mathrm{Fl}, \quad i_{<\bar{w}} : {}_\Theta \mathrm{Fl}_{<\bar{w}} \hookrightarrow {}_\Theta \mathrm{Fl}, \quad i_{\leq \bar{w}} : {}_\Theta \mathrm{Fl}_{\leq \bar{w}} \hookrightarrow {}_\Theta \mathrm{Fl}$$

The stratas are affine spaces of corresponding dimensions. When $\Theta = \emptyset$, this reduces to the plain flag variety case.

Then, consider $U_\Theta U_{L_\Theta}^-$. The orbits of $U_\Theta U_{L_\Theta}^- \backslash G/B$ are still indexed by $w \in W$, and we take

$${}_\Theta \mathrm{Fl}^w = U_\Theta U_{L_\Theta}^- \cdot BwB/B \subset {}_\Theta \mathrm{Fl}, \quad {}_\Theta \mathrm{Fl}^{\overset{\Theta}{\leq} w} = \overline{{}_\Theta \mathrm{Fl}^w}$$

where $s \overset{\Theta}{\leq} t$ means $w_\Theta s \leq w_\Theta t$.

Finally, we take $\widetilde{\mathrm{Fl}} = G/U$, and make the likewise definitions $\tilde{i}_w : \widetilde{\mathrm{Fl}}_w \hookrightarrow \widetilde{\mathrm{Fl}}$ to be the preimages of Fl_w . The other definitions are done likewise.

Construction 3.2.1.11. We construct the big cell $C \subset \mathrm{Fl}$. In the finite dimensional case, this construction coincides with the cell B^-B/B . Again we construct it as an ind-scheme. Recall that we have a projective embedding $S_w \hookrightarrow \mathbb{P}(E_w(\lambda))$, where $E_w(\lambda) = U(\mathfrak{b})L(\lambda)_{w\lambda}$. Then, the contragradient $L(\lambda)^*$ has a lowest weight line $\sigma_{-\lambda}$. We define C_w to be the complement to the hyperplane $\sigma_{-\lambda} = 0$, and $C = \mathrm{colim} C_w$.

Lemma 3.2.1.12. Let $\mu : \mathbb{G}_m \rightarrow H$ be any antidominant regular coweight. Then for any $w \in W$, C_w contracts to the base point $B/B \in \mathrm{Fl}$ under the left action of $\mu(\mathbb{G}_m)$. Hence, for any $v, w \in W$, $vC \cap \mathrm{Fl}_{\leq w}$ contracts to the point $vB/B \in \mathrm{Fl}$ under the action of $(v\mu)\mathbb{G}_m$.

Proof: By definition of C_w . □

3.2.2. A realization result

Definition 3.2.2.1. Consider the category $\mathbf{Vect}_{\overline{\mathbb{Q}}_\ell}^\heartsuit$. We define a Frobenius module M to be an object $M \in \mathbf{Vect}_{\overline{\mathbb{Q}}_\ell}^\heartsuit$ equipped with an automorphism $\mathrm{Fr}_M : M \rightarrow M$. We denote the category of Frobenius modules by $\mathbf{Mod}_{\mathrm{Fr}}$. We define a Frobenius module to be locally finite if it is a union of its finite submodules. For any Frobenius module M , we denote M^f by the union of its finite submodules.

There is clearly a forgetful functor $\mathbf{Mod}_{\mathrm{Fr}}^\heartsuit \rightarrow \mathbf{Vect}_{\overline{\mathbb{Q}}_\ell}^\heartsuit$ given by $M \mapsto M^{\mathrm{Fr}}$, the fixed point functor. Later on, we will use this functor to realize a category enriched over $\mathbf{Mod}_{\mathrm{Fr}}$ to a DG category over $\overline{\mathbb{Q}}_\ell$.

Definition 3.2.2.2. We define a locally finite Frobenius module to be continuous if the action $\mathbb{Z} \curvearrowright M$ through Frobenius extends to an action $\widehat{\mathbb{Z}} \curvearrowright M$. For a locally finite Frobenius module, we may define its weights as in the usual convention. For a general Frobenius module M , we make the above definitions after applying M^f .

From now on, we assume every Frobenius module to be continuous with integral weights.

Definition 3.2.2.3. Let M be a Frobenius module. We define $M^{\mathrm{Fr}\text{-unip}}$ to be the Frobenius submodule of M on which Fr_M acts unipotently. We define M^{op} to denote the Frobenius module with the same underlying vector space as M , but with the inverse Frobenius action.

Definition 3.2.2.4. Let $E \in \mathbf{Alg}(\mathbf{Mod}_{\mathrm{Fr}}^\heartsuit)$. we define $\mathbf{Mod}_{E, \mathrm{Fr}}^\heartsuit$ to be the category of E -modules equipped with a Frobenius automorphism compatible with the Frobenius module structure on E . We define $\mathbf{Mod}_{E, \mathrm{Fr}}$ to be its derived category.

In this section, our main goal is to answer the following question.

Question 3.2.2.5. For a category $\mathcal{D}$ enriched in $\mathbf{Mod}_{\mathrm{Fr}}$, when do we have a fully faithful inclusion $\mathcal{D} \hookrightarrow \mathbf{Mod}_{E, \mathrm{Fr}}$?

We will show this with the following assumptions.

Assumption 3.2.2.6. Let $\mathcal{D}$ to be a $\overline{\mathbb{Q}}_\ell$ -linear category enriched in $\mathbf{Mod}_{\mathrm{Fr}}$ (equivalently, the Hom and Ext groups are Frobenius modules), and is equipped with the following data:

- (1) A finite poset Λ such that $\mathcal{D}$ has a recollement indexed by Λ (that is, a series of open-closed semi-orthogonal decompositions). We define $\mathcal{D}_\lambda = \mathcal{D}_{\leq \lambda} / \mathcal{D}_{< \lambda}$.
- (2) A full additive subcategory $\mathcal{C}_\lambda \subset \mathcal{D}_\lambda$ stable under taking tensor with unipotent Frobenius modules, such that for any objects $c_1, c_2 \in \mathcal{C}_\lambda$, we have

$$\mathrm{Ext}_{\mathcal{D}_\lambda}^i(c_1, c_2)^{\mathrm{Fr}\text{-unip}} = 0$$

- (3) $\mathcal{C}_\lambda$ generates $\mathcal{D}_\lambda$ as a triangulated category. Moreover, there is an object $\mathbb{1}_\lambda \in \mathcal{C}_\lambda$ such that every object in $\mathcal{C}_\lambda$ is of the form $\mathbb{1}_\lambda \otimes M$ for some $M \in D(\text{Fr})$ (however, for arbitrary M , the object $\mathbb{1}_\lambda \otimes M \in \mathcal{D}_\lambda$ may not lie in $\mathcal{C}_\lambda$).
- (4) There exists an object $c_\lambda \in \mathcal{C}_{\leq \lambda}$ such that $i_\lambda^* c_\lambda = \mathbb{1}_\lambda$. Moreover, the induced ring morphism

$$i_\alpha^* : \text{End}(c_\lambda) \rightarrow \text{End}(\mathbb{1}_\lambda)$$

has a nilpotent kernel.

We let $\mathcal{C}$ denote the full subcategory of $\mathcal{D}$ consisting of objects d such that $i_\lambda^* d, i_\lambda^! d \in \mathcal{C}_\lambda$.

In practice, $\mathcal{D}$ will be a category of mixed ℓ -adic sheaves constructible with respect to a certain stratification, and $\mathcal{C}_\alpha$ the weight 0 constant sheaves (normally we would have locally constant, but we require constant in our case) over a strata. Then $\mathcal{C}$ will be the category of weight 0 sheaves.

The main theorem is the following, which would be our main technical tool later on.

Theorem 3.2.2.7. For a category $\mathcal{D}$ satisfying the requirements in [Assumption 3.2.2.6](#), there is a $\overline{\mathbb{Q}_\ell}$ -algebra E with a Frobenius action, such that $\mathcal{D} \hookrightarrow \mathbf{Mod}_{E, \text{Fr}}$ is fully faithful in the sense that both categories are enriched in $\mathbf{Set}$ rather than $D(\text{Fr})$. Moreover, such construction is functorial for functors preserving the subcategory $\mathcal{C}$.

We will begin with a simple lemma on the computation of a homotopy fixed point.

Lemma 3.2.2.8. Let $M \in \mathbf{Mod}_{\text{Fr}}$. Then,

$$0 \longrightarrow H^{i-1}(M)_{\text{Fr}} \longrightarrow H^i(M^{h\text{Fr}}) \longrightarrow H^i(M)^{\text{Fr}} \longrightarrow 0$$

Proof: Consider the following description of homotopy fixed points of a $\mathbb{Z}$ action as a fiber

$$M^{h\text{Fr}} \longrightarrow M \xrightarrow{\text{Fr} - \text{id}} M$$

Then the desired exact sequence follows from the long exact sequence of the fiber sequence. $\square$

Corollary 3.2.2.9. Let $\mathcal{D}$ satisfy the condition (1) and (2) in [Assumption 3.2.2.6](#). Then, for $c_1, c_2 \in \mathcal{C}$, we have

$$\text{Ext}_{\mathcal{D}}^i(c_1, c_2)^{\text{Fr-unip}} = 0, i \neq 0$$

and

$$H^i(R\text{Hom}_{\mathcal{D}}(c_1, c_2)^{h\text{Fr}}) = \begin{cases} \text{Hom}_{\mathcal{C}}(c_1, c_2)^{\text{Fr}} & \text{if } i = 0 \\ \text{Hom}_{\mathcal{C}}(c_1, c_2)_{\text{Fr}} & \text{if } i = 1 \\ 0 & \text{otherwise} \end{cases}$$

Proof: For the first assertion, we apply induction on the stratas. On one strata, this is precisely condition (2). Then consider the excision triangle, which gives rise to the following long exact sequence for $c_1, c_2 \in \mathcal{C}_{\leq \lambda}$

$$\cdots \longrightarrow \mathrm{Ext}_{\mathcal{C}_{<\lambda}}^i(i_{<\lambda}^*c_1, i_{<\lambda}^!c_2)^{\mathrm{Fr-unip}} \longrightarrow \mathrm{Ext}^i(c_1, c_2)^{\mathrm{Fr-unip}} \longrightarrow \mathrm{Ext}_{\mathcal{C}_\lambda}^i(i_\lambda^*c_1, i_\lambda^*c_2)^{\mathrm{Fr-unip}} \longrightarrow \cdots$$

which gives the vanishing result.

For the second assertion, notice that for any $M \in \mathbf{Mod}_{\mathrm{Fr}}^\heartsuit$, we have $M^{\mathrm{Fr}} \subset M^{\mathrm{Fr-unip}}$. The rest follows from the previous lemma. $\square$

Lemma 3.2.2.10. We adopt the decreasing filtration convention. Then, there is an equivalence of categories

$$\widetilde{\mathrm{gr}} : \mathrm{Fil}(\mathcal{C}) \rightarrow \mathrm{Ch}(\mathcal{C})$$

where $\mathrm{Fil}(\mathcal{C})$ is the full subcategory of $\mathrm{Fil}(\mathcal{D})$ such that $\mathrm{gr}^i(F^\bullet d) \in \mathcal{C}[-i]$, and $\mathrm{Ch}(\mathcal{C})$ is the category of chain complexes without any homotopy relations. Here, chain complexes mean the composite of two differentials are homotopic to 0. The equivalence is given by

$$\widetilde{\mathrm{gr}}(F^\bullet d) = \cdots \longrightarrow \mathrm{gr}^i(d)[i] \longrightarrow \mathrm{gr}^{i+1}(d)[i+1] \longrightarrow \cdots$$

where the transition maps are given by the following triangle

$$\mathrm{gr}^{i+1}(F^\bullet d) \longrightarrow \mathrm{gr}^{[i, i+1]}(F^\bullet d) \longrightarrow \mathrm{gr}^i(F^\bullet d) \longrightarrow \mathrm{gr}^{i+1}(F^\bullet d)[1]$$

Proof: This is well-known if $\mathcal{C}$ is a stable ∞ -category. Now, $\mathcal{C}$ is a additive subcategory of such, the proof is likewise. $\square$

Definition 3.2.2.11. The composition $\mathrm{Ch}(\mathcal{C}) \rightarrow \mathrm{Fil}(\mathcal{C}) \rightarrow \mathcal{D}$ (the second functor is the functor sending a filtered object to its underlying object in $\mathcal{D}$) descends to a functor $\tilde{\rho}(\mathcal{C}) : K(\mathcal{C}) \rightarrow \mathcal{D}$. Let $K_{\mathrm{acyc}}(\mathcal{C}) \subset K(\mathcal{C})$ denote the subcategory of the bounded homotopy category spanned by objects sent to $0 \in \mathcal{D}$ by $\tilde{\rho}(\mathcal{C})$.

Lemma 3.2.2.12. For any $c \in \mathcal{C}$, the functors

$$\mathrm{Hom}(c, -)^{\mathrm{Fr-unip}}, \mathrm{Hom}(-, c)^{\mathrm{Fr-unip}} : K_{\mathrm{acyc}}(\mathcal{C}) \rightarrow K_{\mathrm{acyc}}(\mathcal{C})$$

maps any complex to an exact sequence.

Proof: Fix an object $\mathcal{F} \in \mathrm{Fil}(\mathcal{C})$ such that $\widetilde{\mathrm{gr}} \mathcal{F} \in K_{\mathrm{acyc}}(\mathcal{C})$. Consider the spectral sequence

$$E_1^{p,q} = \mathrm{Ext}^{p+q}(c, \mathrm{gr}^p \mathcal{F}) = \mathrm{Ext}^q(c, \mathrm{gr}^p \mathcal{F}[p]) \implies \mathrm{Ext}^{p+q}(c, \mathcal{F})$$

However, since $\widetilde{\mathrm{gr}} \mathcal{F} \in K_{\mathrm{acyc}}(\mathcal{C})$, $\mathcal{F} \simeq 0 \in \mathcal{D}$. Thus, the above spectral sequence converges to 0.

Then, applying $(-)^{\mathrm{Fr-unip}}$ to it, the spectral sequence degenerates at page 1 because q is forced to be 0. Thus, since the spectral sequence also converges to zero, the complex at $q = 0$ is exact. $\square$

Definition 3.2.2.13. We define $\mathcal{C}^{\mathrm{Fr-unip}}$ to be the category with the same objects as $\mathcal{C}$, but with Hom groups replaced by

$$\mathrm{Hom}_{\mathcal{C}^{\mathrm{Fr-unip}}}(c_1, c_2) = \mathrm{Hom}_{\mathcal{C}}(c_1, c_2)^{\mathrm{Fr-unip}}$$

Construction 3.2.2.14. Notice that the forgetful functor $\mathcal{C} \rightarrow \mathcal{C}^{\mathrm{Fr-unip}}$ admits a pro-left adjoint $\mathcal{C}^{\mathrm{Fr-unip}} \rightarrow \mathrm{Pro}(\mathcal{C})$, given by $c \mapsto c[[t]] = \lim c[t]/t^n = c \otimes \overline{\mathbb{Q}}_\ell[[t]]$, where Frobenius acts on $\overline{\mathbb{Q}}_\ell[[t]]$ by multiplying by $\exp t$. Explicitly, we have

$$\mathrm{Hom}(c, c')^{\mathrm{Fr-unip}} = \mathrm{Hom}(c[[t]], c')^{\mathrm{Fr}}$$

We remark that the exponential here is used to ensure unipotency.

Proposition 3.2.2.15. We have a well-defined functor $\rho(\mathcal{C}) : K(\mathcal{C})/K_{\mathrm{acyc}}(\mathcal{C}) \rightarrow \mathcal{D}$. Moreover, it is fully faithful, and is essentially surjective if $\mathcal{C}_\alpha$ generates $\mathcal{D}_\alpha$ as a triangulated category.

Proof: Recall that maps $\mathrm{Hom}_{K(\mathcal{C})/K_{\mathrm{acyc}}(\mathcal{C})}(c_1, c_2)$ are computed by spans $c_1 \leftarrow c \rightarrow c_2$, where $\mathrm{cofib}(c \rightarrow c_1) \in K_{\mathrm{acyc}}(\mathcal{C})$. We may assume that $c_1 \in \mathcal{C}$, and we first show that taking the two term complex $c_1[[t]] \rightarrow c_1[[t]]$ is an initial (pro-) object mapping to c_1 with an acyclic cone. For this, we suppose $c \rightarrow c_1$ has an acyclic cone. Writing

$$c = \cdots \rightarrow c^{-1} \rightarrow c^0 \rightarrow c^1 \rightarrow \cdots$$

the cone is of the form

$$\cdots \rightarrow c^{-1} \rightarrow c^0 \rightarrow c^1 \oplus c_1 \rightarrow \cdots$$

Thus, by [Lemma 3.2.2.12](#), taking $\mathrm{Hom}(c_1, -)^{\mathrm{Fr-unip}}$, we obtain a long exact sequence

$$\cdots \rightarrow \mathrm{Hom}(c_1, c^0)^{\mathrm{Fr-unip}} \rightarrow \mathrm{Hom}(c, c^1 \oplus c_1)^{\mathrm{Fr-unip}} \rightarrow \cdots$$

This amounts to the exact sequence

$$\cdots \rightarrow \mathrm{Hom}(c_1[[t]], c^0)^{\mathrm{Fr}} \rightarrow \mathrm{Hom}(c_1[[t]], c^1 \oplus c_1)^{\mathrm{Fr}} \rightarrow \cdots$$

Note the Frobenius equivariant map $(0, p) : c_1[[t]] \rightarrow c^1 \oplus c_1$ has zero image, thus it comes from a map $\varphi_0 : c_1[[t]] \rightarrow c^0$. Then, since we want $- \times t : c_1[[t]] \rightarrow c_1[[t]]$, we also have $t\varphi_0 : c_1[[t]] \rightarrow c^0$ zero, so it comes from a map $\varphi_{-1} : c_1[[t]] \rightarrow c^{-1}$. Thus, we obtain a commutative square

$$\begin{array}{ccc} c_1[[t]] & \xrightarrow{t} & c_1[[t]] \\ \varphi_{-1} \downarrow & & \downarrow \varphi_0 \\ c^{-1} & \longrightarrow & c^0 \end{array}$$

which in turn exhibits initialness.

Hence, $\mathrm{Hom}_{K(\mathcal{C})/K_{\mathrm{acyc}}(\mathcal{C})}^\bullet(c_1, c_2)^{h\mathrm{Fr}}$ is equivalently the two term complex

$$0 \rightarrow \mathrm{Hom}(c_1[[t]], c_2)^{\mathrm{Fr}} \xrightarrow{(\times t)^*} \mathrm{Hom}(c_1[[t]], c_2)^{\mathrm{Fr}} \rightarrow 0$$

which, by the universal property, is the complex

$$0 \rightarrow \mathrm{Hom}(c_1, c_2)^{\mathrm{Fr-unip}} \xrightarrow{\log \mathrm{Fr}} \mathrm{Hom}(c_1, c_2)^{\mathrm{Fr-unip}} \rightarrow 0$$

which is precisely the complex computing $\mathrm{Hom}_{\mathcal{D}}^{\bullet}(c_1, c_2)^{\mathrm{Fr}}$, by [Corollary 3.2.2.9](#). Thus the functor is fully faithful. The image is clearly generated by the $\mathcal{C}_{\alpha}$'s, in a somewhat inaccurate sense. $\square$

Now, we notice that $\mathcal{C}^{\mathrm{Fr-unip}}$ is in some sense the Frobenius semisimplification, as opposed to $\mathcal{C}^{\mathrm{Fr}}$ is removing the Frobenius structure. This can be shown in the following lemma.

Lemma 3.2.2.16. From now on, we require $\mathcal{D}$ to satisfy all assumptions from [Assumption 3.2.2.6](#). For any $c \in \mathcal{C}$, there are complexes $M_{\lambda} \in \mathbf{Mod}_{\mathrm{Fr}}$, such that

$$c = \bigoplus c_{\lambda} \otimes M_{\lambda} \in \mathcal{C}^{\mathrm{Fr-unip}}$$

Proof: Notice that we have $\mathrm{Ext}_{\mathcal{C}_{<\lambda}}^1(i_{<\lambda}^* c_1, i_{<\lambda}^! c_2)^{\mathrm{Fr-unip}} = 0$, so the restriction map

$$\mathrm{Hom}_{\mathcal{C}}(c_1, c_2)^{\mathrm{Fr-unip}} \twoheadrightarrow \mathrm{Hom}_{\mathcal{C}_{\lambda}}(i_{\lambda}^* c_1, i_{\lambda}^* c_2)^{\mathrm{Fr-unip}}$$

is surjective.

To prove the statement, we apply induction on the support of c . Suppose $c \in \mathcal{C}_{\leq \lambda}$. Then we have $i_{\lambda}^* c = \mathbb{1}_{\lambda} \otimes M_{\lambda}$. Moreover, since the left isomorphism is naturally Frobenius invariant, by the above surjection, the isomorphism lifts to a retraction diagram

$$c_{\lambda} \otimes M_{\lambda} \rightarrow c \rightarrow c_{\lambda} \otimes M_{\lambda}$$

This is a retraction since $\mathrm{End}(c_{\lambda}) \rightarrow \mathrm{End}(\mathbb{1}_{\lambda})$ has nilpotent kernel, so any preimage of an isomorphism is an isomorphism.

Since retractions splits off, we may assume $c = c_{\lambda} \otimes M_{\lambda} \oplus c'$, where c' is supported on $\mathcal{C}_{<\lambda}$. Thus by induction, the proof is finished. $\square$

Corollary 3.2.2.17. Let $c \in \mathcal{C}$ be such that $\mathrm{End}(c)$ is Fr-semisimple. Then,

$$c \simeq \bigoplus c_{\lambda} \otimes M_{\lambda} \in \mathcal{C}$$

such that M_{λ} is a Frobenius semisimple complex.

Proof: The idempotents corresponding to $c_{\lambda} \otimes M_{\lambda}$ in $\mathcal{C}^{\mathrm{Fr-unip}}$ is actually Frobenius invariant because $\mathrm{End}(c)$ is semisimple, so the direct sum decomposition occurs in $\mathcal{C}$. Then, obviously each M_{λ} would be semisimple, otherwise $\mathrm{End}(M_{\lambda})$ would not be semisimple. $\square$

Now, suppose there is a family of objects $d_{\lambda} \in \mathcal{C}_{\leq \lambda}$ also pulling back to $\mathbb{1}_{\lambda}$. For example, $d_{\lambda} = c_{\lambda}$ is a valid choice.

Lemma 3.2.2.18. $\mathcal{D}$ is generated by $\{d_{\lambda}\}_{\lambda \in \Lambda}$.

Proof: Again we apply induction on the stratas. Suppose $\mathcal{D}_{<\lambda}$ is generated by d_{μ} s for $\mu < \lambda$. Then, by adjunction, we have a map $i_{\lambda!} \mathbb{1}_{\lambda} \rightarrow d_{\lambda}$, whose cofiber lies in $\mathcal{D}_{<\lambda}$, so is generated by d_{μ} s for $\mu < \lambda$. Thus, $i_{\lambda!} \mathbb{1}$ is generated by d_{μ} s for $\mu \leq \lambda$. But $\mathcal{D}_{\leq \lambda}$ is generated by $i_{\lambda!} \mathbb{1}_{\lambda}$ and $\mathcal{D}_{<\lambda}$ by assumption (3). Thus, we are done. $\square$

Proof: Now we are ready to prove [Theorem 3.2.2.7](#). We let $d = \bigoplus d_\lambda$, and take the $\overline{\mathbb{Q}}_\ell$ -algebra E to be

$$E = \bigoplus_{i \in \mathbb{Z}} \text{Ext}_{\mathcal{D}}^i(d, d)^{\text{op}}$$

In particular, this is concentrated at degree 0, so not a DG algebra. We will prove that we have a fully faithful functor $\mathcal{C} \hookrightarrow \mathbf{Mod}_{E, \text{Fr}}^\heartsuit$, and then take derived categories, and prove $K(\mathcal{C})/K_{\text{acyc}}(\mathcal{C}) \hookrightarrow \mathbf{Mod}_{E, \text{Fr}}$ is fully faithful. Finally, by [Proposition 3.2.2.15](#), we obtain the desired fully faithful map.

Firstly, the functor $\mathcal{C} \hookrightarrow \mathbf{Mod}_{E, \text{Fr}}^\heartsuit$ is via the obvious Yoneda functor

$$h_d = \bigoplus_{i \in \mathbb{Z}} \text{Ext}_{\mathcal{D}}^i(d, -) : \mathcal{C} \rightarrow \mathbf{Mod}_{E, \text{Fr}}^\heartsuit$$

Note that by definition

$$\text{Hom}_{\mathbf{Mod}_{E, \text{Fr}}} (h_d(c_1), h_d(c_2)) = \text{Hom}_E(h_d(c_1), h_d(c_2))^{\text{Fr}}$$

so we only need

$$\text{Hom}_{\mathcal{C}}(c_1, c_2)^{\text{Fr-unip}} \xrightarrow{\cong} \text{Hom}_E(h_d(c_1), h_d(c_2))^{\text{Fr-unip}}$$

For this, if we have $c_1 = d$, then

$$\text{Hom}_E(h_d(d), h_d(c_2))^{\text{Fr-unip}} = \text{Hom}_E(E, h_d(c_2))^{\text{Fr-unip}} = h_d(c_2)^{\text{Fr-unip}} = \text{Hom}_{\mathcal{C}}(d, c_2)^{\text{Fr-unip}}$$

And then, by projecting to direct summands in $\mathcal{C}^{\text{Fr-unip}}$, we see the above map is an isomorphism for d_λ s. Finally, since the d_λ s generates $\mathcal{C}^{\text{Fr-unip}}$ under direct sums, we have the functor fully faithful.

Then passing to homotopy categories, we first prove $K_{\text{acyc}}(\mathcal{C})$ is null-homotopic in $K(\mathcal{C}^{\text{Fr-unip}})$. This is done by [Lemma 3.2.2.12](#), and explicitly constructing the contracting homotopy via the exactness of Hom functors. Thus, we have a well-defined map $K(\mathcal{C})/K_{\text{acyc}}(\mathcal{C}) \rightarrow \mathbf{Mod}_{E, \text{Fr}}$. However, in this case, the Ext groups are computed likewise as above, so we are done.

Finally, for the functoriality, for a map $\Phi : \mathcal{D} \rightarrow \mathcal{D}'$ with $\Phi|_{\mathcal{C}} : \mathcal{C} \rightarrow \mathcal{C}'$, we claim that we have the following commutative diagram:

$$\begin{array}{ccccc} \mathcal{D} & \xleftarrow{\rho(\mathcal{C})} & K(\mathcal{C})/K_{\text{acyc}}(\mathcal{C}) & \xrightarrow{h_d} & \mathbf{Mod}_{E, \text{Fr}} \\ \Phi \downarrow & & \downarrow K(\Phi|_{\mathcal{C}}) & & \downarrow B_\Phi \otimes_E (-) \\ \mathcal{D}' & \xleftarrow{\rho(\mathcal{C}')} & K(\mathcal{C}')/K_{\text{acyc}}(\mathcal{C}') & \xrightarrow{h_{d'}} & \mathbf{Mod}_{E', \text{Fr}} \end{array}$$

where $B_\Phi = \text{Hom}_{\mathcal{C}'}(d', \Phi(d))$. The left square obviously commutes, and for the right square, it suffices to show the following square commutes:

$$\begin{array}{ccc} \mathcal{C} & \xrightarrow{h_d} & \mathbf{Mod}_{E, \text{Fr}}^\heartsuit \\ \Phi|_{\mathcal{C}} \downarrow & & \downarrow B_\Phi \otimes_E (-) \\ \mathcal{C}' & \xrightarrow{h_{d'}} & \mathbf{Mod}_{E', \text{Fr}}^\heartsuit \end{array}$$

for this square, note we have a natural transformation

$$B_\Phi \otimes_E (-) = \mathrm{Hom}_{\mathcal{C}'}(d', \Phi(d)) \otimes_E \mathrm{Hom}_{\mathcal{C}}(d, -) \rightarrow \mathrm{Hom}_{\mathcal{C}'}(d', \Phi(-))$$

via the composition map. It suffices to prove this map is an isomorphism for all $c \in \mathcal{C}$. Obviously, it is an isomorphism for $c = d$ hence d_λ for any λ . By [Lemma 3.2.2.16](#), c_λ is a direct summand of d_λ in $\mathcal{C}^{\mathrm{Fr}\text{-unip}}$, hence the above map is an isomorphism for all c_λ . But since c_λ generates the whole category as in [Lemma 3.2.2.16](#), we are done. $\square$

Finally, we will also be able to compute the image of the above fully faithful functor.

Corollary 3.2.2.19. The image of the functor $\mathcal{D} \hookrightarrow \mathbf{Mod}_{E, \mathrm{Fr}}$ is the full subcategory generated by the twists of $\mathrm{Hom}(d, d_\lambda)$ s as a triangulated category.

Proof: Obvious. $\square$

Another first corollary is the computation of A -equivariant derived category over a point, where A is any torus.

Definition 3.2.2.20. Let A be any torus over a finite field. Then we define its ℓ -adic Tate module

$$T_\ell(A) = \varprojlim_{n \rightarrow \infty} A[\ell^n]$$

where $A[\ell^n]$ is the ℓ^n -torsion points of A , and $V_A = T_\ell(A) \times_{\mathbb{Z}_\ell} \overline{\mathbb{Q}_\ell} = H_1(A, \overline{\mathbb{Q}_\ell})$. By definition, this is a Fr -module of weight -2 .

We then define $\check{S}_A = \mathrm{Sym} V_A^\vee$, which is a $\overline{\mathbb{Q}_\ell}$ -algebra with Fr -action. We can also define the graded version $\check{S}_A^\bullet$, where we put $V_A^\vee$ in degree 2.

remark 3.2.2.1. Notice that the Tate module $T_\ell(A)$ is equivalently the pro- ℓ part of $\pi_1(A)^{\mathrm{tame}}$, since $A[\ell^k]$ can be identified as the Galois group of the ℓ^k -th power covering $A \rightarrow A$.

Corollary 3.2.2.21. Take any torus A , we have an equivalence of categories

$$D_m(\mathbb{B}A) \simeq \mathbf{Mod}_{\check{S}_A, \mathrm{Fr}}^\omega$$

The pullback functor $D_m(\mathbb{B}A) \rightarrow D_m(\mathrm{pt}) = \mathbf{Mod}_{\mathrm{Fr}}$ corresponds to the functor $(-) \otimes_{\check{S}_A} \overline{\mathbb{Q}_\ell}$ using the natural augmentation of $\check{S}_A$.

Proof: By [Theorem 3.2.2.7](#), it suffices to verify that $D_m(\mathbb{B}A)$ satisfies the assumptions

[Assumption 3.2.2.6](#). For (1), we simply pick the trivial decomposition. For (2), we take $\mathcal{C}_\lambda$ to be the very pure of weight 0 complexes (see [Definition 3.2.3.7](#)) such that they are constant over $\bar{k} = \overline{\mathbb{F}_q}$. Then, the $\mathrm{Ext}_{D_\lambda}^i(c_1, c_2)$ would be pure of weight i by the same argument as in [Proposition 3.2.3.10](#), whose Frobenius unipotent part obviously vanish. They are clearly stable under taking tensors with unipotent Frobenius modules.

For (3), we take $\mathbb{1}_\lambda$ to be the constant sheaf. Since the torus A is connected, the space $\mathbb{B}A$ is simply connected, so $D_m(\mathbb{B}A)$ is generated by the object $\mathbb{1}_\lambda$, let along the subcategory $\mathcal{C}$ of very

pure of weight 0 complexes. By definition, every constant very pure of weight 0 complex is the tensor product of $\mathbb{1}_\lambda$ with some complex of Frobenius module M . The condition (4) is obvious since there is only 1 stratum.

Finally, the essential image of the fully faithful embedding is given by [Corollary 3.2.2.19](#), which is $D(\check{S}_A, \text{Fr})^\omega$, or equivalently, the smallest triangulated subcategory generated by the constant sheaf.

The second assertion follows from the functoriality of $\check{S}_A$ by [Theorem 3.2.2.7](#). □

3.2.3. Application 1: The equivariant (irreducible) side

In this section, we will consider the properties of the category $D_m(B \backslash G/B)$ and its parabolic variant $D_m(P_\Theta \backslash G/B)$.

Definition 3.2.3.1. For each $\bar{w} \in [W_\Theta \backslash W]$, we define the following objects in $D_m(P_\Theta \backslash G/B)$:

$$\Delta_{\bar{w}} = i_{\bar{w},!} \overline{\mathbb{Q}}_\ell[\ell(\bar{w})](\ell(\bar{w})/2), \quad \nabla_{\bar{w}} = i_{\bar{w},*} \overline{\mathbb{Q}}_\ell[\ell(\bar{w})](\ell(\bar{w})/2), \quad \text{IC}_{\bar{w}} = i_{\bar{w},!*} \overline{\mathbb{Q}}_\ell[\ell(\bar{w})](\ell(\bar{w})/2)$$

The projection ${}_\Theta \pi : \text{Fl} \rightarrow {}_\Theta \text{Fl}$ is proper, so we obtain two pairs of adjunctions

$$\begin{array}{ccc} & \xleftarrow{{}_\Theta \pi^*} & \\ & \perp & \\ D_m(P_\Theta \backslash G/B) & \xrightarrow{{}_\Theta \pi_*} & D_m(B \backslash G/B) \\ & \perp & \\ & \xrightarrow{{}_\Theta \pi^!} & \end{array}$$

We will first investigate the relations between the categories $D_m(P_\Theta \backslash G/B)$ through the projection ${}_\Theta \pi$.

Lemma 3.2.3.2. For $w \in W$, we write $w = u\bar{w}$ for $u \in W_\Theta$ and $\bar{w} \in [W_\Theta \backslash W]$. Then, we have

$${}_\Theta \pi_*(\Delta_w) = \Delta_{\bar{w}}[-\ell(u)](-\ell(u)/2), \quad {}_\Theta \pi_*(\nabla_w) = \nabla_{\bar{w}}[\ell(u)](\ell(u)/2)$$

Proof: Firstly, notice that the map of cells $\text{Fl}_w \rightarrow {}_\Theta \text{Fl}_{\bar{w}}$ is the projection

$${}_\Theta \pi_w : \mathbb{A}^{\ell(w)} = \mathbb{A}^{\ell(u)} \times \mathbb{A}^{\ell(\bar{w})} \twoheadrightarrow \mathbb{A}^{\ell(\bar{w})}$$

Then, we have the computation

$$\begin{aligned} {}_\Theta \pi_*(\Delta_w) &= {}_\Theta \pi_!(\Delta_w) = {}_\Theta \pi_! i_{w,!} \overline{\mathbb{Q}}_\ell[\ell(w)](\ell(w)/2) \\ &= i_{\bar{w},!} {}_\Theta \pi_{w,!} \overline{\mathbb{Q}}_\ell[\ell(w)](\ell(w)/2) \\ &= i_{\bar{w},!} \overline{\mathbb{Q}}_\ell[\ell(w) - 2\ell(u)]((\ell(w) - 2\ell(u))/2) \\ &= (i_{\bar{w},!} \overline{\mathbb{Q}}_\ell[\ell(\bar{w})](\ell(\bar{w})/2))[-\ell(u)](-\ell(u)/2) \\ &= \Delta_{\bar{w}}[-\ell(u)](-\ell(u)/2) \end{aligned}$$

The pushforward of ∇_w is computed likewise. □

The computation of pushforward of IC_w is much more harder. We postpone this to the later part of this section (Lemma 3.2.3.18). In fact, we can not give a complete characterization of this pushforward, but only a few things on its components.

Lemma 3.2.3.3. We have a natural isomorphism of functors ${}_{\Theta}\pi^* {}_{\Theta}\pi_*(-) \simeq \mathbb{1}_{\Theta} *_B (-)$. Here, $\mathbb{1}_{\Theta}$ is the constant sheaf on $\mathrm{Fl}_{\leq w_{\Theta}}$ (w_{Θ} is the maximal length element in W_{Θ}), so is equivalently $\mathrm{IC}_{w_{\Theta}}[-\ell_{\Theta}](-\ell_{\Theta}/2)$.

Proof: Firstly note that we have the following Cartesian diagram

$$\begin{array}{ccc} (B_{\Theta} \backslash L_{\Theta}) \times^{B_{\Theta}} (U_{\Theta} \backslash G) & \xrightarrow{m} & B \backslash G \\ p_2 \downarrow \lrcorner & & \downarrow {}_{\Theta}\pi \\ B \backslash G & \xrightarrow{{}_{\Theta}\pi} & P_{\Theta} \backslash G \end{array}$$

This square is Cartesian because we can explicitly construct a map

$$(B_{\Theta} \backslash L_{\Theta}) \times^{B_{\Theta}} (U_{\Theta} \backslash G) \rightarrow (B \backslash G) \times_{P_{\Theta} \backslash G} (B \backslash G), (B_{\Theta} l, U_{\Theta} g) \mapsto (Blg, Bg)$$

and this is an isomorphism by the decomposition $B = U_{\Theta} B_{\Theta}$, and hence we have an inverse map $(Bg_1, Bg_2) \mapsto (B_{\Theta} l, U_{\Theta} g_2)$, where l is the unique element such that $Bg_1 = Blg_2$.

By the proper base change along the Cartesian diagram, we have ${}_{\Theta}\pi^* {}_{\Theta}\pi_*(-) \simeq m_* p_2^*(-)$. Then, in the coordinates $(B_{\Theta} \backslash L_{\Theta}) \times^{B_{\Theta}} (U_{\Theta} \backslash G)$, we can write $p_2^*(\mathcal{F}) = \mathbb{1}_{\Theta} \boxtimes^{B_{\Theta}} \mathcal{F}$, where $B_{\Theta} \backslash L_{\Theta}$ is identified as ${}_{\Theta}\pi^{-1}(P_{\Theta} \backslash P_{\Theta}) \subset B \backslash G$. Finally, $m_*(\mathbb{1}_{\Theta} \boxtimes^{B_{\Theta}} \mathcal{F}) = \mathbb{1}_{\Theta} *_B \mathcal{F}$, because

$$(B_{\Theta} \backslash L_{\Theta}) \times^{B_{\Theta}} (U_{\Theta} \backslash G) = (B_{\Theta} \backslash L_{\Theta}) \times^B G = {}_{\Theta}\pi^{-1}(P_{\Theta} \backslash P_{\Theta}) \times^B G \subset \mathrm{Fl} \times^B G$$

□

Definition 3.2.3.4. Besides the tensor product, there is a convolution product on $D_m(B \backslash G/B)$, given by the diagram

$$B \backslash G/B \times B \backslash G/B \xleftarrow{\delta} B \backslash G \times^B G/B \xrightarrow{m} B \backslash G/B$$

where δ sends (g_1, g_2) to (g_1, g_2) is well-defined, since the middle quotient splits into 2 quotients. We then define $\mathcal{F}_1 *_B \mathcal{F}_2 = m_! \delta^*(\mathcal{F}_1 \boxtimes \mathcal{F}_2)$. Note that here m is ind-proper and δ is a quotient by B hence smooth.

Lemma 3.2.3.5. If $u, v \in W$ is such that $\ell(u) + \ell(v) = \ell(uv)$, then

$$\Delta_u *_B \Delta_v = \Delta_{uv}, \quad \nabla_u *_B \nabla_v = \nabla_{uv}$$

Proof: Follows from that $\mathrm{Fl}_u \times^B \mathrm{Fl}_v \rightarrow \mathrm{Fl}_{uv}$ is an isomorphism.

□

Proposition 3.2.3.6. For $\mathcal{F}_1, \mathcal{F}_2 \in D_m(B \backslash G/B)$ very pure of weight 0, we have $\mathcal{F}_1 *_B \mathcal{F}_2$ very pure of weight 0.

Proof: Note that the category of very pure complexes of weight 0 is

$$\langle \Delta_w | w \in W \rangle \cap \langle \nabla_w | w \in W \rangle$$

where $\langle - \rangle$ denote the additive category generated by some objects under extensions. Then, we have

$$\Delta_u *_B \Delta_v \in \langle \Delta_w \langle \leq 0 \rangle | w \in W \rangle$$

and

$$\nabla_u *_B \nabla_v \in \langle \nabla_w \langle \geq 0 \rangle | w \in W \rangle$$

by writing the standard and costandard sheaves into convolutions of standard and costandard sheaves associated to simple reflections using [Lemma 3.2.3.5](#). The convolution $\Delta_s *_B \Delta_s$ is a very classical computation using excision on $\mathbb{P}^1$, and we invite the readers not familiar with the result to compute it by themselves.

Thus, we have

$$\mathcal{F}_1 *_B \mathcal{F}_2 \in \langle \Delta_w \langle \leq 0 \rangle | w \in W \rangle \cap \langle \nabla_w \langle \geq 0 \rangle | w \in W \rangle$$

which implies $\mathcal{F}_1 *_B \mathcal{F}_2$ is pure of weight 0. Hence by [Lemma 3.2.3.9](#), it is very pure of weight 0. $\square$

Definition 3.2.3.7. Let $X = \coprod X_\lambda$ be a stratified scheme, and suppose $\mathcal{F} \in D_m(X)$ is constructible with respect to the stratification. Then, let $i_\lambda : X_\lambda \hookrightarrow X$ denote the embeddings, we say $\mathcal{F}$ is $*$ -pure or $!$ -pure of weight n if for each λ and each $m \in \mathbb{Z}$, we have $H^m(i_\lambda^*(\mathcal{F}))$ or $H^m(i_\lambda^!(\mathcal{F}))$ is a sheaf pointwise pure of weight $m + n$. We say $\mathcal{F}$ is very pure of weight 0 if it is both $!$ -pure and $*$ -pure of weight 0.

Lemma 3.2.3.8. Let V be a variety over k , equipped with a $\mathbb{G}_m$ -action contracting V to a point $v \in V$. Then, we have for any complex of $\mathbb{G}_m$ -equivariant constructible ℓ -adic sheaf $\mathcal{F}$, the cohomology $H^*(V, \mathcal{F}) = \mathcal{F}_v$.

Proof: By the spectral sequence computing the hypercohomology, it suffices to prove that for any $\mathbb{G}_m$ -equivariant ℓ -adic sheaf $\mathcal{F}$, the cohomology $H^*(V, \mathcal{F}) = \mathcal{F}_v$ (in particular, the higher cohomologies vanishes). Then, by replacing $\mathcal{F} = \ker(\mathcal{F} \rightarrow v_* v^* \mathcal{F})$, we may assume $\mathcal{F}_v = 0$. We now prove $\mathcal{F}$ has trivial cohomology.

We denote $\mu : \mathbb{G}_m \times V \rightarrow V$, and extend μ naturally to $\mu : \mathbb{A}^1 \times V \rightarrow V$, where $(0, u) \mapsto v$ for any $u \in V$. Then, consider the shearing map $\tau : \mathbb{A}^1 \times V \rightarrow \mathbb{A}^1 \times V$, which sends $(t, u) \mapsto (t, \mu(t, u))$. Note that τ is an $\mathbb{A}^1$ -morphism, where $\mathbb{A}^1 \times V$ is equipped with the $\mathbb{A}^1$ -scheme structure by the map $\pi_1 : \mathbb{A}^1 \times V \rightarrow \mathbb{A}^1$.

Since $\mathcal{F}$ is $\mathbb{G}_m$ -equivariant, we have $\pi_2^*(\mathcal{F})|_{\mathbb{G}_m \times V} = \tau^* \pi_2^*(\mathcal{F})|_{\mathbb{G}_m \times V}$. Then, $\tau^* \pi_2^*(\mathcal{F})|_{0 \times V} = 0$ by assumption, so if we let $j : \mathbb{G}_m \times V \hookrightarrow \mathbb{A}^1 \times V$ denote the inclusion, we have

$$\tau^* \pi_2^*(\mathcal{F}) = j_! i^* \pi_2^*(\mathcal{F})$$

Applying the counit $i_! i^* \rightarrow \text{id}$ and the derived projection to $\mathbb{A}^1$, we obtain a map

$$\alpha : \pi_{1,*} \tau^* \pi_2^*(\mathcal{F}) \rightarrow \pi_{1,*} \pi_2^*(\mathcal{F})$$

On the other hand, since we have the commutative triangle

$$\begin{array}{ccc} \mathbb{A}^1 \times V & \xrightarrow{\tau} & \mathbb{A}^1 \times V \\ & \searrow \pi_1 & \swarrow \pi_1 \\ & V & \end{array}$$

we have a natural transformation $\pi_{1,*} \rightarrow \pi_{1,*} \tau^*$ given by $\pi_{1,*} = \pi_{1,*} \tau_*$, so

$$\pi_{1,*} \xrightarrow{\eta} \pi_{1,*} \tau_* \tau^* \xrightarrow{\simeq} \pi_{1,*} \tau^*$$

Applying this to $\pi_2^*(\mathcal{F})$, we obtain a map

$$\beta : \pi_{1,*} \pi_2^*(\mathcal{F}) \rightarrow \pi_{1,*} \tau^* \pi_2^*(\mathcal{F})$$

Notice that $\pi_{1,*} \pi_2^*(\mathcal{F})$ is the constant sheaf with values $\pi_{2,*}(\mathcal{F})$ by smooth base change. Also, note the stalk of $\alpha \circ \beta$ at $1 \in \mathbb{A}^1$ is bijective, and the stalk of this map at 0 is 0. Since it is a constant sheaf, it must be 0 everywhere. $\square$

Lemma 3.2.3.9. Suppose $\mathcal{F} \in D_m(P_\Theta \backslash G/B)$ is pure of weight 0. Then it is very pure of weight 0.

Proof: Firstly, we may assume $\Theta = \emptyset$, because the projection ${}_\Theta \pi$ is smooth and ${}_\Theta \pi^*$ preserves weights. By definition, we have for any $x \in B \backslash G/B$, the $*$ -stalk $x^*(\mathcal{F})$ having weight ≤ 0 . So it suffices to prove that the $*$ -stalks have weights ≥ 0 . For this, we think $\mathcal{F}$ as an equivariant sheaf over Fl , and first assume $\text{Supp } \mathcal{F} \subset \text{Fl}_{\leq w}$. Consider $v \leq w$, and we abuse notations to let $v = vB/B \in \text{Fl}$. Then, it suffices to prove $v^*(\mathcal{F})$ has weight ≥ 0 for all v , since they are representatives of all the orbits. But this is because

$$v^*(\mathcal{F}) = H^*(vC \cap \text{Fl}_{\leq w}, \mathcal{F})$$

by Lemma 3.2.1.12 and Lemma 3.2.3.8. Then, since H^* is the $*$ -pushforward to the point, it increases weight, so $v^*(\mathcal{F})$ has weight ≥ 0 . Hence the assertion follows. $\square$

Proposition 3.2.3.10. If $\mathcal{F}_1 \in D_m(B \backslash G/B)$ is $*$ -pure of weight 0, and $\mathcal{F}_2 \in D_m(B \backslash G/B)$ is $!$ -pure of weight 0, then $\text{Ext}^i(\mathcal{F}_1, \mathcal{F}_2)$ is a Frobenius module of weight i , and it is free over the left or right copy of the graded algebra $\check{S}^\bullet$. Here, the $\check{S}^\bullet$ bimodule structure is given by the action

$$D_m(H \backslash \text{pt}/H) \subset D_m(B \backslash G/B)$$

Proof: We apply induction on the support of $\mathcal{F}_1$ and $\mathcal{F}_2$. Suppose the statement is true for $\mathcal{F}_1$ and $\mathcal{F}_2$ supported on $\text{Fl}_{<w}$. Then, suppose now they are supported on $\text{Fl}_{\leq w}$, consider the excision triangle

$$i_{<w,!} i_{<w}^!(\mathcal{F}_2) \rightarrow \mathcal{F}_2 \rightarrow i_{w,*} i_w^*(\mathcal{F}_2)$$

and applying $\text{Hom}(\mathcal{F}_1, -)$, we obtain a long exact sequence

$$\cdots \rightarrow \mathrm{Ext}^i(i_{<w}^*(\mathcal{F}_1), i_{<w}^!(\mathcal{F}_2)) \rightarrow \mathrm{Ext}^i(\mathcal{F}_1, \mathcal{F}_2) \rightarrow \mathrm{Ext}^i(i_w^*(\mathcal{F}_1), i_w^*(\mathcal{F}_2)) \rightarrow \cdots$$

Now, note i_w is open, and by assumption $i_w^*(\mathcal{F}_1)$ and $i_w^*(\mathcal{F}_2) = i_w^!(\mathcal{F}_2)$ are both pure of weight 0. We now prove that we have $\mathrm{Ext}^i(i_w^*(\mathcal{F}_1), i_w^*(\mathcal{F}_2))$ pure of weight i . Firstly, we write $\mathcal{G}_i = i_w^*(\mathcal{F}_i)$. Then, since $\mathcal{G}_i \in D_m(B \backslash BwB/B) = D_m(B \backslash \mathrm{Fl}_w)$, we have the stabilizer of the actino $B \subset \mathrm{Fl}_w$ given by $B_w = B \cap wBw^{-1}$, which is constant. Thus, the underlying sheaf of $\mathcal{G}_i$ in $D_m(\mathrm{Fl}_w)$ is constant. We may assume $\mathcal{G}_i = \mathcal{L}_w \otimes M_i$ non-canonically, where $\mathcal{L}_w = \overline{\mathbb{Q}}_\ell[d](d/2)$ is the constant sheaf on the strata and M_i is a Frobenius module pure of weight 0. Moreover, we may assume M_i is dualizable. Hence, we have the following computation

$$\mathrm{Hom}^\bullet(\mathcal{G}_1, \mathcal{G}_2) = \mathrm{Hom}^\bullet(\mathcal{L}_w, \mathcal{L}_w \otimes M_2 \otimes M_1^\vee) = H^\bullet(B \backslash \mathrm{Fl}_w) \otimes \mathrm{Hom}^\bullet(M_1, M_2)$$

Now, by definition, $\mathrm{Ext}^i(M_1, M_2)$ is pure of weight i , and since $B \backslash \mathrm{Fl}_w$ is equivalent to $\mathbb{B}H$ for the torus $H \subset G$, $H^i(B \backslash \mathrm{Fl}_w) = H^i(\mathbb{B}H)$ is also pure of weight i . Hence, taking tensor products gives the desired purity result.

On the other hand, $\mathrm{Ext}^i(i_{<w}^*(\mathcal{F}_1), i_{<w}^!(\mathcal{F}_2))$ is pure of weight i by induction hypothesis. Then, since we have an actual map of Frobenius modules

$$\mathrm{Ext}^i(i_w^*(\mathcal{F}_1), i_w^*(\mathcal{F}_2)) \rightarrow \mathrm{Ext}^{i+1}(i_{<w}^*(\mathcal{F}_1), i_{<w}^!(\mathcal{F}_2))$$

by weight reasons, it must be 0. Hence, the long exact sequence splits into short exact sequences

$$0 \rightarrow \mathrm{Ext}^i(i_{<w}^*(\mathcal{F}_1), i_{<w}^!(\mathcal{F}_2)) \rightarrow \mathrm{Ext}^i(\mathcal{F}_1, \mathcal{F}_2) \rightarrow \mathrm{Ext}^i(i_w^*(\mathcal{F}_1), i_w^*(\mathcal{F}_2)) \rightarrow 0$$

so $\mathrm{Ext}^i(\mathcal{F}_1, \mathcal{F}_2)$ is pure of weight i . Then, it remains to prove $\mathrm{Ext}^\bullet(i_w^*(\mathcal{F}_1), i_w^*(\mathcal{F}_2))$ is free both as a left $\check{S}^\bullet$ -module and as a right $\check{S}^\bullet$ -module. This is also by the computation above, since by definition $H^\bullet(B \backslash \mathrm{Fl}_w) = H^\bullet(\mathbb{B}H)$. $\square$

Then we will introduce the hypercohomology functor for the convenience for certain computations.

Definition 3.2.3.11. By identifying $B \backslash G/B = H \backslash (U \backslash G/U)/H$, we can project $U \backslash G/U$ to a point, and obtain the equivariant cohomology functor

$$\Gamma_{H \times H}(U \backslash G/U, -) : D_m(B \backslash G/B) \rightarrow D_m(\mathbb{B}(H \times H)) = \mathbf{Mod}_{\check{S} \otimes \check{S}, \mathrm{Fr}}^\omega$$

where $\check{S} = \check{S}_H$. We denote this functor by $\mathbb{H}$.

Lemma 3.2.3.12. The functor $\mathbb{H}$ has a natural monoidal structure with respect to the convolution on $D_m(B \backslash G/B)$ and the tensor product $- \otimes_{\check{S}} -$ on $\mathbf{Mod}_{\check{S} \otimes \check{S}, \mathrm{Fr}}^\omega$.

Proof: It suffices to prove that the tensor product $- \otimes_{\check{S}} -$ on $D(\check{S} \times \check{S}, \mathrm{Fr})^\omega$ is the same as the following pull-push:

$$H \backslash \mathrm{pt}/H \times H \backslash \mathrm{pt}/H \xleftarrow{\delta_H} H \backslash \mathrm{pt} \times^H \mathrm{pt}/H \xrightarrow{m_H} H \backslash \mathrm{pt}/H$$

then, clearly the convolutions (here we use $m_* \delta^*$, which makes no difference on $D_m(B \backslash G/B)$ by ind-properness) on the corresponding diagrams commutes with the usual pushforward functor $\pi_* : D_m(B \backslash G/B) \rightarrow D_m(H \backslash \mathrm{pt}/H)$ (essentially by smooth base change and ignoring trivial unipotent actions).

On the level of derived categories, we have the following identifications:

$$\begin{array}{ccccc}
D_m(H \backslash \text{pt}/H \times H \backslash \text{pt}/H) & \xrightarrow{\delta_H^*} & D_m(H \backslash \text{pt} \times^H \text{pt}/H) & \xrightarrow{m_{H,*}} & D_m(H \backslash \text{pt}/H) \\
\downarrow \simeq & & \downarrow \simeq & & \downarrow \simeq \\
\mathbf{Mod}_{\check{S}^{\otimes 4}, \text{Fr}} & \xrightarrow{F} & \mathbf{Mod}_{\check{S}^{\otimes 3}, \text{Fr}} & \xrightarrow{U} & \mathbf{Mod}_{\check{S} \otimes \check{S}, \text{Fr}}
\end{array}$$

Here, by [Corollary 3.2.2.21](#) F is $-\otimes_{\check{S} \otimes \check{S}} \check{S}$, where the bottom $\check{S} \otimes \check{S}$ is the two in the middle. The question now is what is U . We claim that this is forgetting the middle $\check{S}$ -action, hence identifying $U \circ F$ as the tensor product.

We first identify the functor $\pi_* : D_m(\mathbb{B}A) \rightarrow D_m(\text{pt})$. While identifying $D_m(\mathbb{B}A)$ with $D(\check{S}_A, \text{Fr})$, we used the functor $\text{Hom}(\mathbb{1}, -)$, which is precisely taking the cohomology. Thus π_* is simply the forgetful functor $\mathbf{Mod}_{\check{S}_A, \text{Fr}} \rightarrow \mathbf{Mod}(\text{Fr})$. Hence in the above case, this is also the forgetful functor, but only forgetting the middle factor by the Künneth formula. $\square$

Lemma 3.2.3.13. For $\mathcal{F} \in D_m(B \backslash \overline{BwB}/B)$ $*$ -pure or $!$ -pure of weight 0, we have exact sequences

$$0 \rightarrow \mathbb{H}(j_! j^! \mathcal{F}) \rightarrow \mathbb{H}(\mathcal{F}) \rightarrow \mathbb{H}(i_* i^*(\mathcal{F})) \rightarrow 0$$

$$0 \rightarrow \mathbb{H}(i_! i^! \mathcal{F}) \rightarrow \mathbb{H}(\mathcal{F}) \rightarrow \mathbb{H}(j_* j^*(\mathcal{F})) \rightarrow 0$$

here $i : B \backslash \text{Fl}_{<w} \hookrightarrow B \backslash \text{Fl}_{\leq w}$, and $j : B \backslash \text{Fl}_w \hookrightarrow B \backslash \text{Fl}_{\leq w}$ are the closed and open part. We temporarily denote them by $i : Z \hookrightarrow X$ and $j : U \hookrightarrow X$.

Proof: It suffices to prove the statements for the second exact sequence, since the others follows from taking Verdier duality. For $\mathcal{F}$ $!$ -pure of weight 0, this case follows from [Proposition 3.2.3.10](#) and weight reasons.

Now, for $\mathcal{F}$ $*$ -pure of weight 0, suppose $\mathcal{F} = i_{\leq v}^* \mathcal{K}$. By the long exact sequence, it suffices to show that $j^* : H^\bullet(X, \mathcal{F}) \rightarrow H^\bullet(U, \mathcal{F})$ is degreewise surjective. Let v_H denote the inclusion of stacks $v_H : H \backslash v \hookrightarrow H \backslash \text{Fl}$. By $\mathbb{A}^1$ -invariance, we have an isomorphism of cohomology groups

$$H^\bullet(B \backslash \text{Fl}_G, \mathcal{K}) \simeq H^\bullet(H \backslash \text{Fl}_G, \mathcal{K})$$

Then, one has a commutative diagram

$$\begin{array}{ccccccc}
H^\bullet(B \backslash \text{Fl}, \mathcal{K}) & \longrightarrow & H^\bullet(X, \mathcal{F}) & \xrightarrow{j^*} & H^\bullet(U, \mathcal{F}) & \xlongequal{\quad} & H^\bullet v_H^* \mathcal{F} \\
\parallel & & & & & & \parallel \\
H^\bullet(H \backslash \text{Fl}, \mathcal{K}) & \xrightarrow{\alpha^*} & H^\bullet(H \backslash vC, \mathcal{K}) & \xlongequal{\quad} & H^\bullet v_H^* \mathcal{F} & &
\end{array}$$

Here, the rightmost horizontal arrows are taking stalks at the contraction point, so by [Lemma 3.2.3.8](#) is an isomorphism. Now, to prove j^* surjective, it suffices to prove α^* surjective, where $\alpha : H \backslash vC \hookrightarrow H \backslash \text{Fl}$ is an open immersion of stacks. We take β to be the reduced complement closed of α , we have

$$\dots \rightarrow H^k(H \backslash \text{Fl}, \mathcal{K}) \rightarrow H^k(H \backslash vC, \mathcal{K}) \rightarrow H^{k+1}(H \backslash (\text{Fl} - vC), \beta^! \mathcal{K}) \rightarrow \dots$$

Note that we may assume $\mathcal{K}$ is pure of weight 0. Hence, $\beta^! \mathcal{K}$ is of weight ≥ 0 . Thus, the Fr-module $H^{k+1}(H \backslash (\text{Fl} - vC), \beta^! \mathcal{K})$ is of weight $\geq k+1$. On the other hand, as a Fr-module we

have $H^k(H \setminus vC, \mathcal{K}) = H^k(v_H^*(\mathcal{K}))$, so this group has weight k . Hence the connecting morphism is 0, so we obtain the surjectivity of α^* . $\square$

The main evidence that $\mathbb{H}$ is useful for detecting data in $D_m(B \setminus G/B)$ is given by the following proposition.

Proposition 3.2.3.14. For $\mathcal{F}_1, \mathcal{F}_2 \in D_m(B \setminus G/B)$ very pure of weight 0, the natural map

$$\mathrm{Ext}^\bullet(\mathcal{F}_1, \mathcal{F}_2) \rightarrow \mathrm{Hom}_{\check{S} \otimes \check{S}}(\mathbb{H}(\mathcal{F}_1), \mathbb{H}(\mathcal{F}_2))$$

is an isomorphism of Fr-modules.

Proof: We first prove that for $\mathcal{F}_1, \mathcal{F}_2 \in D_m(B \setminus G/B)$ supported on $B \setminus \overline{BwB}/B$ very pure of weight 0, we have $\mathrm{Hom}^\bullet(\mathcal{F}_1, \mathcal{F}_2) \simeq \mathrm{Hom}_{A_{\leq w}}^\bullet(\mathbb{H}(\mathcal{F}_1), \mathbb{H}(\mathcal{F}_2))$. Here $A_{\leq w} := H^\bullet(B \setminus \mathrm{Fl}_{\leq w})$. We apply induction on the Bruhat order, and prove that for any v , we have

$$\mathrm{Hom}^\bullet(i_{\leq v,*} i_{\leq v}^*(\mathcal{F}_1), i_{\leq v,*} i_{\leq v}^!(\mathcal{F}_2)) \simeq \mathrm{Hom}_{A_{\leq v}}(\mathbb{H}(i_{\leq v,*} i_{\leq v}^*(\mathcal{F}_1)), \mathbb{H}(i_{\leq v,*} i_{\leq v}^!(\mathcal{F}_2)))$$

The assertion for $v = e$ is trivial. Then, for the induction step, we should apply [Lemma 3.2.3.13](#). We claim that we have a morphism of exact sequences

$$\begin{array}{ccc} 0 & & 0 \\ \downarrow & & \downarrow \\ \mathrm{Hom}^\bullet(i_* i^*(\mathcal{F}_1), i_* i^!(\mathcal{F}_2)) & \longrightarrow & \mathrm{Hom}_{A_{<v}}(\mathbb{H}(i_* i^*(\mathcal{F}_1)), \mathbb{H}(i_* i^!(\mathcal{F}_2))) \\ \downarrow & & \downarrow \\ \mathrm{Hom}^\bullet(\mathcal{F}_1, \mathcal{F}_2) & \longrightarrow & \mathrm{Hom}_{A_{\leq v}}(\mathbb{H}(\mathcal{F}_1), \mathbb{H}(\mathcal{F}_2)) \\ \downarrow & & \downarrow \\ \mathrm{Hom}^\bullet(j_* j^*(\mathcal{F}_1), j_* j^*(\mathcal{F}_2)) & \longrightarrow & \mathrm{Hom}_{H^\bullet(U)}(\mathbb{H}(j_* j^*(\mathcal{F}_1)), \mathbb{H}(j_* j^*(\mathcal{F}_2))) \\ \downarrow & & \downarrow \\ 0 & & 0 \end{array}$$

Now, we know that the left line is an exact sequence, the right line is exact at top and bottom, and that the top and bottom horizontal arrows are isomorphisms, either by the induction hypothesis or by the computation on the case of a torus (note $U = B \setminus BwB/B \simeq H \setminus \mathrm{pt}$ up to factors of $\mathbb{A}^1$'s). Now, it suffices to prove that the right line is exact in the middle, so we can finish the induction step.

Firstly, note that we have an exact sequence

$$0 \longrightarrow H_c^\bullet(U) \longrightarrow A_{\leq v} \longrightarrow A_{<v} \longrightarrow 0$$

By the above equivalence, $H^\bullet(U)$ is a free $\check{S}$ -module of rank 1, so we may choose a generator $[U] \in H_c^{2\ell(v)}(U)$. Then, the action of $[U]$ on $\mathbb{H}(\mathcal{F}_i)$ would then kill the closed part, hence factors as

$$\mathbb{H}(\mathcal{F}_i) \twoheadrightarrow \mathbb{H}(j_* j^*(\mathcal{F}_i)) \xrightarrow{u} \mathbb{H}(j_! j^*(\mathcal{F}_i))[2\ell(v)](\ell(v)) \hookrightarrow \mathbb{H}(\mathcal{F}_i)[2\ell(v)](\ell(v))$$

Now, suppose there is some $\varphi : \mathbb{H}(\mathcal{F}_1) \rightarrow \mathbb{H}(\mathcal{F})_2$ induces the zero map $\mathbb{H}(j_*j^*(\mathcal{F}_1)) \rightarrow \mathbb{H}(j_*j^*(\mathcal{F}_2))$, then we have $[U] \circ \varphi = 0$ by the above factoring. Hence, $\varphi \circ [U] = 0$ since φ is an $A_{\leq v}$ -module map. Thus $\varphi = 0$ on the image of $[U]$, which is $\mathbb{H}(j_!j^*(\mathcal{F}_1))$. Thus it comes from an $A_{< v}$ -module map $\mathbb{H}(i_*i^*(\mathcal{F}_1)) \rightarrow \mathbb{H}(i_*i^*(\mathcal{F}_2))$, hence the claim follows. $\square$

Then, we can compute the equivariant cohomology of the standard, costandard, and intersection cohomology sheaves.

Lemma 3.2.3.15. For each $w \in W$, we have an isomorphism in $\mathbf{Mod}_{\check{S}^\bullet \otimes \check{S}^\bullet, \text{Fr}}^{\text{gr}}$

$$\mathbb{H}(\Delta_w) = \mathcal{O}_{\Gamma(w)}[-\ell(w)](-\ell(w)/2)$$

$$\mathbb{H}(\nabla_w) = \mathcal{O}_{\Gamma(w)}[\ell(w)](\ell(w)/2)$$

Here, $\Gamma(w) \subset V_H \times V_H = \text{Spec } \check{S} \otimes \check{S}$ is the graph of the map $w : V_H \rightarrow V_H$, given by the action $W \curvearrowright V_H$.

Proof: Note that the projection $B \backslash G/B \rightarrow \text{pt}$ is ind-proper, so the hypercohomologies in question are actually given by

$$\mathbb{H}(\Delta_w)[- \ell(w)](-\ell(w)/2) = H_c^\bullet(B \backslash BwB/B, \overline{\mathbb{Q}}_\ell), \quad \mathbb{H}(\nabla_w)[\ell(w)](\ell(w)/2) = H^\bullet(B \backslash BwB/B, \overline{\mathbb{Q}}_\ell)$$

We first compute the cohomology of the costandard object. Notice that by Lemma 3.2.3.8 we have an isomorphism $H^\bullet(B \backslash BwB/B, \overline{\mathbb{Q}}_\ell) = H^\bullet(H \backslash HwH/H, \overline{\mathbb{Q}}_\ell)$. We then suffices to identify it with the ring of functions over $\mathcal{O}_{\Gamma(w)}$. Note the $H \times H$ -action on HwH is given by $(h_1, h_2) \cdot x = h_1 x h_2^{-1}$, so the stabilizer is given by $K_w = \{(whw^{-1}, h) | h \in H\}$. Thus, we can furthermore identify the stack $H \backslash HwH/H = \mathbb{B}K_w$. However, by the identifying K_w with H , we have $H \simeq K_w \hookrightarrow H \times H$ induces $V_H \hookrightarrow V_H \times V_H$, $v \mapsto (wv, v)$ on the level of Tate modules, hence its image is precisely $\Gamma(w)$. Hence the claim for the second part follows.

For the first part, note that

$$H_c^\bullet(B \backslash BwB/B, \overline{\mathbb{Q}}_\ell) = H^\bullet(H \backslash HwH/H, \iota^! \overline{\mathbb{Q}}_\ell)$$

where $\iota : H \backslash HwH/H \hookrightarrow B \backslash BwB/B$ is the inclusion of the left H -contraction point, and by Lemma 3.2.3.8, we have $H^\bullet(-) = \iota^*(-)$, so by taking Verdier dual $H_c^\bullet(-) = \iota^!(-)$. This time, by purity for the section of an affine bundle, we have $\iota^! \overline{\mathbb{Q}}_\ell = \overline{\mathbb{Q}}_\ell[-2\ell(w)](-\ell(w))$. Hence the conclusion follows. $\square$

remark 3.2.3.1. The cohomologies are actually isomorphism to each other if we forget the Tate twists, cohomological grading, and $X_*(H)$ -gradings. Indeed, we have $\mathcal{O}_{\Gamma(w)} = \mathcal{O}_{V_H}$ regarded as a plain $\check{S}$ -module.

The cohomologies of IC_w 's are much more interesting. They can be described as the algebra of functions over a union of the orbits, which gives a algebro-geometric description of Soergel bimodules. For example, it is well-known that in the case $G = \text{SL}_2$, $\mathbb{H}(\text{IC}_s)$ is the Soergel bimodule $\check{S} \otimes_{\check{S}^s} \check{S}$. This can be interpreted by definition as $\mathcal{O}(\Gamma(e) \cup \Gamma(s))[1](1/2)$.

Lemma 3.2.3.16. For $w \in W$, we have $\mathbb{H}_w := \mathbb{H}(\mathrm{IC}_w)$ is a direct sum of Fr-modules $\overline{\mathbb{Q}}_\ell[n](n/2)$ for $n \equiv \ell(w) \pmod{2}$.

Proof: For the case where w is a simple reflection, it follows from the computation of SL_2 . In general, take a reduced word presentation $w = s_{i_1} \cdots s_{i_m}$, and we claim

$$\mathrm{IC}_w \subset \mathrm{IC}_{s_{i_1}} *_B \cdots *_B \mathrm{IC}_{s_{i_n}}$$

is a direct summand. Note

$$\mathrm{End}\left(\mathrm{IC}_{s_{i_1}} *_B \cdots *_B \mathrm{IC}_{s_{i_n}}\right) \subset \mathrm{End}_{\check{S} \otimes \check{S}}\left(\mathbb{H}_{s_{i_1}} \otimes_{\check{S}} \cdots \otimes_{\check{S}} \mathbb{H}_{s_{i_n}}\right)$$

is the 0-th graded direct summand and the latter is Frobenius semisimple since it is an endomorphism Fr-module of a semisimple Fr-module, and by [Corollary 3.2.2.17](#) $\mathrm{IC}_w \otimes M_w$ would be a direct summand. However M_w must be $\overline{\mathbb{Q}}_\ell$ by counting the dimension of $H^{\ell(w)}(-)$. So IC_w is indeed a direct summand.

Hence $\mathbb{H}_w$ is a direct summand of $\mathbb{H}_{s_{i_1}} \otimes_{\check{S}} \cdots \otimes_{\check{S}} \mathbb{H}_{s_{i_n}}$, and in the latter, only $\overline{\mathbb{Q}}_\ell[n](n/2)$ appears for $n \equiv \ell(w) \pmod{2}$ □

remark 3.2.3.2. I believe the proof using Bott-Samuelson resolutions is also available.

Lemma 3.2.3.17. For $w_1, w_2 \in W$, the convolution $\mathrm{IC}_{w_1} *_B \mathrm{IC}_{w_2}$ is a direct sum of $\mathrm{IC}_w[n](n/2)$'s where $n \equiv \ell(w) - \ell(w_1) - \ell(w_2) \pmod{2}$. In particular, if $\ell(w)_1 + \ell(w_2) = \ell(2)$, then IC_w appears with multiplicity 1.

Proof: Let $\mathcal{F} = \mathrm{IC}_{w_1} *_B \mathrm{IC}_{w_2}$. Again, we have

$$\mathrm{End}(\mathcal{F}) \subset \mathrm{End}_{\check{S} \otimes \check{S}}\left(\mathbb{H}_{w_1} \otimes_{\check{S}} \mathbb{H}_{w_2}\right)$$

is the 0-th graded direct summand. Hence $\mathcal{F}$ decomposes into $\mathrm{IC}_w \otimes M_w$ for Fr-semisimple modules. Then, we have $\mathbb{H}(\mathcal{F}) = \mathbb{H}_{w_1} \otimes_{\check{S}} \mathbb{H}_{w_2}$ is a direct sum of $\mathbb{H}_w \otimes M_w$. Hence by [Lemma 3.2.3.16](#), we can compute M_w and see that it is a direct sum of $\overline{\mathbb{Q}}_\ell[n](n/2)$ with the desired parity.

In the case $\ell(w_1) + \ell(w_2) = \ell(w)$, the map $\overline{Bw_1B} \times^B \overline{Bw_2B} \rightarrow \overline{BwB}$ is birational, hence the multiplicity 1 follows. □

Lemma 3.2.3.18. For $w \in W$, the complex ${}_{\Theta}\pi_*(\mathrm{IC}_w)$ is a direct sum of $\mathrm{IC}_{\bar{v}}[n](n/2)$ with $n \equiv \ell(\bar{w}) - \ell(\bar{v}) \pmod{2}$. In particular, if $w \in [W_{\Theta} \setminus W]$, we have $\mathrm{IC}_{\bar{w}}$ a direct summand of ${}_{\Theta}\pi_*(\mathrm{IC}_w)$ with multiplicity 1.

Proof: Again we compute the endomorphism algebra

$$\mathrm{End}_{D_m(P_{\Theta} \setminus G/B)}({}_{\Theta}\pi_*(\mathrm{IC}_w)) \simeq \mathrm{Hom}_{D_m(B \setminus G/B)}(\mathbb{1}_{\Theta} *_B \mathrm{IC}_w, \mathrm{IC}_w)$$

and the latter is again Frobenius semisimple by applying $\mathbb{H}$. Hence ${}_{\Theta}\pi_*(\mathrm{IC}_w)$ is a direct sum of $\mathrm{IC}_{\bar{v}} \otimes M_{\bar{v}}$, hence ${}_{\Theta}\pi^* {}_{\Theta}\pi_*(\mathrm{IC}_w) = \mathbb{1}_{\Theta} *_B \mathrm{IC}_w$ is a direct sum of ${}_{\Theta}\pi^*(\mathrm{IC}_{\bar{v}}) \otimes M_{\bar{v}}$. Applying $\mathbb{H}$ and

again use [Lemma 3.2.3.16](#) we have the statement for the parity. The multiplicity 1 again follows from birationality. $\square$

Finally, we show that we can apply [Theorem 3.2.2.7](#) to give a computation of $D_m(B \backslash G/B)$, but we will postpone the actual computation to later parts.

Proposition 3.2.3.19. Suppose X is any scheme with an action of a reductive group G having finitely many orbits X_λ . Moreover, assume each orbit inclusion $X_\lambda \hookrightarrow X$ is affine. Then, the category $D_m(G \backslash X)$ satisfies [Assumption 3.2.2.6](#).

Proof:

- (1) The finite poset is given by Λ , with the order given by taking boundaries. Each strata is given by $\mathcal{D}_\lambda = D_m(G \backslash X_\lambda)$.
- (2) For each $D_m(G \backslash X_\lambda)$, we take $\mathcal{C}_\lambda$ to be the subcategory of constant pure of weight 0 complexes. Thus by a similar argument as [Proposition 3.2.3.10](#), the Frobenius unipotent part of the higher Ext groups vanish.
- (3) We can take $\mathbb{1}_\lambda$ to be the constant sheaf, and since $\mathcal{C}_\lambda$ consists of constant complexes, every $c \in \mathcal{C}_\lambda$ is of the form $\mathbb{1}_\lambda \otimes M$. The generation is because $G \backslash X_\lambda$ is of the form $\mathbb{B}G_\lambda$ where G_λ is the stabilizer of some point. Then, G_λ is connected so $\pi_1(\mathbb{B}G_\lambda) = 0$. Hence $D_m(\mathbb{B}G_\lambda)$ is generated by the constant sheaf.
- (4) We take $c_\lambda = \text{IC}_\lambda$. By irreducibility, we have the map $\text{End}(c_\lambda) = \text{End}(\mathbb{1}_\lambda)$ is an isomorphism.

$\square$

Corollary 3.2.3.20. $D_m(P_\Theta \backslash \overline{P_\Theta \overline{w}B}/B)$ satisfies [Assumption 3.2.2.6](#).

3.2.4. Basics on (completed) monodromic categories

We first give a recap on the definition of monodromic sheaves. In this part, all monodromic means unipotently monodromic.

Definition 3.2.4.1. For convenience, we introduce the following variant of $!$ -pushforwards and pullbacks: we define the adjoint pair $\pi_+ \dashv \pi^\dagger := \pi_![r] \dashv \pi^\dagger[-r]$, where π is of relative dimension r . In our purpose, π is often the quotient functor of a torus of rank r . Note in particular that there are no weight shifts here.

Definition 3.2.4.2. Let X be a variety equipped with an action by a torus A . We denote $\pi : X \rightarrow X/A := Y$ the projection. We define $D_m^{\text{mon}}(X)$ to be the full subcategory generated by the image of $\pi^\dagger : D_m(Y) \rightarrow D_m(X)$ under finite colimits. We define $\mathbf{Perv}_m^{\text{mon}}(X) = \mathbf{Perv}_m(X) \cap D_m^{\text{mon}}(X)$.

However, this category is not what we need. Rather, we need a certain completion of this category. We can see this from the following example.

Example 3.2.4.1. Take $G = \mathbb{G}_m$. Then, the category $D_m^{\text{mon}}(\mathbb{G}_m)$ is the subcategory spanned by the constant sheaves. We let $\text{Loc}_u(\mathbb{G}_m)$ denote the Abelian category of unipotent local systems over $\mathbb{G}_m$ (i.e., local systems $\mathcal{L}$ over $\mathbb{G}_m$ equipped with a filtration $0 = \mathcal{L}_0 \subset \dots \subset \mathcal{L}_n = \mathcal{L}$ such that $\mathcal{L}_i/\mathcal{L}_{i-1}$ is constant). Then, we claim that we have the following characterization:

$$D_m^{\text{mon}}(\mathbb{G}_m) = \{\mathcal{F} \in D_m(\mathbb{G}_m), \mathcal{H}^i(\mathcal{F}) \in \text{Loc}_u(\mathbb{G}_m)\}$$

Indeed, the right hand side is closed under taking kernels, cokernels, extensions, and direct summands, and contains the constant sheaf. Conversely, every unipotent local system must lie in $D_m^{\text{mon}}(\mathbb{G}_m)$ by considering the filtration, so the two categories are equivalent.

Then, we can identify $\text{Loc}(\mathbb{G}_m)$ with ℓ -adic representations of $\pi_1(\mathbb{G}_m)^{\text{tame}}$. Such representations then factors through its pro- ℓ part $T_\ell(\mathbb{G}_m) = \mathbb{Z}_\ell(1)$, so we have $\text{Loc}(\mathbb{G}_m) = \mathbf{Rep}(\mathbb{Z}_\ell(1), \overline{\mathbb{Q}}_\ell^n)^{\heartsuit}$. The unipotent local systems hence corresponds to unipotent representations.

Now, for unipotent representations over a field of characteristic 0, we should pass to the nilpotent monodromy operator. Namely, for a representation $\rho : T_\ell(\mathbb{G}_m) \rightarrow \text{GL}(\overline{\mathbb{Q}}_\ell^n)$, we have a linear map

$$\log \rho : V_{\mathbb{G}_m} \rightarrow \text{End}(\overline{\mathbb{Q}}_\ell^n), \quad v \mapsto \log \rho(v), v \in T_\ell(\mathbb{G}_m)$$

If we choose a basis $v_0 \in V$ (V is 1-dimensional), clearly the data of $\log \rho$ is completely determined by $N := \log \rho(v_0)$. Hence, $\text{Loc}_u(\mathbb{G}_m) = \mathbf{Mod}_{\overline{\mathbb{Q}}_\ell[[t]], \text{Fr}}^{f.l., \heartsuit}$, where $\text{Fr}(t) = q^{-1}t$, and $f.l.$ denote finite length. Thus, we have by Morita theory $D_m^{\text{mon}}(\mathbb{G}_m) = \mathbf{Mod}_{\overline{\mathbb{Q}}_\ell[[t]], \text{Fr}}^{f.l.}$.

Notice that the whole category $\mathbf{Mod}_{\overline{\mathbb{Q}}_\ell[[t]], \text{Fr}}$ is a completion of $\mathbf{Mod}_{\overline{\mathbb{Q}}_\ell[[t]], \text{Fr}}^{f.l.}$. So we see that we need a completion $\hat{D}_m^{\text{mon}}(\mathbb{G}_m)$ instead. In particular, we will need the object corresponding to the module $\overline{\mathbb{Q}}_\ell[[t]]$.

For convenience of conventions, we denote $\varinjlim$ by the limit taken in the pro-category.

Construction 3.2.4.3. Recall that $\text{Pro}(D_m(X))$ admits a natural stable ∞ -structure. We define $\hat{D}_m^{\text{mon}}(X)$ to be the full subcategory of $\text{Pro}(D_m^{\text{mon}}(X))$ spanned by objects $\varinjlim \mathcal{F}_i$ satisfying the following conditions:

- (1) π -constant, that is, $\varinjlim \pi_* \mathcal{F}_i \in \text{Pro}(D_m(Y))$ is constant.
- (2) Uniformly bounded in degrees, that is, there exists a number N such that $\mathcal{F}_i$ are in perverse degrees $[-N, N]$
- (3) Uniformly bounded above in weights, that is, objects such that there exists a number N with each $\mathcal{F}_i$ of weight at most N .

For $X = A$, $Y = \text{pt}$, there is a typical element in $\hat{D}_m^{\text{mon}}(X)$ given as follows. We define $\tilde{\mathcal{L}} \in \hat{D}_m^{\text{mon}}(A)$ to be $\varinjlim \mathcal{L}_n$, where $\mathcal{L}_n$ is the local system over A corresponding to the representation $\text{Sym } V_A / (V_A^{n+1})$, and with the obvious transition maps.

Then, we investigate the functorialities of this construction.

Lemma 3.2.4.4. Let $\mu : A \times X \rightarrow X$ denote the action map. Then, there is a natural isomorphism for $\mathcal{F} \in \hat{D}_m^{\text{mon}}(X)$ given by $\mu_!(\tilde{\mathcal{L}} \boxtimes \mathcal{F}) \simeq \mathcal{F}[-2r](-r)$.

Proof: Since $D_m^{\text{mon}}(X)$ generate $\hat{D}_m^{\text{mon}}(X)$ under limits, it suffices to prove the isomorphism for $\mathcal{F} \in D_m^{\text{mon}}(X)$ (note that we will still have to prove a pro-isomorphism for the presence of $\tilde{\mathcal{L}}$). Since μ is smooth, we will first construct a map $\tilde{\mathcal{L}} \boxtimes \mathcal{F} \rightarrow \mu^! \mathcal{F}[-2r](-r) = \mu^* \mathcal{F}$.

By purity, if we denote $e : \text{pt} \rightarrow A$ the identity point, we have

$$e^! \mathcal{L}_n = e^* \mathcal{L}_n[-2r](-r) \simeq \text{Sym}(V_A) / (V_A^{n+1})[-2r](-r)$$

Hence, take the inclusion $1 : \overline{\mathbb{Q}}_\ell \rightarrow \text{Sym}(V_A)/(V_A^{n+1})$, we have a morphism

$$\overline{\mathbb{Q}}_\ell[-2r](-r) \rightarrow e^! \mathcal{L}_n$$

Then, since $e_! = e_* \dashv e^!$, we have

$$e_! \overline{\mathbb{Q}}_\ell[-2r](-r) \rightarrow \mathcal{L}_n$$

Taking exterior tensor with $\mathcal{F}$ and applying $\mu_!$, we have

$$\mu_!(e_! \overline{\mathbb{Q}}_\ell[-2r](-r) \boxtimes \mathcal{F}) = \mu_!(e_{X,!}(\mathcal{F}[-2r](-r))) = \mathcal{F}[-2r](-r)$$

where $e_X : X \hookrightarrow A \times X$ is the inclusion at the identity. Thus, we get a map

$$\mathcal{F}[-2r](-r) \rightarrow \mu_!(\mathcal{L}_n \boxtimes \mathcal{F})$$

Then, note that it suffices to prove that the above is an isomorphism for $\mathcal{F} = \pi^! \mathcal{K}$, where $\mathcal{K} \in D_m(Y)$, since by definition $D_m^{\text{mon}}(X)$ is generated by these objects under extensions. Note that we have $\pi \circ p_2 = \pi \circ \mu : A \times X \rightarrow Y$, so $\mu^*(\mathcal{F}) = p_2^*(\mathcal{F})$.

$$\mu_!(\mathcal{L}_n \boxtimes \mathcal{F}) = \mu_!(p_1^* \mathcal{L}_n \otimes \mu^* \mathcal{F}) = \mu_! p_1^* \mathcal{L}_n \otimes \mathcal{F} = \mathcal{F} \otimes R\Gamma_c(A, \mathcal{L}_n)$$

where the last equivalence is by base change.

Thus, it suffices to show that

$$\overline{\mathbb{Q}}_\ell[-2r](-r) \rightarrow \varinjlim R\Gamma_c(A, \mathcal{L}_n)$$

is a pro-equivalence. We first take $A = \mathbb{G}_m$. Then, we claim that for any unipotent local system $\mathcal{L}$ with fiber M , the cohomology is given by

$$R\Gamma(\mathbb{G}_m, \mathcal{L}) = \left[M \xrightarrow{t} M(-1) \right]$$

where t is the logarithm of the monodromy automorphism of M . Note that local systems over $\mathbb{G}_m$ are equivalent to $\pi_1(\mathbb{G}_m)^{\text{tame}}$ -representations, the unipotent local systems corresponds to modules over the completed logarithmic group algebra $\overline{\mathbb{Q}}_\ell[[t]]$, where t has grade -1 . Thus, for the local system $\mathcal{L}$ which corresponds to M , we have a resolution for $\overline{\mathbb{Q}}_\ell$ by

$$0 \longrightarrow \overline{\mathbb{Q}}_\ell[[t]](1) \xrightarrow{t} \overline{\mathbb{Q}}_\ell[[t]] \longrightarrow \overline{\mathbb{Q}}_\ell \longrightarrow 0$$

applying $\text{Hom}(-, M)$ gives the computation for cohomology. Applying Poincaré duality, we have

$$H_c^1(\mathbb{G}_m, \mathcal{L}_M) = \ker(t : M \rightarrow M(-1)), \quad H_c^2(\mathbb{G}_m, \mathcal{L}_M) = (M/tM)(-1)$$

Thus, for $\mathcal{L}_M = \mathcal{L}_n$, we have $H_c^1 = \overline{\mathbb{Q}}_\ell t^n$, $H_c^2 = \overline{\mathbb{Q}}_\ell(-1)$. Then, note that the transition maps between H_c^1 is 0 and between H_c^2 is the identity. Hence, for $\mathbb{G}_m$ we have the desired pro-isomorphism. For $A = \mathbb{G}_m^n$, the conclusion holds by Künneth formula. $\square$

Lemma 3.2.4.5. Let $\pi_i : X_i \rightarrow Y_i$ be A -torsors. Then, suppose we have a commutative diagram

$$\begin{array}{ccc}
D_m^{\text{mon}}(X_1) & \xrightarrow{\Phi} & D_m^{\text{mon}}(X_2) \\
\pi_{1,\dagger} \downarrow & & \downarrow \pi_{2,\dagger} \\
D_m(Y_1) & \xrightarrow{\bar{\Phi}} & D_m(Y_2)
\end{array}$$

then Φ extends naturally to $\hat{\Phi} : \hat{D}_m^{\text{mon}}(X_1) \rightarrow \hat{D}_m^{\text{mon}}(X_2)$.

Proof: It suffices to show that $\text{Pro}(\Phi)$ preserves the objects with the three requirements. For π -constance, it follows from that the above diagram is commutative, and $\text{Pro}(\bar{\Phi})$ sends constant diagrams to constant diagrams. For uniformly bounded in degrees and uniformly bounded above in weights, it follows from the following argument of devissage.

Take $\mathcal{G} := \varinjlim \mathcal{G}_m$ with $\pi_{\dagger} \mathcal{G} = K$ for an ordinary object $K \in D_m(Y)$. We show that $\mathcal{G}$ can be disintegrated into pieces with each piece of the form $\pi^{\dagger} K \otimes \text{Sym}^j(V_A)$. For each n , we define the sheaf

$$\mathcal{E}_n = \varinjlim_m \mu_!(\mathcal{L}_n \boxtimes \mathcal{G}_m)[2r](r) \in \text{Pro}(D_m(X))$$

Note that $\mathcal{L}_n$ is the local system corresponding to $\text{Sym}(V_A)/(V_A^{n+1})$, so admits a filtration by augmentation, which has graded pieces given by $\text{Sym}^j(V_A)$. Then, choosing an identification $X \times_Y X = X \times A$, proper base change gives

$$\mu_!(\bar{\mathbb{Q}}_{\ell} \boxtimes \mathcal{G}_m)[2r](r) = \pi^* \pi_{\dagger} \mathcal{G}[2r](r) = \pi^{\dagger} K$$

Hence, the filtration on $\mathcal{L}_n$ by $\text{Sym}^j(V_A)$ gives a filtration on $\mathcal{E}_n$ with graded pieces $\pi^{\dagger} K \otimes \text{Sym}^j(V_A)$. Moreover, by [Lemma 3.2.4.4](#), we have $\mu_!(\tilde{\mathcal{L}} \boxtimes \mathcal{G}_m)[2r](r) = \mathcal{G}_m$, so we have $\varinjlim \mathcal{E}_n = \mathcal{G}$. To sum up briefly what we have done above, we chose a convenient pro-presentation for $\mathcal{G}_m$ and hence gave another pro-presentation for $\mathcal{G}$.

We finally prove that uniformly bounded in degrees and uniformly bounded above in weights are preserved by $\text{Pro}(\Phi)$. Now, suppose K is in perverse degrees $[-N, N]$, so does $\pi^{\dagger} K$. Since $\mathcal{E}_n$ are extensions of $\pi^{\dagger} K$'s, the $\mathcal{E}_n$'s are bounded by $[-N, N]$ and hence so is $\mathcal{G} = \varinjlim \mathcal{E}_n$. For weights, note V_A has weight -2 , so the weight of $\pi^{\dagger}(K) \otimes \text{Sym}^j(V_A) \leq w - r - 2j \leq w - r$ where K has weight $\leq w$. Thus, every $\mathcal{E}_n$ hence $\mathcal{G}$ have weight $\leq w - r$. $\square$

Corollary 3.2.4.6. Let $f : X_1 \rightarrow X_2$ be an A -equivariant morphism. Then, the functors $f^*, f_*, f_!, f^!$ extend to functors between $\hat{D}_m^{\text{mon}}(X_i)$.

Proof: We apply [Lemma 3.2.4.5](#) to various $\bar{\Phi}$ s. For $f_!$, we directly have $\pi_{2,\dagger} f_! = \bar{f}_! \pi_{1,\dagger}$, and for f^* , by base change we have $\pi_{1,\dagger} f^* = \bar{f}^* \pi_{2,\dagger}$, so these two cases are done.

For f_* , we only have a natural transformation $\pi_{2,\dagger} f_* \rightarrow \bar{f}_* \pi_{1,\dagger}$, which we claim to be an isomorphism when we restrict the source to $D_m^{\text{mon}}(X_1)$. We check this on objects of the form $\pi_1^{\dagger}(\mathcal{K})$. Then, we have

$$\pi_{2,\dagger} f_* \pi_1^{\dagger}(\mathcal{K}) = \pi_{2,\dagger} \pi_2^{\dagger} \bar{f}_*(\mathcal{K}) = \bar{f}_*(\mathcal{K}) \otimes H_c^*(A)$$

On the other hand

$$\bar{f}_* \pi_{1,\dagger} \pi_1^\dagger(\mathcal{K}) = \bar{f}_*(\mathcal{K} \otimes H_c^*(A)) = \bar{f}_*(\mathcal{K}) \otimes H_c^*(A)$$

The readers may draw the diagrams themselves and check that the induced map between the objects is indeed a natural isomorphism.

Finally, for $f^!$, again we have only a natural transformation $\pi_{1,\dagger} f^! \rightarrow \bar{f}^! \pi_{2,\dagger}$, and it is an isomorphism restricted to $D_m^{\text{mon}}(X_2)$ by the same reason as above. $\square$

Corollary 3.2.4.7. The adjunction

$$\pi_\dagger : D_m^{\text{mon}}(X) \rightleftarrows D_m(Y) : \pi^\dagger$$

extends to an adjunction

$$\pi_\dagger : \hat{D}_m^{\text{mon}}(X) \rightleftarrows D_m(Y) : \pi^\dagger$$

and the functor $\pi_\dagger : \hat{D}_m^{\text{mon}}(X) \rightarrow D_m(Y)$ is conservative.

Proof: The lifting of the adjunctions is a direct corollary of the general theory of pro-categories. Then, since $\pi_\dagger$ is exact, to show it is conservative suffices to show that it kills no object. Suppose $\pi_\dagger(\varinjlim \mathcal{F}_n) = 0$. Then, for any $\mathcal{G} \in D_m(Y)$ we have all maps $\varinjlim \mathcal{F}_n \rightarrow \pi^\dagger \mathcal{G}$ are zero. However $\pi^\dagger \mathcal{G}$ generates $\hat{D}_m^{\text{mon}}(X)$ under extensions and limits, which implies that $\varinjlim \mathcal{F}_n$ is itself zero. $\square$

The main example we will be concerned with is the following.

Example 3.2.4.2. Take $\widetilde{\text{Fl}} := G/U$ the enhanced flag variety. We define $D_m^{\text{mon}}(U \backslash G/U)$ to be the category of monodromic sheaves with respect to the projection $U \backslash G/U \rightarrow U \backslash G/B$. Equivalently, this is the category of monodromic sheaves with respect to the projection $U \backslash G/U \rightarrow B \backslash G/B$. We may apply the above construction and define $\hat{D}_m^{\text{mon}}(U \backslash G/U)$.

More precisely, we should not take the full category at once, but first treat the case with finitely many strata

$$U \backslash \widetilde{\text{Fl}}_{\leq w} \twoheadrightarrow U \backslash \text{Fl}_{\leq w}$$

and then take the colimit to glue together the original category.

From now on, we assume that $X \rightarrow Y$ is a trivial A -torsor, since it is the case of our interest.

Definition 3.2.4.8. Let $\mathcal{F} \in D_m(Y)$. We define $\text{Free}(\mathcal{F}) \in \hat{D}_m^{\text{mon}}(X)$ to be the object $\mathcal{F} \boxtimes \tilde{\mathcal{L}}r$, which is monodromic because the torsor $X \rightarrow Y$ is trivial.

Definition 3.2.4.9. We define $\check{S}_A^\wedge$ to be the completion of $\check{S}_A$ along V_A , regarded as a pro-object. Namely, we define $\check{S}_A^\wedge := \varprojlim \text{Sym}(V_A)/V_A^n$.

Lemma 3.2.4.10. Take any section $\varepsilon : Y \rightarrow X$. Then, we have an adjunction

$$\mathrm{Hom}_{\hat{D}_m^{\mathrm{mon}}(X)}(\mathrm{Free}(\mathcal{F}), \mathcal{G}) = \mathrm{Hom}_{\mathrm{Pro}(D_m(Y))}(\mathcal{F}, \varepsilon^\dagger(\mathcal{G})), \mathcal{F} \in D_m(Y), \mathcal{G} \in D_m^{\mathrm{mon}}(X)$$

where $\varepsilon^\dagger = \varepsilon^! [r]$.

Proof: We construct the unit map for the adjunction. Firstly, note $\varepsilon^! \tilde{\mathcal{L}}[2r](r) = \varepsilon^* \tilde{\mathcal{L}}$ by purity, and the latter is equivalent to $\check{S}_A^\wedge \in \mathrm{Pro}(D_m(\mathrm{pt}))$. Let $s : \overline{\mathbb{Q}}_\ell \rightarrow \check{S}_A^\wedge$ be the unit, so by the compatibility of pullbacks with exterior tensors, we have a map

$$\mathcal{F} \rightarrow \mathcal{F} \otimes \check{S}_A^\wedge = \mathcal{F} \boxtimes \check{S}_A^\wedge = \mathrm{id}^!(\mathcal{F}) \boxtimes e^!(\tilde{\mathcal{L}}[2r](r)) = \varepsilon^!(\mathcal{F} \boxtimes \tilde{\mathcal{L}})[2r](r) = \varepsilon^\dagger \mathrm{Free}(\mathcal{F})$$

then, to show that this is indeed a unit for an adjunction, we check that it is a unit for $\mathcal{G} = \pi^\dagger \mathcal{K}$. In this case, note $\varepsilon^\dagger \mathcal{G} = \mathcal{K}$, and so the right hand side is simply $\mathrm{Hom}(\mathcal{F}, \mathcal{K})$. On the left hand side, using the computation of $R\Gamma_c(A, \mathcal{L}) = \overline{\mathbb{Q}}_\ell[-2r](-r)$ as in [Lemma 3.2.4.4](#), we have $\pi_! \mathrm{Free}(\mathcal{F}) = \mathcal{F}$, so $\mathrm{Hom}(\mathrm{Free}(\mathcal{F}), \pi^\dagger(\mathcal{K})) = \mathrm{Hom}(\mathcal{F}, \mathcal{K})$. $\square$

Construction 3.2.4.11. Let $\mathcal{C}$ be a symmetric monoidal category, and let $\mathrm{Pro}(\mathcal{C})$ be equipped with the natural symmetric monoidal structure. Then, take $A \in \mathbf{CAlg}(\mathrm{Pro}(\mathcal{C}))$. We define $\mathbf{Mod}_A^{\mathrm{nilp}}$ to be the full subcategory of $\mathbf{Mod}_A(\mathrm{Pro}(\mathcal{C}))$ spanned by constant objects.

Proposition 3.2.4.12. The functor $\varepsilon^\dagger$ lifts to an equivalence of Abelian categories

$$\sigma : \mathbf{Perv}_m^{\mathrm{mon}}(X) \rightarrow \mathbf{Mod}_{\check{S}_A^\wedge, \mathrm{Fr}}^{\mathrm{nilp}}(\mathbf{Perv}_m(Y))$$

Proof: Firstly, note $\varepsilon^\dagger$ indeed has image in $\check{S}_A^\wedge$ -modules, given by the nilpotent monodromy action

$$\varepsilon^\dagger(\mu_{\mathcal{F}}) : V_A \otimes \varepsilon^\dagger(\mathcal{F}) \rightarrow \varepsilon^\dagger(\mathcal{F})$$

and by choosing a basis of V_A , we see that this action is uniformly nilpotent, hence defines an $\check{S}_A^\wedge$ -action. Then, we mimic the Barr-Beck theorem and construct an inverse functor. We consider the functor

$$\mathbf{Mod}_{\check{S}_A^\wedge, \mathrm{Fr}}^{\mathrm{nilp}}(\mathbf{Perv}(Y)) \rightarrow \mathbf{Perv}_m^{\mathrm{mon}}(X), \mathcal{F} \mapsto \mathrm{coker} \left(V_A \otimes (\mathcal{F} \boxtimes \tilde{\mathcal{L}})r \xrightarrow{m(\mathcal{F}) \boxtimes \mathrm{id} - \mathrm{id} \boxtimes m(\tilde{\mathcal{L}})} \mathcal{F} \boxtimes \tilde{\mathcal{L}}r \right)$$

where $m(\mathcal{F})$ and $m(\tilde{\mathcal{L}})$ are the action of V_A on $\mathcal{F}$ and $\tilde{\mathcal{L}}$ respectively. Then, the two functors are clearly inverse to each other. $\square$

Corollary 3.2.4.13. The adjunction

$$H^0 \pi_! : \mathbf{Perv}_m^{\mathrm{mon}}(X) \rightleftarrows \mathbf{Perv}_m(Y) : \pi^\dagger$$

is equivalent to

$$- \otimes_{\check{S}_A^\wedge} \overline{\mathbb{Q}}_\ell : \mathbf{Mod}_{\check{S}_A^\wedge, \mathrm{Fr}}^{\mathrm{nilp}}(\mathbf{Perv}_m(Y)) \rightleftarrows \mathbf{Perv}_m(Y) : \mathrm{triv}$$

Proof: Note $\varepsilon^\dagger \pi^\dagger \mathcal{F} = \mathcal{F}$, so $\pi^\dagger$ can be identified with triv . The left adjoint hence can be identified. □

Proposition 3.2.4.14. Assume that $\mathbf{Perv}_m(Y)$ is graded highest weight (recall from Definition 3.1.1.16 the definition of graded highest weight categories). Then, we have the free monodromic objects $\text{Free}(\mathcal{F})$ generate $\hat{D}_m^{\text{mon}}(X)$.

Proof: We will use two core properties: firstly, every object in $\mathbf{Perv}_m(Y)$ admits a finite resolution by objects projective in $\mathbf{Perv}(Y)$; secondly, the realization functor

$$\rho_{Y,m} : D(\mathbf{Perv}_m(Y)) \rightarrow D_m(Y)$$

is an equivalence.

Now, given an object $\varinjlim \mathcal{F}_n \in \hat{D}_m^{\text{mon}}(X)$, by π -constance we have $\varinjlim \pi_+ \mathcal{F}_n$ represented by some object $\mathcal{G} \in D_m(Y)$. By assumption, $\mathcal{G}$ is represented by a complex

$$\dots \rightarrow \mathcal{K}^{-1} \rightarrow \mathcal{K}^0 \rightarrow \dots$$

where each $\mathcal{K}^i \in \mathbf{Perv}_m(Y)$ is projective in $\mathbf{Perv}(Y)$. We now construct a weight filtration on $\mathcal{K}^i$, inverse to the usual weight filtration. Namely, we construct a decreasing filtration

$$0 \subset \dots \subset W^{\leq v} \mathcal{K}^i \subset W^{\leq v-1} \mathcal{K}^i \subset \dots \subset \mathcal{K}^i$$

where the usual weight filtration is given by the inclusions $w^{\geq v} \mathcal{K}^i \subset \mathcal{K}^i$. By Lemma 3.2.2.8, suppose P, Q are projective in $\mathbf{Perv}(Y)$ with P has weight $\leq v$ and Q has weight $< v$. Moreover, suppose P is indecomposable in $\mathbf{Perv}(Y)$ with a simple quotient of weight v . Hence

$$\underline{\text{Ext}}^1(P, Q) = \text{Hom}(P, Q)^{\text{Fr}} = 0$$

Hence, the above filtration exists, and is unique.

We define the width of $\varinjlim \pi_+ \mathcal{F}_n$ to be the smallest possible numbers of pairs (v, i) where $\text{gr}_W^v \mathcal{K}^i \neq 0$, smallest among all representing complexes $\mathcal{K}^\bullet$. Now, we apply induction on the width of such object, and prove that they are generated by free objects.

For the base case, if $\varinjlim \pi_+ \mathcal{F}_n$ has width 0, then $\varinjlim \pi_+ \mathcal{F}_n = 0$, and by conservativity, $\varinjlim \mathcal{F}_n = 0$. Now, suppose that for $\varinjlim \pi_+ \mathcal{F}_n$ of width $< N$, the object $\varinjlim \mathcal{F}_n$ is an extension of free monodromic objects. Take $\varinjlim \pi_+ \mathcal{F}_n$ of width N . Noticing that weight shifting and cohomological shifting does not affect anything, we may assume that the representing complex terminates at $\mathcal{K}^0$, and $W^{\geq 1} \mathcal{K}^0 = 0$ but $\text{gr}_W^0 \mathcal{K}^0 \neq 0$.

Now, we show that we may assume that each $\mathcal{F}_n \in D_m(X)^{\leq 0}$, with $H^0(\mathcal{F}_n)$ having weight ≤ 0 . Note that we have $\varinjlim \pi_+ \mathcal{F}_n \in D_m(Y)^{\leq 0}$. Then, by uniform boundedness, we may assume $\mathcal{F}_n \in D_m(X)^{\leq d}$ for some d . If $d \leq 0$ we are done with the first part. If not, we have $\varinjlim H^d(\pi_+ \mathcal{F}_n) = 0$, which shows that the transition $H^d(\pi_+ \mathcal{F}_n) \rightarrow H^d(\pi_+ \mathcal{F}_m)$ is zero for large n, m . Thus, by Corollary 3.2.4.13, the image of the $\mathcal{F}_n \rightarrow \mathcal{F}_m$ must land in $V_A \mathcal{F}_m$, so the projective system $\varinjlim H^d(\mathcal{F}_n)$ is itself 0. The proof on the weight assertion on H^0 is proven likewise.

We then construct a map $\text{Free}(W^{\geq 0} \mathcal{K}^0) \rightarrow \varinjlim \mathcal{F}_n$. We first construct $\text{Free}(W^{\geq 0} \mathcal{K}^0) \rightarrow H^0(\mathcal{F}_n)$, equivalent to a map $W^{\geq 0} \mathcal{K}^0 \rightarrow H^0(\varepsilon^\dagger \mathcal{F}_n)$. For this, note $w_{\geq 0}(\varepsilon^\dagger H^0(\mathcal{F}_n)) = w_{\geq 0}(H^0(\pi_+ \mathcal{F}_n))$, because V_A has weight -2 , and $H^0(\pi_+ \mathcal{F}_n) = H^0(\mathcal{F}_n)/V_A H^0(\mathcal{F}_n)$. Then, by construction, we have a map

$$W^{\geq 0} \mathcal{K}^0 \rightarrow H^0(\pi_+ \mathcal{F}_n) \rightarrow w_{\geq 0}(H^0(\pi_+ \mathcal{F}_n)) = w_{\geq 0}(\varepsilon^\dagger H^0(\mathcal{F}_n))$$

This map then factors through $W^{\geq 0}\mathcal{K}^0 \rightarrow \varepsilon^\dagger H^0(\mathcal{F}_n) \rightarrow w_{\geq 0}(\varepsilon^\dagger H^0(\mathcal{F}_n))$ by the previous computation of vanishing of Ext^1 (the lifting obstruction). Again by vanishing of Ext^1 , we can lift $W^{\geq 0}\mathcal{K}^0 \rightarrow \varepsilon^\dagger H^0(\mathcal{F}_n)$ to $W^{\geq 0}\mathcal{K} \rightarrow \varepsilon^\dagger \mathcal{F}_n$ (here we must use $\mathcal{F}_n$ has cohomological degree ≤ 0).

Finally, take $\varinjlim \mathcal{F}'_n$ to be the cone of the map $\text{Free}(W^{\geq 0}\mathcal{K}^0) \rightarrow \varinjlim \mathcal{F}_n$, which has π_+ represented by the complex

$$\dots \rightarrow \mathcal{K}^1 \rightarrow \mathcal{K}^0/W^{\geq 0}\mathcal{K}^0 \rightarrow \dots$$

which is of width $< N$. This finishes the induction. $\square$

Corollary 3.2.4.15. Assume that $\mathbf{Perv}_m(Y)$ is graded highest weight. We have the following:

(1) The realization functor $\rho_{X,m}^{\text{mon}} : D(\mathbf{Perv}_m^{\text{mon}}(X)) \rightarrow D_m^{\text{mon}}(X)$ is an equivalence. Hence, we have

$$D_m^{\text{mon}}(X) = D\left(\mathbf{Mod}_{\check{S}_A, \text{Fr}}^{\text{nilp}}(\mathbf{Perv}_m(Y))\right)$$

(2) Suppose $D_m(Y) = \mathbf{Mod}_{E, \text{Fr}}$ for some E , there is an equivalence of categories

$$\hat{D}_m^{\text{mon}}(X) = \mathbf{Mod}_{E \otimes \check{S}_A, \text{Fr}}$$

Under this equivalence, the adjunction $\pi_+ \dashv \pi^\dagger$ becomes

$$- \otimes_{\check{S}_A} \overline{\mathbb{Q}}_\ell : \mathbf{Mod}_{E \otimes \check{S}_A, \text{Fr}} \rightleftarrows \mathbf{Mod}_{E, \text{Fr}} : \text{triv}$$

Proof: Firstly, $\rho_{X,m}^{\text{mon}}$ is essentially surjective, because both sides are generated by objects of the form $\pi^\dagger \mathcal{F}$ for $\mathcal{F} \in \mathbf{Perv}_m(Y)$ by definition. Then, for full faithfulness, it suffices to compute the Ext groups between objects of the form $\pi^\dagger \mathcal{F}$. On one hand, we have by the projection formula

$$\text{Ext}_{D_m^{\text{mon}}(X)}(\pi^\dagger \mathcal{F}, \pi^\dagger \mathcal{G}) = \text{Ext}_{D_m(Y)}(\mathcal{F}, \mathcal{G}) \otimes H^*(A)$$

On the other hand, we have

$$\text{Ext}_{D(\mathbf{Perv}_m^{\text{mon}}(X))}(\pi^\dagger \mathcal{F}, \pi^\dagger \mathcal{G}) = \text{Ext}_{\mathbf{Perv}_m(Y)}(\mathcal{F}, \mathcal{G}) \otimes \text{Ext}_{\check{S}_A}(\overline{\mathbb{Q}}_\ell, \overline{\mathbb{Q}}_\ell)$$

Now, we have $\text{Ext}_{D_m(Y)}(\mathcal{F}, \mathcal{G}) = \text{Ext}_{\mathbf{Perv}_m(Y)}(\mathcal{F}, \mathcal{G})$ since $\mathbf{Perv}_m(Y)$ is graded highest weight, and $H^*(A) = \text{Ext}_{\check{S}_A}(\overline{\mathbb{Q}}_\ell, \overline{\mathbb{Q}}_\ell)$ by the equivalence $\text{Ext}_{\check{S}_A}(\overline{\mathbb{Q}}_\ell, \overline{\mathbb{Q}}_\ell) = \bigwedge^*(V_A^\vee[-1]) = H^*(A)$.

For the second assertion, firstly note that under the assertion $D_m(Y) = \mathbf{Mod}_{E, \text{Fr}}$, we have

$$\mathbf{Mod}_{\check{S}_A, \text{Fr}}^{\text{nilp}}(\mathbf{Perv}_m(Y)) = \mathbf{Mod}_{\check{S}_A, \text{Fr}}^{\text{nilp}}(\mathbf{Mod}_{E, \text{Fr}}^\heartsuit) = \mathbf{Mod}_{(E \otimes \check{S}_A, \text{Fr})}^{\text{nilp}, \heartsuit}$$

we extend the equivalence $D_m^{\text{mon}}(X) = D\left(\mathbf{Mod}_{\check{S}_A, \text{Fr}}^{\text{nilp}}(\mathbf{Perv}_m(Y))\right)$ to the pro-categories. Then, we can identify $\mathbf{Mod}_{E \otimes \check{S}_A, \text{Fr}}$ as a full subcategory of $\text{Pro}(D_m^{\text{mon}}(X))$. On one hand, any object in $\mathbf{Mod}_{E \otimes \check{S}_A, \text{Fr}}$ is isomorphic to a complex of free objects, which shows that it is contained in $\hat{D}_m^{\text{mon}}(X)$. On the other hand, $\hat{D}_m^{\text{mon}}(X)$ is generated by the free objects by [Proposition 3.2.4.14](#), so the two categories are equivalent. The identification of the adjunction follows from

[Corollary 3.2.4.13](#). $\square$

To sum up, taking the category of completed monodromic sheaves over a trivial A -torsor is somehow taking the category of $\check{S}_A^\wedge$ -modules. We end this section with a treatment of the stratified space, which also gives the perverse t -structure over $\hat{D}_m^{\text{mon}}$.

Assumption 3.2.4.16. Now, we assume Y is finitely stratified by affine spaces, indexed by Λ . Then, suppose there is a map $\pi : X \rightarrow Y$, and we denote $X_\lambda = \pi^{-1}(Y_\lambda)$, and suppose each $X_\lambda \rightarrow Y_\lambda$ is a trivial A -torsor for some torus A . In this case, we define $\mathbf{Perv}_m(Y) = \mathbf{Perv}_{m, Y_\Lambda}(Y)$ to be the category of stratified perverse sheaves, where we additionally require to be constant along each strata (later, in our application, we will see that this condition is naturally satisfied).

We denote $i_\lambda : Y_\lambda \rightarrow Y$ and $\tilde{i}_\lambda : X_\lambda \rightarrow X$ denote the locally closed inclusions of strata.

Definition 3.2.4.17. Under [Assumption 3.2.4.16](#), we denote $\mathcal{L}_\lambda \in \mathbf{Perv}_m(Y_\lambda)$ to be the unique weight 0 object forgetting to the constant perverse sheaf in $\mathbf{Perv}(Y_\lambda)$. Furthermore, we denote

$$\tilde{\mathcal{L}}_\lambda = \text{Free}(\mathcal{L}) \in \hat{D}_m^{\text{mon}}(X_\lambda)$$

by the free monodromic sheaf. Moreover, we define

$$\Delta_\lambda = i_{\lambda,!} \mathcal{L}_\lambda, \nabla_\lambda = i_{\lambda,*} \mathcal{L}_\lambda$$

$$\tilde{\Delta}_\lambda = \tilde{i}_{\lambda,!} \tilde{\mathcal{L}}_\lambda, \tilde{\nabla}_\lambda = \tilde{i}_{\lambda,*} \tilde{\mathcal{L}}_\lambda$$

by the standard and costandard objects.

Lemma 3.2.4.18. The category $\hat{D}_m^{\text{mon}}(X)$ is generated separately by $\tilde{\Delta}_\lambda$'s and $\tilde{\nabla}_\lambda$'s.

Proof: By the definition of a recollement, any $\mathcal{F} \in \hat{D}_m^{\text{mon}}(X)$ is an extension of $\tilde{i}_{\lambda,*} \tilde{i}_\lambda^! \mathcal{F}$'s (or $\tilde{i}_{\lambda,!} \tilde{i}_\lambda^* \mathcal{F}$ respectively). On the other hand, by [Proposition 3.2.4.14](#) we have $\tilde{i}_\lambda^! \mathcal{F}$ or $\tilde{i}_\lambda^* \mathcal{F}$ generated by free objects, hence by copies of $\tilde{\mathcal{L}}_\lambda$. Thus $\mathcal{F}$ is generated by the $\tilde{\Delta}_\lambda$'s or $\tilde{\nabla}_\lambda$'s separately. $\square$

Lemma 3.2.4.19. There is a natural perverse t -structure on $\hat{D}_m^{\text{mon}}(X)$.

Proof: Note each $\hat{D}_m^{\text{mon}}(X_\lambda) \simeq \mathbf{Mod}_{\tilde{\mathcal{S}}_\lambda, \text{Fr}}$ by [Corollary 3.2.4.15](#), so admits a t -structure. Then, by the general theory of gluing t -structures, these t -structures glue to a perverse t -structure on $\hat{D}_m^{\text{mon}}(X)$. By general categorical nonsense, this t -structure can also be identified with the restriction of the t -structure on $\text{Pro}(D_m^{\text{mon}}(X))$. $\square$

We denote the heart of this t -structure by $\widehat{\mathbf{Perv}}_m^{\text{mon}}(X)$.

Definition 3.2.4.20. We say $\mathcal{F} \in \hat{D}_m^{\text{mon}}(X)$ is a free monodromic tilting sheaf if for all $\lambda \in \Lambda$, we have $\tilde{i}_\lambda^* \mathcal{F}$ and $\tilde{i}_\lambda^! \mathcal{F}$ a direct sum of $\tilde{\mathcal{L}}_\lambda$'s. We say $\mathcal{F} \in \hat{D}_m^{\text{mon}}(X)$ is a free monodromic tilting sheaf if its underlying sheaf in $\hat{D}^{\text{mon}}$ is.

Lemma 3.2.4.21. An object $\mathcal{T} \in \hat{D}_m^{\text{mon}}(X)$ is free monodromic tilting iff $\pi_{\dagger} \mathcal{T} \in D_m(Y)$ is a tilting sheaf.

Proof: By the identification [Corollary 3.2.4.15](#), $\mathcal{T}$ is perverse iff $\pi_+ \mathcal{T}$ is. The tilting part is obvious. $\square$

Proposition 3.2.4.22. By [Proposition 3.1.1.14](#), if $\mathbf{Perv}_m(Y)$ is graded highest weight, we have tilting objects $\mathcal{T}_\lambda$ whose restriction to Y_λ is $\mathcal{L}_\lambda$, and is indecomposable. Then, there exists free monodromic tilting sheaves $\tilde{\mathcal{T}}_\lambda \in \hat{D}_m^{\text{mon}}(X_{\leq \lambda})$ such that $\pi_+ \tilde{\mathcal{T}}_\lambda = \mathcal{T}_\lambda$.

Proof: We proceed by induction on strata. Assume that X has a minimal stratum Z , and suppose that the required sheaf has been constructed on $X - Z = U$. We temporarily denote this object by $\tilde{\mathcal{T}}_U$, where we have $\pi_+ \tilde{\mathcal{T}}_U = \mathcal{T}_\lambda|_U$. We denote the inclusions $\tilde{j} : U \hookrightarrow X$ and $\tilde{i} : Z \hookrightarrow X$.

We consider the object $\text{cofib}(\tilde{j}_! \tilde{\mathcal{T}}_U \rightarrow \tilde{j}_* \tilde{\mathcal{T}}_U)$. Note by applying π_+ , it becomes $\text{cofib}(j_! \mathcal{T}_\lambda|_U \rightarrow j_* \mathcal{T}_\lambda|_U)$. Since this object is zero on U , it is represented by some $i_* \mathcal{F}[1] \oplus i_* \mathcal{G}$ for $\mathcal{F}, \mathcal{G} \in \mathbf{Perv}_m(\pi(Z))$ (it is in fact a cofiber, but one may check that the differential is 0). We must now add $\mathcal{F}$ and $\mathcal{G}$ into $\tilde{\mathcal{T}}_U$.

Note that the above implies $\text{cofib}(\tilde{j}_! \tilde{\mathcal{T}}_U \rightarrow \tilde{j}_* \tilde{\mathcal{T}}_U)$ is represented by $\tilde{i}_* \tilde{\mathcal{K}}^{-1} \rightarrow \tilde{i}_* \tilde{\mathcal{K}}^0$, where $\tilde{\mathcal{K}}^i \in \widehat{\mathbf{Perv}}_m^{\text{mon}}(Z)$, and $\pi_+ \tilde{\mathcal{K}}^{-1} = \mathcal{F}$, $\pi_+ \tilde{\mathcal{K}}^0 = \mathcal{G}$. Thus, consider the extension class

$$\tilde{i}_* \tilde{\mathcal{K}}^0 \rightarrow \text{cofib}(\tilde{j}_! \tilde{\mathcal{T}}_U \rightarrow \tilde{j}_* \tilde{\mathcal{T}}_U) \rightarrow \tilde{j}_! \tilde{\mathcal{T}}_U[1]$$

We take $\tilde{\mathcal{T}}_\lambda$ to be the fiber of the above map (i.e. the extension class). By applying π_+ to the above diagram, the desired equivalence holds. $\square$

Proposition 3.2.4.23. Let $\tilde{\mathcal{T}}_1, \tilde{\mathcal{T}}_2 \in \text{Tilt}(\widehat{\mathbf{Perv}}_m^{\text{mon}}(X))$. Then, $\text{Hom}_{\hat{D}_m^{\text{mon}}(X)}(\tilde{\mathcal{T}}_1, \tilde{\mathcal{T}}_2)$ is a free $\hat{S}_A$ -module, and there is a Fr-equivariant isomorphism

$$\text{Hom}_{\hat{D}_m^{\text{mon}}(X)}(\tilde{\mathcal{T}}_1, \tilde{\mathcal{T}}_2) \otimes_{\hat{S}_A} \overline{\mathbb{Q}}_\ell = \text{Hom}_{D_m(Y)}(\pi_+ \tilde{\mathcal{T}}_1, \pi_+ \tilde{\mathcal{T}}_2)$$

Proof: We prove a stronger version where $\tilde{\mathcal{T}}_1$ is only assumed to have a standard filtration and $\tilde{\mathcal{T}}_2$ a costandard filtration.

For the case where there is only one stratum, the assertion follows directly from

[Corollary 3.2.4.15](#). For the general case, we proceed by induction on strata. Assume that X_λ is an open stratum and the statement holds for $X_{< \lambda}$. Then, we have the exact sequence

$$0 \longrightarrow \text{Hom}_{\hat{D}_m^{\text{mon}}(X_{< \lambda})}(\tilde{i}_{< \lambda}^* \tilde{\mathcal{T}}_1, \tilde{i}_{< \lambda}^! \tilde{\mathcal{T}}_2) \longrightarrow \text{Hom}_{\hat{D}_m^{\text{mon}}(X_{\leq \lambda})}(\tilde{\mathcal{T}}_1, \tilde{\mathcal{T}}_2) \longrightarrow \text{Hom}_{\hat{D}_m^{\text{mon}}(X_\lambda)}(\tilde{i}_\lambda^* \tilde{\mathcal{T}}_1, \tilde{i}_\lambda^* \tilde{\mathcal{T}}_2) \longrightarrow 0$$

Now, the left and right Hom-spaces satisfy the relation required by the induction hypothesis, hence so does the middle, by drawing the same exact sequences for $\pi_+ \tilde{\mathcal{T}}_i$'s. $\square$

3.2.5. Application 2: The monodromic (tilting) side

4. The Bezrukavnikov-Arkhipov equivalence

See my thesis.

5. Mixed hodge modules

5.1. Pure and mixed hodge structures

5.1.1. Preliminaries on filtered objects

Definition 5.1.1.1. We fix a category $\mathcal{C}$. We define the category of filtered to be the category of functors $F : \mathbb{Z} \rightarrow \mathcal{C}$ such that each morphism is injective. A filtration on an object $c \in \mathcal{C}$ is a functor $\mathbb{Z} \rightarrow \text{Sub}_c$, the category of subobjects of c .

We define a morphism of filtered vector spaces to be $f : c \rightarrow d$, such that $f(F_i c) \subset F_i d$. We say the morphism is strictly compatible with the filtration if we have an exact sequence

$$0 \rightarrow \text{gr ker } f \rightarrow \text{gr } V \rightarrow \text{gr } W \rightarrow \text{gr coker } f \rightarrow 0$$

Equivalently, we must have $f(F_n c) = f(c) \cap F_n d$.

We define a decreasing filtered object to be $\mathbb{Z}^{\text{op}} \rightarrow \mathcal{C}$. We denote decreasing filtrations by $F^\bullet c$, while increasing filtrations by $F_\bullet c$. The translation between increasing filtered objects and decreasing filtered objects is given by the convention $F^n c = F_{-n} c$.

Construction 5.1.1.2. Let $F_\bullet c \in \mathbf{Alg}(\text{Fil}(\mathcal{C}))$, then we define the Rees construction of c to be the graded ring

$$\tilde{c} = \bigoplus_{p \in \mathbb{Z}} F_p c \lambda^p$$

which is a graded $\mathbb{1}_{\mathcal{C}}[\lambda^{\pm 1}]$ -algebra.

Warning 5.1.1.3. For an Abelian category $\mathcal{A}$, the category of filtered objects $\text{Fil}(\mathcal{A})$ is generally not Abelian. This can be seen in the example in **FilVect** $_{\mathbb{C}}$: take $F^\bullet V$ with $F^i V = \mathbb{C}$ for $i < 0$ and 0 for $i \geq 0$, and $F^\bullet W$ with $F^i W = \mathbb{C}$ for $i \leq 0$ and 0 for $i > 0$. Then, $f : F^\bullet V \rightarrow F^\bullet W$ corresponding to id on the underlying space has trivial kernel and cokernel, but is not an isomorphism.

However, the above example is the only part where Abelianess fails. If we take the wide subcategory $\text{Fil}(\mathcal{A})^{\text{str}}$ with morphisms strictly compatible with filtrations, we get an Abelian category. In fact by [Sta26] 05SI, we have a morphism is strict iff its coimage coincide with the image, and other axioms for Abelian categories holds already for filtered objects by [Sta26] 0122.

5.1.2. Basic definitions

Definition 5.1.2.1. We define a pure hodge structure of weight w to be a finite dimensional $\mathbb{C}$ -vector space $\mathcal{H}$ equipped with two decreasing filtrations $F'^\bullet \mathcal{H}$ and $F''^\bullet \mathcal{H}$, such that

$$F'^p \mathcal{H} \oplus F''^{w-p+1} \mathcal{H} = \mathcal{H}$$

Equivalently, $\mathcal{H}$ admit a bigraded direct sum decomposition

$$\mathcal{H} = \bigoplus_{p \in \mathbb{Z}} \mathcal{H}^{p, w-p}$$

The two definitions can be translated to the other version as follows:

$$\mathcal{H}^{p,w-p} = F'^p \mathcal{H} \cap F''^{w-p} \mathcal{H}, \quad F'^p \mathcal{H} = \bigoplus_{q \leq p} \mathcal{H}^{p,w-p}, \quad F''^{w-p} \mathcal{H} = \bigoplus_{q \geq p} \mathcal{H}^{p,w-p}$$

We define a morphism of Hodge structures to be a linear map between the underlying vector spaces compatible with both filtrations. We denote the category of Hodge structures pure of some weight to be $\text{HS}(\mathbb{C})$, and the category of Hodge structures pure of weight w to be $\text{HS}(\mathbb{C}, w)$.

Construction 5.1.2.2. $\text{HS}(\mathbb{C})$ admits a closed monoidal structure. For $\mathcal{H}_1$ of weight w_1 and $\mathcal{H}_2$ of weight w_2 , we may define $\mathcal{H}_1 \otimes \mathcal{H}_2$ to be the pure Hodge structure of weight $w_1 + w_2$, with

$$(\mathcal{H}_1 \otimes \mathcal{H}_2)^{k, w_1 + w_2 - k} = \bigoplus_p (\mathcal{H}_1^{p, w_1 - p} \otimes \mathcal{H}_2^{k-p, w_2 - k + p})$$

Then, for $\mathcal{H}$ pure of weight w , we may define the dual $\mathcal{H}^\vee$ pure of weight $-w$ by

$$(\mathcal{H}^\vee)^{-k, -w+k} = (\mathcal{H}^{k, w-k})^\vee$$

Thus, we have $\text{Hom}(\mathcal{H}_1, \mathcal{H}_2) = \mathcal{H}_1^\vee \otimes \mathcal{H}_2$, which is pure of weight $w_2 - w_1$.

Then, we define the complex conjugate of $\mathcal{H}$ to be a Hodge structure of the same weight, with $\overline{\mathcal{H}}^{k, w-k} = \overline{\mathcal{H}^{k, w-k}}$. For the Hermitian dual, we define $\mathcal{H}^* = \overline{\mathcal{H}^\vee} = \overline{\mathcal{H}}^\vee$.

Proposition 5.1.2.3. $\text{HS}(\mathbb{C}, w)$ is a semisimple Abelian category. $\text{HS}(\mathbb{C})$ is not Abelian.

Proof: We prove a morphism in $\text{HS}(\mathbb{C}, w)$, compatible with both filtrations, is strictly compatible with both filtrations. Notice first that $f : \mathcal{H}_1 \rightarrow \mathcal{H}_2$ preserves the direct sum decomposition, namely $f(\mathcal{H}_1^{p, w-p}) \subset \mathcal{H}_2^{p, w-p}$. This is equivalent to the strict compatibility. $\square$

Hence, the above caveat shows that $\text{HS}(\mathbb{C})$ is not the correct category to consider. Instead, we must pass to the Rees construction to give the correct definition.

6. Nori motives

7. Appendix

7.1. Filtered objects

Bibliography

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